

Bio 101 Basics of Biology

Numbered Transcript

BIO 101 • Dr. Benjamin Bikman

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Lecture 1: The Cell: Life's Basic Unit

At a glance — Bikman introduces the cell as life's basic unit.

- 1.1** Hello, and thank you so much for joining me. I'm Professor Ben Bikman. Today, the material that we are covering is biology. And I've entitled this course Basics of Biology, because biology is a topic that is so substantial, that is so broad and deep, that we would need an enormous amount of time in order to get through all of it. So in the time that we have together, what I want to cover are the basics, the fundamental pieces that go into biology. But before we go too far, let's take a moment and understand what biology is.
- 1.2** Biology, as a scientist, I define it as the study of life and living organisms. And within that definition is this inherent understanding that there is an investigation. There are cell biologists, actual biological scientists like me, where we are understanding, we're seeking to understand, the complex interactions that make life possible, at whatever level it is that we are studying. Whether we are starting at the cell, which is the most fundamental unit of life, or whether it's the unit or the level of the cells coming together to form a tissue, or whether it is the tissue working within a physiological system, or whether it's the entire organism, biology encompasses all of that.
- 1.3** Now, the first thing we're going to get into then is actually starting at that base unit of the cell. But in the time that we have together, we're going to cover four fundamental aspects or basics in biology. The first course or lecture that we're going to go through is the cell itself. So by the end of this immediate lecture, you're going to understand the cell, the nature of the cell, what it is, how do we define the life and the growth of the cell. Next, the second topic of the basics we'll cover is central dogma. This is this grand idea that explains how we go from a genetic piece of information in the DNA through how that genetic information flows through this process that actually gives life to an organism and makes us who we are.
- 1.4** The third of the basics that we're going to cover is cell cycle, the living and even the dying of the cell, how a cell will grow, how it will divide, and this is what makes us have the organism that we call our bodies. And then last, and I confess, my favorite topic, is the energy within a cell. How does a cell get energy? After all, it needs energy. Like a locomotion, like an automobile that needs some kind of fuel to move it, the cell needs fuel to give it its life and its work, its ability to get something done.
- 1.5** All right. So this first lecture, I have a few key learning outcomes that I'm going to want to get through with you. The first thing, by the end of this lecture, I want to make sure that you are capable of describing the aspects of cell theory. Second, I want you to appreciate the different families of life at a cellular level, the difference between prokaryotes and eukaryotes. And then third, I want you to understand the cellular compartments or the components that make up the cells that give life to every animal

and plant on the planet. And then, fourth and final within this general topic of the cell, I want to make sure that you have an appreciation for osmosis, diffusion, and gradients, because these are central aspects that enable a cell, that not only define the cell, but enable it to actually do stuff to get work done.

1.6 Now, firstly, before we go too far, let's just pause for a moment. Again, I've already said that one time, but let's define life. What does it mean for something to be alive? Think about the room you're in, the space that you're in right now. Look around and consider, of the things that I'm seeing, what is alive? For example, is a cell alive? Is a virus alive? Is an animal alive or a seed alive? Think about these things. We're going to revisit some of these in just a moment.

1.7 But because I'm a scientist, I want some criteria. I want some list of things that we could go through and check and say, if all of these criteria are met, then we can answer yes or no, depending on whatever the item is that we're trying to determine whether it's alive or not. So let's go through seven specific criteria that you could use to define or to determine whether something is actually living.

1.8 First of all, does it have some aspect of a cell to it? For something to be alive, it must have at least one cell. So one or more cells is going to be the first of the criteria to define whether something is living. Second, does it have some aspect of metabolism? Is there some need, is there an ability within this thing that we're trying to determine whether it's alive or not to take in energy, to metabolize, to transform that molecule, that energy molecule, into something that the cell can use to get some work done? And then can it eliminate waste? That's, after all, a very important part of metabolism.

1.9 Third, the third of the seven criteria, is it capable of growing and developing? Does it follow some kind of genetic instruction for growth and have a programmed process for determining that growth? When is it time to grow? When is it not time to grow? And then fourth, is it capable of reproduction? Can it make new versions of itself? That is an essential aspect of whether something is living. Everything that is living is determined to reproduce, indeed, to the point that that is an essential criterion.

1.10 Fifth, is it capable of adapting? And by that, I mean, is it capable of evolving in response to its environment? Can it sense the environment and then start to adapt in order to ensure its own survival? Because after all, everything that is alive wants to live, and adaptation is an essential component to that. Related to that one, the sixth is determining whether this thing is capable of responding to stimuli. If something happens around it, is it capable of determining this, and responding? And that would be, of course, lead to the adaptation over time.

1.11 And then the final component, and this is really high-level, but so important, is homeostasis. So if something is alive, it must be able to, in response to these stimuli, have this ability to change some aspect of itself in order to maintain an internal constancy. It wants to maintain its current function. That is homeostasis. Now,

homeostasis is a little bit of an ambiguous point, or it can be a little difficult to understand. So I want to take a moment and define it a little further. Just think about it at the level of the human body. We know we are alive. We meet all of these criteria; the human body does.

- 1.12** So how does a human body respond to changes in temperature? If an environment is cold or if it's hot, you get into a hot—let's say a cold bath. Everyone loves to cold-plunge these days. It's so popular. You get into that cold bath, it is going to be substantially lower temperature than you. If your body matched that temperature, you would die. So you must have an ability to keep your body back at a normal range, at a normal temperature.
- 1.13** Similarly, I am currently now filming this in Florida, but I'm coming from the mountains of Utah. That is a substantial change in elevation, especially going from Florida to Utah, where the oxygen is relatively more modest in the atmosphere. The body makes certain adaptations in order to get all of the oxygen it needs from a relatively thinner environment, atmospherically. So homeostasis is that essential process of maintaining—the living thing being able to maintain an internal constancy or keep itself constant.
- 1.14** Now, an obvious example of this around us or in our daily living, an obvious example of homeostasis would be our home having a thermostat. The thermostat is the most obvious or simplest example of something that's trying to seek homeostasis. Now, I'm not suggesting the thermostat is alive, but rather the thermostat provides a good metaphor, where if the temperature within the room starts to get too cold, the thermostat will sense this, and it will kick on the furnace, thereby bringing the temperature back up to where the thermostat wants it to be.
- 1.15** In contrast, certainly here in the muggy hot environment of Florida, if things start to get too hot, then the thermostat will sense this departure from the norm, and it will kick on the air conditioner. Then the air conditioner will seek to bring the temperature back down. So that's just an example, of course, of homeostasis. Now, let's bring that example or the metaphor of the thermostat and then assume some aspect of it in the body or wonder how it could apply to something like the human body. I'd mentioned the cold bath earlier. If the body were immersed in this really, really icy cold water, that temperature change would be lethal. And so the body has a mechanism to respond in order to maintain homeostasis or the temperature set point.
- 1.16** And that would be, it's going to start to shiver. So shivering would be a mechanism as the temperature is starting to drop. No, let's snap it right back up and maintain homeostasis. In contrast, if we are in a very warm environment, things are getting too hot. The best way for the body to cool down is to sweat. So we start to sweat, and then the sweat begins to evaporate, and the evaporation will create a chilling effect on the skin, thereby allowing the body to try to keep that temperature from climbing too high.

- 1.17** One last metaphor would be imagining a conductor. I had the great fortune in my childhood to play in lots of bands, and my mom was determined to have all of her children learn musical instruments. And boy, I fought it at the time, but I'm very grateful now. That's so stereotypical. But I remember looking at the conductor, and I was a trumpet player, and boy, the trumpets can get so loud. Well, if the trumpet section, if the brass section is getting too loud, the conductor will respond to that, knowing things are getting a little too rowdy in the brass section. And so there may be some sign to turn the volume down.
- 1.18** And so the brass section would quiet down, and maybe he needs more of the woodwinds. And so there'd be some kind of gesture that would coordinate all of these processes to work together. All of the band members and all of their distinct sections would know what to do, whether to play faster or slower, louder or quieter, based on what the conductor was telling them, the conductor seeking some set point or homeostasis.
- 1.19** Now, let's revisit some of the examples I mentioned at the beginning when we were wondering how to define life, keeping these seven criteria in mind. So first of all, a cell. Is a cell alive? Well, it obviously meets the first of the criteria, which is that it must be composed of at least one cell. One cell is one cell, and so it meets that criteria. And in fact, it does meet all of them. It is capable of reproducing. It's capable of responding to its environment. It's capable of adapting, etc. So a cell is alive.
- 1.20** And then that's no surprise, of course, to then say, as we expand on that cell or put a bunch of cells together, whether it is an animal or whether it is a plant, all of those things are going to be alive as well. Even let's start at the very base or the first evidence of a plant or the first instance of it, which would be a seed. Is a seed alive? Think about that for a moment. The seed is capable of reproducing. It is capable of responding to its environment after all. If it's not planted or watered, it won't grow. Once it meets these certain criteria, then it will know. Once it has the proper environment, it can respond to that and grow accordingly. So yes, even a seed is considered alive.
- 1.21** What about a virus? A virus seems to be alive, right? We sometimes think that it might be similar to a bacteria, which is its own little cell, and we'll define bacteria in just a moment. But in fact, a virus is not alive. It does not meet all of the criteria. They don't grow. They do not maintain—a virus does not maintain homeostasis. It cannot reproduce on its own. The only reason we talk about viruses at all is because of their ability to get into a cell and hijack the cell, and the cell is alive. So in that sense, the virus seems to be alive, but on its own, it is not. It is not a living thing. It does not meet the criteria.
- 1.22** And this, of course, would be similar to things like a rock. A rock is not alive. The walls are not alive. These are not things that are responding to their environment. They're not adapting. They don't have any kind of metabolic process. So at this point, you should be fairly comfortable with understanding the criteria of what goes into

determining whether something is alive. And with this, let's transition to the next subtopic here, which is the cell theory.

- 1.23** Given that everything that is alive must be composed of at least one cell, the cell theory is a fundamental concept in biology that essentially articulates the properties and significance of cells within every living organism. In other words, it's just emphasizing the relevance of the cell. And within this cell theory, there are three main principles. One, and this is redundant, that all living organisms must be composed of at least one cell. That's something we already discussed.
- 1.24** The second aspect of cell theory is that cells are the smallest units of life. That if we get down to the most fundamental or basic aspect of life, we come to the cell. And beyond that, there's going no further. There's nothing more fundamental to life. There's no unit that is considered more basic or smaller than the cell. That's the second aspect. Again, the cells are the smallest units of life. And then lastly, the third aspect of cell theory is that all cells, all new cells, come from pre-existing cells. And that's the aspect of cell division, which is something we're going to talk about when we discuss the basics of mitosis and meiosis in the third of these lectures, as we get through the basics in biology.
- 1.25** Now, very briefly, I've mentioned, within that cell theory, it's that all living organisms are composed of at least one cell. I've already alluded to this, but I really want to bring the point home that organismal function, so the whole, whatever the unit of life we're looking at, a tissue or an animal or the human body, that they are composed of very organized or specialized cells that have lumped together to form tissues. For example, the kidney cells. Those are very unique types of cells that have grown together in a particular structure. The liver cells, those are cells that are all the same that have grown together. Muscle cells, skin cells, etc.
- 1.26** So the organism functions at the level of the tissues, which are then composed of cells. So to understand the organismal function, you have to understand the role of the specific cells that are involved. And then cells do have very specific functions, whether it's something as unique as sperm, and we'll talk all about the gametes or these germ cells, sperm and ova, or eggs; whether it's the cells of the kidney, which have an incredibly unique structure, enable them to filter out certain molecules and neurons and other cell types.
- 1.27** Now, because the cell is the smallest unit of life and the tissues are composed of those cells, they all must learn to work together. And that's another aspect of the cell theory. So at this point in the lecture so far, you should be able to understand or describe what it means for something to be alive, as well as appreciating and understanding the three basic aspects of cell theory and why they're important.
- 1.28** With that information in mind, let's transition into this next level of understanding life by breaking up the cells into two main families, the prokaryotes and the eukaryotes.

Now, as a little mnemonic or a teaching device, the eukaryotes are what you are made of. Now, let's get into it then. So two categories, prokaryotes and eukaryotes. Within the prokaryote family, so all of this you could imagine through this theory of evolution, emphasis on this theory, we don't exactly know how life started. We don't exactly know how humans came to be humans. The idea, the prevailing idea, is that it is a theory of evolution, that if we take all life, we could go backwards in time, and then the tree of life would start to get narrower and narrower and narrower, coming down to one singular point.

1.29 Within that one singular point, if we then reverse direction and start going back up, we would immediately get into two distinct families. And the first being prokaryotes. Now, the prokaryotes are plural. There are two lines or two smaller family lines that fit within prokaryotes. Bacteria, which you've all heard of, and we'll discuss very briefly, and then the archaea. Archaea is a type of prokaryote that most people don't really appreciate. And then lastly, we have the eukaryotes. And again, the eukaryotes are what we're focusing on the most throughout this series, because that's what you are made of. And we want to know more about ourselves in the role of biology within the human body, or at least I want you to.

1.30 Now, one interesting thing about the eukaryote, and we will dive into this in much more detail very soon, the eukaryote family is very unlike the prokaryotes. Yes, they're both alive. Both families meet all of the criteria of life. But the eukaryotes have this—if you zoom in, and I'm looking at a prokaryote, it would just look like a plain little sac. There's nothing really to identify in it. However, with the eukaryotes, we see these very distinct sections, these organelles, as they're known, and we'll define them more in just a moment.

1.31 But there's an interesting theory that I just wanted to teach you because it is so fascinating, which is our ability or our effort to try to understand, where did the eukaryotes come from? What was responsible for such a significant departure in these cells? Why do some cells have such a basic appearance and function, and another cell type is much more complicated? This is the theory of endosymbiosis. So the endosymbiotic theory, again, theory, is that at some point a prokaryotic cell, a prokaryote, a very simple, just big bubble, with nothing really to identify it or make it unique if we looked at it under a microscope—the prokaryote, at some point in its development, would have had some kind of invagination or some pulling in of the membrane itself.

1.32 So we have, the cell is surrounded by a membrane, and then the membrane started to collapse in on itself, resulting in these little folds and grooves. That became part of the nucleus and other organelles that we'll discuss. But then at the same time, and this is really where the endosymbiosis comes into play, endo meaning it's taking it in, symbiotic meaning it's two things working well together, this then is a prokaryotic cell that pulled in a bacteria. And this bacterium then became the mitochondria, or the

energy-producing parts of the cell. The mitochondria, and as we discuss later on in this series in the cellular energy part, the chloroplast, which is kind of a unique plant version of a mitochondria.

- 1.33** So the mitochondria represents a totally unique part of the eukaryotic cell. And that idea, the origin of how that came to be there, with its own unique DNA, is that this was a prokaryote that engulfed another prokaryote, and them finding, "Hey, we get along really well together," this is a symbiotic relationship. Let's just keep this going. And thus we gave birth or we have the birth of the eukaryotic cell, which makes up you. All right, now let's back up for just a moment.
- 1.34** I'd mentioned the archaea, and the archaea are the most unknown and certainly worldwide, the smallest in abundance. But the archaea, basically, are these small little cells, single-cell organisms, that only live in environments where nothing else can live. In other words, even bacteria couldn't live here. And these are things where it is, for example, blisteringly hot in certain areas, where you may have hot springs that are temperatures where no other bacteria could live. Or it is areas where the salt content is so high, like the Dead Sea or the Great Salt Lake in Utah, that is so salty that it would kill bacteria. But these archaea are built, they're designed to survive in these very hostile environments.
- 1.35** Now, bacteria, of course, is something we're much more familiar with, and rightly so, because if we look at all of the trillions of cells that make up our body, and we have of our own body type cells, about 30 trillion. But for almost one to one, for every one of our cells, eukaryotic animal cells, there is a bacterium or one bacteria cell. And the bacteria are everywhere. You certainly, everyone has heard about the microbiome. That's a very popular topic nowadays. So our digestive tract is loaded with bacteria, but so too, not only on the inside, but even the outside. Our skin is covered in bacteria.
- 1.36** Now, lest you think that's a problem, it is a good thing. It's a healthy thing, both outside and inside. Once upon a time, there was an idea that the number of bacteria on or in the human body was 10 times higher, and that idea has been really changed and challenged. And we generally nowadays accept it as about a 1 to 1. So about 30 trillion of us cells and then 30 trillion of them, but certainly worked in to the normal physiology or the normal biology of the human animal. So we have bacteria everywhere. Unlike the archaea, which are only living in hostile environments, bacteria are everywhere else, on the planet and on the animals and plants.
- 1.37** Now, speaking of animals and plants, that gets us to the eukaryotes. So the eukaryotes are the form of life that you most recognize. These are animals, these are plants, these are little things living in the pond if you get a scoop of pond water and you see little things moving around; those are all going to be eukaryotes. So most of what you think of when you think of life are going to be eukaryotic cells. Now, let's define. I've been alluding to some of the differences between prokaryotes and eukaryotes. Prokaryotes are very simple cell structures. I've been using this analogy of a bubble. We need to get

no more complicated than that. There's no nucleus. There are none of these other components or compartments within the cell. If we looked at them under a microscope and really zoomed in, it would look like it's just a small bubble. Now, there can be different shapes and sizes of it, but there'd be little substance to the center of it.

1.38 In contrast, the eukaryote is a much more complex cell structure. It has DNA, which is, rather than just floating around, the DNA floating around, and we'll define DNA soon, the DNA is just floating around within the prokaryote. There's no nucleus to encapsulate it or envelope it. Within the eukaryotes, there's a well-defined nucleus that is protecting and holding on to the DNA, keeping it very enclosed. There are also other organelles that we would see surrounding or in some other part of the cell. All right, so at this point now, you should be able to identify and understand some of the differences between prokaryotic and eukaryotic cells.

1.39 As I mentioned, one of the key distinguishing features between them is the presence of what's called organelles. So now let's talk about the organelles. I want to make sure that you understand the various components of the eukaryotic cell, which we call the organelles. It's almost as if, within the human body, we have distinct organs. Within the cell, there are this cellular microscopic version. There are different parts of it that are doing different things. The heart is doing something different from the liver, which is different from the kidneys. So too, the nucleus has a different role than the mitochondria has, which itself has a different role from the endoplasmic reticulum. And you'll know all about those in a moment.

1.40 Now, cells are little factories. They have, just like within a factory, the cell has some instructions or protocols, a list of orders of things to get done. It has tools that it needs to use in order to build machines within the cell. And then it has the machines itself, the things that are organizing and working to get the cell to do what it needs to do. It has an energy source. A factory needs some energy source to fuel all of the working or to power all the work. So too does a cell. It has some form of internal management of the energy, storing it or burning it.

1.41 It also has a compartmentalization. In other words, it's all the distinct little sections that I've been talking about. Within a factory, there'd be some piece of machinery over here producing some component. That component may then go to this other big production line or some other piece of machinery. And then lastly, everything is built on site, just like a good factory that is capable of going from base product to finished product, the cell has the same capability.

1.42 Now, within the cell itself, and again, remember, we're talking about eukaryotic cells, there are some distinct parts of this that I want to cover with you for just a moment over the next few minutes. The nucleus, the mitochondria, the endoplasmic reticulum, the centrosome, which is very important to a process later on, peroxisome, and the Golgi apparatus. Let's define each of these briefly. The nucleus is something we're going to be revisiting later because it is so important in the central dogma or the

movement of genetic information, the flow of this genetic information that ends up creating life as we know it and giving definition and unique qualities to each of us as humans.

- 1.43 So the function of the nucleus is to contain DNA, and the DNA itself stores our genetic code. And also within the nucleus, we have the beginnings of the replication or the reproduction capability. Remember, that was an essential quality of life. The thing must be able to reproduce. So DNA gets transcribed into RNA, and you will understand that process very soon. And that's a necessary step, this central dogma or the movement of genetic information. In this case, quite literally, from within the nucleus as a DNA genetic code and then moving out.
- 1.44 Now, when we move out along the nucleus itself, in addition, the nucleus is more important than just holding on to DNA. One interesting part of the anatomy of the nucleus is that it has these teeny little pores. And the pores themselves are much too small for the DNA to get out, which is good. It's small enough that it will only let out what can meet the physical constraints of such a small little pore. And that's going to be the RNA, as we discuss, the central dogma in the next lecture among these basics.
- 1.45 Now, if we go out of the nucleus, we get into the part of the cell that we can call the cytoplasm. The cytoplasm incorporates two main components. It's not only the water that is within the cell. A cell does have water. It's like a little water balloon. It does have water. That's called cytosol. But then it has all of the other organelles that are non-nucleus. So everything that we're about to get into now, all the other components of the cell, all the other organelles, are part of what's called the cytoplasm. So as we've left the nucleus, we've now entered the cytoplasm. And within that is the swimming water of the cytosol. But then there are all these other components collectively that constitute the cytoplasm.
- 1.46 So the first of these, in fact, in order that we would bump into in this case, as we come right out of the nucleus, the nucleus is surrounded by these flattened, squishy sacs. And it is a little awkward, where imagine if we had a long tube and then we started to press it together. But as we were pressing it together, it started to have these little folds. It was folding in on itself. That and then it was surrounding the nucleus. So we had this tube. We start to compact it. That is the endoplasmic reticulum. So the endoplasmic reticulum are these flattened membrane-wrapped or membrane-bound, membrane-wrapped organelles. And again, it surrounds the nucleus, surrounding the nucleus on purpose, because it is a necessary part of taking the genetic information from the nucleus, and it must pass through the ER.
- 1.47 The ER itself has two parts, even though it's all connected; it is all one connected tube or network, but it has a smooth part of it itself, and it has a rough part of itself. And they do have different functions, even though it's all part of the same compressed, flattened, pancakey, stacked structure. It's all connected, but they have two different parts. The smooth part is involved in producing fats, and we'll talk all about this in the

cellular energy component. It is storing calcium, and cells need calcium. Some need them very importantly, like muscle cells or nerve cells. They have this specialized version of the smooth ER that is basically just acting as these big calcium sacs.

- 1.48** And also the smooth ER is very important in detoxification. There may be some molecules within the cell that the cell needs to get rid of or it needs to neutralize, and the smooth ER can help with that. Then the rough ER, we call it rough because when you look at it under a microscope, it looks like it kind of has a fuzzy edge. It doesn't actually have a fuzzy edge. It's just covered with these little things called ribosomes. Now, let's talk about the ribosome for just a moment. We're going to revisit the ribosome when we talk about central dogma in the flow of genetic information. This is how we get from DNA to actual protein, which is the final product. The ribosome is an essential part of this. So the ribosome is lining the rough ER. It's in the perfect position to get this genetic information from RNA and then to start to turn it into what will become a protein.
- 1.49** Adjacent next to the ER, so the ER is the compressed, pancake, stacked network around the nucleus, adjacent to the ER is something called the Golgi apparatus. It is outside and around in distinct sections, the ER, because it's also part of moving things out. If something is originating from the nucleus and it's moving out on its way to get out of the cell, it then has to pass through the Golgi. And the Golgi is going to be involved in some of what's called post-translational processing. Now, the word translation in the context of biology at the moment doesn't make a lot of sense, but it will very soon. But translation, suffice it to say, is just part of the movement of the genetic information from the nucleus on its way out of the cell.
- 1.50** So it's essential in packaging things up, to make it suitable for getting to the edge of the cell and then coming out. Now, the most famous of the organelles may be second only to the nucleus. The nucleus is probably the most famous. But for a guy who studies energy like me, as an obesity scientist, I focus mostly on the mitochondria. So the mitochondria is the next organelle to talk about, and that is the essential powerhouse of the cell. That's how we always learn about it in basic biology.
- 1.51** It produces the main molecules of energy that enable the cell to get work done. So all energy is coming from the mitochondria. And within the mitochondria are some essential components, which we will really explore in the cellular energy topic. But this is things like the citrate or Krebs cycle. It is also the electron transport system. But all of these things are working in concert in order to produce energy for the cell.
- 1.52** Then we have a couple of other organelles that are important, but we're not going to really explore their role in the Basics of Biology course. We have the lysosome, which is another part of the cell that's essential for just cleaning things, but also is the peroxisome. The peroxisome is similarly important in breaking down stuff, including some toxins and things like that. Now, the last couple of things I want to mention with regards to the organelles or the structures within the eukaryotic cells are important

because they're very relevant to the upcoming processes of mitosis. Later on, we're going to talk about how a cell grows and then divides, which, remember, is an essential criterion of life.

- 1.53** The centrosome is an essential part of it. The centrosomes are these little organelles that start to orient themselves. They literally are moving through the cell in order to then start pulling the cell apart. It's almost like they have this similar charge, like a magnetic pole wanting to push something apart or pull it apart. So the centrosome is essential in actually building or structuring out these things called filaments. And these little filaments are these kind of string-like structures within a cell that not only—we'll talk more about them in just a moment, actually, because they're a formal part of the cell. But the microtubules are these thread-like structures that the centrosome is coordinating. So the centrosome is basically determining the orientation and then the eventual splitting and the directionality of the cell pulling apart.
- 1.54** And then, related to this, are things called the cytoskeleton and the microfilaments. And these are the last parts of this that we'll talk about within animal cells. So the cytoskeleton is actually this small little network. I've mentioned these little thread-like molecules that give the cell its particular structure or shape. Not all cells are shaped the same. The cytoskeleton, just like the skeleton is giving a shape and structure to my body, the cytoskeleton, cyto meaning cell, the cellular skeleton is giving shape and structure to the cell itself.
- 1.55** And very related to that are these things called microfilaments. And a filament, if you haven't, make sure that you watch my Fundamentals in Physiology series, and we talk a lot about muscle contraction. Within muscle contraction, we have a lot of these filaments, but the filament is necessary for a cell to move. Whatever movement you're thinking of, whether it is, in fact, a muscle contracting, whether it is the sperm swimming, that is a movement that requires some degree of microfilaments. It's also important in the actual splitting apart of the cell, this process called cytokinesis, which is the last step of mitosis, which we'll talk all about. So the cell is moving apart, and it eventually pulls apart entirely. The microfilaments are going to be an important part of that process.
- 1.56** Now, one last thing then to discuss, and I've alluded to it before, or one last organelle to mention, is an organelle that's unique primarily to plants. Now, everything I've just been describing is very relevant, of course, to animal cells. But with plant cell, we have one more, which is kind of a unique plant version of the mitochondria, which we will discuss in the cell energy topic, but that is the chloroplast. So the chloroplast is the unique version, a plant version, of the mitochondria that's capable of getting energy from the sun. Rather than getting energy from food, from calories like we think of when we eat calories, that's what the mitochondria needs to burn to produce energy. The chloroplast, if you will, is burning the sunlight in order to create energetic molecules or food for us to then eat.

- 1.57** All right. So with all of this in mind, to wrap up this little subsection within the cell topic here, let's just revisit the idea that the cell is a little machine with all of these things in mind. You'll recall that we mentioned that the cell, as a little factory or this machinery, it has some instructions or protocols that it needs to follow. It has some tools that it needs to build machines, and it has the machines themselves that are enabling the cell to do whatever it needs to do. And the machines can be quite specific. Muscle cells will have different machines than kidney cells, for example.
- 1.58** It must have an energy source, and it must have an ability to manage the energy, to know when to store it and how to store it, to know when to burn it and at what degree to burn it. And then compartmentalization within the little organelles themselves, like a factory has different sections where different work is getting done. So too does the cell. And then it lastly has the ability to build everything it needs on site. It doesn't need to offload any part of its work.
- 1.59** All right. Up till now, then, with this little subsection behind us, you should be able to understand the various components of the cells. Now, an important part of all of this, and I've used this word membrane very often, so let's talk about cell membranes, because this is an essential part of allowing a cell to survive. So at the end of this little topic, I want you to be able—you should be able to detail or explain to a family member or friend the utility or the usefulness of the cell membrane and the cell wall in general.
- 1.60** And within that earlier example that I used, of looking at the cell as a little factory, one of the final components of that was the ability of the cell to compartmentalize. Just like the factory has different sections doing different work, all of which is seeking one overall level of productivity, the cell also has the ability to compartmentalize. Now, it doesn't have the benefit of having a distinct machine in one singular section. It must be walled off. So the compartment must have some physical structure dividing it. The factory doesn't need a fence to close it off, but somewhat like a fence, the cell must have the ability, some barrier.
- 1.61** But the barrier must be functional. It can't just be totally walled off. Otherwise, it serves no purpose. It must keep out some things, but allow other things to come in. So with this idea of compartmentalization, it allows us to subdivide the cell into distinct regions or compartments. And then these compartments are capable of doing whatever work they need to do, and to do so separately, yet complementarily, with other parts of it. Another aspect of the compartment is that, these compartments, is that they are called membrane-bound organelles. By membrane-bound, it means that the structure is enclosed within some membrane, and that is defining or giving shape and then purpose to whatever the organelle or the structure it is that we're talking about.
- 1.62** But as I mentioned, it cannot be something that closes it off entirely. Otherwise, the cell would have no purpose for it. So it must have this aspect of permeability or a semi-permeability. So it can't let everything through. If everything could come in and out, it

serves no purpose. Similarly, if nothing can come in or out, it serves no purpose. It must be able to have the ability to discern, OK, you are something I need to pull in. In contrast, here is something I need to dump out. But you other thing, you don't come in to me. You're not supposed to come here. Or if you do, I need to have a way to regulate what you do or how much of you is coming in.

1.63 So the first thing that I want to explore with you is the actual cell membrane itself. So before we zoom in any further, let's just talk about the cell membrane. There are actually a handful of names for this. What I'm calling the cell membrane is also referred to sometimes as the plasma membrane. And then even more precise is something called the phospholipid bilayer. And the phospholipid bilayer is another word, it is synonymous with the cell membrane and plasma membrane. But it's helpful because it actually explains the composition of the membrane. Because if we zoomed in super, super deep into the cell membrane, we would see that it is actually these two opposing layers of a type of fat called phospholipid.

1.64 Now we're going to talk about phospholipids later, so we don't need to worry about their structure at the moment. But what is so essential about these is that they are what's called amphipathic. Amphipathic meaning that there is what's called a hydrophilic side to it, which in this case is my wrist, hydrophilic, and then there is these hydrophobic tails. In fact, technically it's two. It's like this. Imagine this phospholipid bilayer, which is creating the membrane of the cell. And the top part, again, is what I called, is hydrophilic.

1.65 And you can hear some words in there. One, you hear, of course, the word water or hydro. And then philic means loving. So the top part, the head, loves water, which is good because the body is mostly made of water. The same thing with cells. Most of what is our mass is water, up to 60% of the human body's water out of the cell and in the cell. But we can't have it just freely mingling. Otherwise, we'd just be a big blob of water and mess. So the phospholipid bilayer has a water-loving section, which is facing out to the water, both the water outside the cell and the water inside the cell.

1.66 But if it was just this phospholipid itself, which is a type of fat that creates the membrane, it has this, of course, hydrophobic tail to it or the tails. Well, that can't face the watery side of the cell, the cytosol. So it has another phospholipid stacked at its bottom. So it has its two, the tails, which are hydrophobic, who don't want the water, are pushed in together so they can both take a breath of relief and avoid the water. And then the hydrophilic heads are also enjoying life because they're able to face the water, both the water outside the cell and the water inside the cell—giving rise to the term, again, amphipathic.

1.67 So amphipathic describes molecules that have both hydrophilic, the head, and hydrophobic, the tails, parts of it. So water-loving and water-repelling parts. So that is the phospholipid bilayer, or the cell membrane, that encompasses or encloses every

single cell of the body. Every cell of the body has that phospholipid bilayer, identifying and giving shape and structure to itself.

- 1.68** Now, an alternative form, in addition to the phospholipid bilayer of the cell membrane, you can also, in some instances, get what is called a micelle. In my Fundamentals of Physiology lecture, we talked a little bit about this, but a micelle is an important part of the movement of fats within the body. So when we're moving fats around in the body, it's typically taking in the form of a micelle. But in this case, it's this funny little structure, whereas, as opposed to having the phospholipids against each other like this, and then enclosing the cell, it's all of these phospholipids stacked in a sphere.
- 1.69** Imagine, if we could just multiply all of my little awkwardly shaped phospholipids here and start pinning them against each other, but keeping the hydrophobic tails together. So the hydrophobic tails are all facing inwards, and then the hydrophilic heads are all facing outwards, giving us this small ball-shaped structure called a micelle. And that is, again, how fats move through the body.
- 1.70** Now, let's go back to the phospholipid bilayer for a moment, which is the cell membrane. So within the cell membrane, we have these phospholipids that have two tails. Each of these tails is what's called a fatty acid or just a distinct fat. Now, you've heard of these terms called saturated fat; you've heard of monounsaturated fat; and you've heard of polyunsaturated fat. Now, I don't want you to bring any bias to this part of the conversation. Don't be thinking any one is better than the other. There's a lot of, I would say, bad science that has gone into our understanding of dietary fats, but we can classify fats chemically into those three areas. Saturated, monounsaturated, mono being one unsaturated bond, and then the poly. And it is all about the carbon bonds.
- 1.71** So if we look at a fat, it is basically a long string of carbons bound together in this zigzaggy chemical structure. Carbon, carbon, carbon, carbon, carbon. So all these carbons bound together. If it is a saturated fat, it is just a straight line of carbons. It's that simple. Every bond is linked. In other words, fully saturated. However, if it's a monounsaturated fat, like the fat from olive oil, then it has one single carbon-carbon bond that is not fully saturated. It has basically an empty spot, and that takes the linear straight saturated fat and introduces a kink, but only one kink.
- 1.72** But having that one kink does change. If you have the phospholipid, one is a straight saturated fat and the other has a little kink, it starts to change the degree to which it can stack up to other phospholipids. And so if we then look at the phospholipid bilayer of the cell, it starts to change the fluidity or the rigidity of the cell membrane. Now, before I get into that in too much detail, let's just introduce the final, the third and final fat, which is polyunsaturated. As the name suggests, the polyunsaturated fat is going to have not only the one unsaturated bond that the monosaturated has, but it can have another one or two. So it can have two or three of these unsaturated bonds.

- 1.73** Now the shape is even more bent. There are even more kinks in the shape of these fatty acid tails of the phospholipid. And with all of that in mind, let's then put it in the context of the cell membrane. If you have phospholipids that are primarily composed of saturated fats, they can stack together very nice and tightly. However, if the cell consisted entirely of saturated fats within its phospholipid tails, it would be too rigid. It's too tightly packed. And now the cell is too dense, and it's not permeable enough. It can't be fluid and dynamic, and maybe it can't even allow things in. It can't make room for anything to slip through.
- 1.74** But remember, in order for compartmentalization to work, it must be semi-permeable. It must have the ability to let some things in or out. So if the cell membrane is composed of too much saturated fat, things get too rigid and too tight. Of course, in contrast, if the cell was only composed of very kinky polyunsaturated fats, these would be the types of fats that we get from seed oils like soybean oil or corn oil, so-called vegetable oil. There's nothing vegetable about them; that is totally a misnomer. They are more accurately simply defined as refined seed oils.
- 1.75** In that case, with the multiple kinks, they're so kinky, they don't mesh together very well. If the phospholipid bilayer was composed of these multiple kinked polyunsaturated fatty acid tails of the phospholipid, it would be too loose. They're not close enough. In fact, indeed, it would actually just burst apart, but it's too permeable. There wouldn't be enough structure to it. So thankfully, the cell actually has the ability to regulate the composition of the fatty acids. Enough saturated fat to give it some structure, but enough poly or monounsaturated fats in order to give it a little bit of dynamic aspect and fluidity.
- 1.76** And again, because of all of this, not only the nature of the phospholipid bilayer and its fatty acids, but also other components in the cell membrane, it allows the cell membrane to be semi-permeable. So some things can come out and some things can come in—but only some things—and some things can go out. And a cell can be specific to allowing some things to come in or out, depending on the nature of the cell. Not all cells will have the same need for the same molecules. And because of this, embedded within the phospholipid bilayer, are other structures. So let's just discuss some of them for just a few minutes.
- 1.77** So we have now adequately discussed the phospholipid bilayer. But woven within, or sprinkled within this phospholipid bilayer, are other structures that allow some degree of interaction with other cells or other molecules. Firstly, it has the ability, or the cell has on it, transporters, and we'll talk more about them in a moment. But a transporter is just as the name suggests. It allows molecules to move in or out that wouldn't otherwise be able to get through the membrane. So this is a very structured way of controlling. It's like a gatekeeper. OK, I need some of you, but I only need two of you. So two of you can come in now, no more can come in. That's what a transporter would do.

- 1.78** Also, embedded in the cell membrane could be an enzyme. So a little protein structure that may be embedded on the inside or the inner side of the cell membrane. And it can take one molecule, and we'll talk about enzymes soon, and change it into another molecule. Enzymes can be freely floating in the cytosol, the watery part of the water balloon of the cell, or in this case, bound to a structure or bound to the membrane. Also, on the other side of the cell could be a cell surface receptor. This is an essential aspect, enabling a cell to respond to any other signals that may be coming from another cell.
- 1.79** Over the next few minutes, we're going to talk a little bit about glucose and insulin. And remember, I study diabetes as a scientist, so this is an example I'm always quick to come to. On the outside of the cell, the cell may not know. The cell is going to be thinking, boy, I sure could use some glucose right now. I'd love to eat some blood sugar right now. But it needs to know, well, is there actually enough for me to take in? Is it my turn to take in some glucose? I don't want to starve everybody else. The cell wants to play well with other cells, but it needs to know whether it's time to pull in the glucose. So there would be a receptor on the surface of the cell that would sense, for example, the hormone insulin.
- 1.80** So insulin would come and bind its receptor, and that receptor itself is embedded within the membrane, in fact, spanning the entire membrane. So across this phospholipid bilayer, all through it would be this insulin receptor. So insulin comes and essentially knocks on the door of the cell. And then the signal transmits through this. And then right next to it would actually be what I just mentioned, which is an enzyme, a membrane-bound enzyme. And so it starts to send this signal within the cell. And the ultimate effect, actually, to go back to the first structure I even mentioned now, would be opening up a transporter. So then the transporter would open, allowing the glucose to come in. And we're going to revisit that example more in a moment, but it's just an interesting way to bring all three of those into one thing.
- 1.81** Now, other structures on the cell that are bound to the membrane also give it an identity. It's a way for the cell to recognize itself. You, of course, are familiar with autoimmune diseases. Sometimes what goes wrong in an autoimmune disease is that immune cells fail to recognize other cells as part of itself. Rather than looking at this and saying, oh, I recognize that signal on your membrane, so I'm going to leave you alone, it forgets, or the cell doesn't have the right signal. And so the immune cell will think, you are foreign and I need to destroy you. That is really the nature of an autoimmune disease. And again, that's based on some component that is embedded in the structure of the cell, like a flag basically poking up and saying, hey, I'm on the same team.
- 1.82** And then other aspects, a couple of final ones are cell-cell adhesion molecules. So on the surface of a cell, there can be essentially the two different components of Velcro, which allow cells to bind together, which is, of course, very important for every single

organ in the body. What keeps the kidney all lumped together while it has this ability to adhere to itself. So almost all cells of the body need to be adhered in order to grow well. And so there are cell-to-cell adhesion molecules that lock things in place.

1.83 All right. Now, let's talk about going back to the membrane and its ability to be semi-permeable, or to discern what should come in or what should go out. Let's just discuss briefly the aspects or the molecules that can pass, what can go in and what isn't. Well, there are some things that can go in very, very readily, and these are small or non-polar molecules. Non-polar being that they don't have an electrical charge to them. They're neither positively charged nor negatively charged. They're neutral, so non-polar, or again, very, very small.

1.84 Or they can be lipid-based or fat, because like attracts like. And I just got done describing the fact that the cell membrane, the phospholipid, lipid meaning fat, the phospholipid bilayer of the cell membrane is fat. So what encompasses every cell is primarily fat. And so if there is a fat coming, it has an easier time, not always, not limitlessly, but it has an easier time simply moving through the membrane. It's basically saying, hey, I'm like you. You're a fat, I'm a fat. I can slip through. And this is why steroid hormones are able to get into a cell very readily. Steroid hormones being not only, of course, the famous ones of the sex hormones, all estrogen and all androgens, the most famous androgen, of course, being testosterone—they're able to freely move through the cell membrane. They come to the membrane and say, hey, fats, I'm also a fat. Let me through. And they're able to get through.

1.85 But also water. Water is very, very small and is able to generally pass through the cell membrane fairly well. But if a molecule is not only much, much larger, but is charged, it has a charge to it, positive or negative, now it's going to have a very hard time getting through. It has to be regulated. So this is going to be things like not only a glucose molecule, like a sugar molecule will have some degree of regulation. It's simply too big, but also electrolytes. So things like sodium and potassium and chloride, sodium and potassium having very positive charges, chloride having a very negative charge. These are molecules that are too polar or too charged to just slip through. They have to be regulated, and they are through specific molecules that can open and close to let them through responsibly.

1.86 Now, the last part about the cell membrane that I want to mention is just how important cholesterol is. We, unfortunately, live in a day and age when cholesterol is vilified. Indeed, most clinical practitioners focus more on cholesterol than probably any other aspect of a patient's health. They want to measure your blood, and they want to look at your cholesterol levels, usually so they can put you on a cholesterol-lowering medication. But this paradigm has really vilified an utterly essential part of just life at the level of every single cell.

1.87 So we don't even need to mention the steroid hormones that I just discussed. All steroid hormones, not only the sex hormones, but many, many more, are all built on

cholesterol as a base. So the cell will take cholesterol in and then turn it into whatever that steroid hormone is going to be. Again, many more hormones than just the sex hormones.

- 1.88** So not only is cholesterol essential to that, but cholesterol is also embedded throughout the phospholipid bilayer. It is essential for not only structuring other structures, or giving shape and place to other structures, like the insulin receptor I mentioned earlier, where you have this protein-heavy receptor spanning the length of the phospholipid bilayer, it forms in areas where there is what's called a lipid raft, which is this cholesterol-heavy area.
- 1.89** So cholesterol is not only clumping around to tell receptors where to be located, but it also is an essential part of the dynamic fluidity of the cell membrane. It gets in the middle of some of these phospholipids and say, hey, you're too close, you're too tight. Mind you, even if it is with mono or polyunsaturated fatty tails, the cholesterol gets inserted just to give it some more dynamic fluidity, enabling the cell to move and enabling some components to move around.
- 1.90** So if we looked at the surface of a cell, if we were standing on the surface of the cell, and the cell was the size of the moon, we would see these components, these various structures that I talked about earlier. They're not just rigidly locked in place, they're flowing around. It's almost like we would be standing on an ocean and we would see something moving. So the cell is not a rigid set structure. As much as its shape is generally staying the same, within that shape is this readily moving, these readily moving parts, and cholesterol enables that to happen. So thank goodness for cholesterol.
- 1.91** Now, everything I've been describing so far is perfectly suitable for understanding animal cells. But within the eukaryotic family of cells, lest we forget, there's also plants. Plants are eukaryotic. They meet every definition of life. In talking about the dynamic phospholipid bilayer of the animal cells, that stands in stark contrast to the very rigid, yet still semi-permeable walls, of the plant cell. So the plant cell still meets all of the necessity or the necessary aspects that allow the cell membrane to be permeable to selective things. But it is also composed very heavily of a structure called cellulose.
- 1.92** And so cellulose, rather than the cell being a blobby fluid thing, the plant cell is very rigid, much more rigid, but still semi-permeable because of the presence of cellulose wrapping it around. So rather than a phospholipid bilayer, it has this cellulose heavy layer, which makes the cell more tough. All right, so by this point now, within this subtopic, you should be able to understand the utility or the beneficial role of the cell membrane, and the cell wall in general.

Lecture 2: Crossing the Cell Wall

At a glance — Crossing the cell wall — membrane transport.

- 2.1** Now, with the cell wall in mind, we can move on to another essential aspect of the functioning of the cell biology, which is the diffusion of molecules and the movement of molecules across or through a gradient. So in the next few minutes, and this is a relatively brief little subtopic, I want to describe diffusion, and I want to describe concentration gradients, which helps us to understand the movement of molecules.
- 2.2** Now, there's a little terminology that we've got to be comfortable with in order to be conversant in this topic. Firstly, I want you to know what the word solvent is. The solvent is the liquid that something is going to be dissolved in or that something is found in. So the solvent is the liquid, like water. Water is a solvent, and indeed, the most relevant solvent within eukaryotic animal-based life and plant-based. So within all eukaryotes, the main solvent of interest is going to be water.
- 2.3** But within the water, within that solvent that we're identifying as water now, will be solutes or molecules that are dissolved in that solvent. So the solute is the molecule that is dissolved in the solvent. And within that space, there's going to be a concentration. So that's another essential word. Concentration being any given, an amount of a given solute within a particular volume of solvent. So if I have a container of solvent and I've added some solute into that, it will dissolve and have a certain concentration within that volume of solvent or water. If I add water or add solvent, the concentration, of course, has gone down. I'm making it more dilute.
- 2.4** And then if I have one part of this container, and I have a container, let's say I have a container right next to it, where I didn't dilute it, this one was diluted with more water, so I added more solvent. It's more dilute. It has a lower concentration. But this container right here, I didn't dilute it with more solvent, so it has a higher concentration. And the two of them, if we could push them together now, but still have them separated with a little membrane— and we'll talk more about types of membranes in a moment, and we have been—now there is a concentration gradient.
- 2.5** This side got more water, and it is more diluted than this side. This side is more concentrated. And the concentration gradient is such, as we'll talk about, the things always want to move down the concentration gradient. And again, we'll talk more about that in just a moment. So within the, within the cell and indeed all of nature, there's this physical principle of something always being in motion. And this is referred to commonly as Brownian movement. Brownian movement is just a term to refer to the chaotic bouncing and moving of all molecules. Not only the gas molecules in the atmosphere around me right now, the nitrogen, the carbon dioxide, the oxygen, bumping into each other; that is a Brownian movement.

- 2.6** But even within biology, within the body, there's a Brownian movement to molecules that are diluted. In other words, to solutes that are found within a solvent. So it's the random erratic movement. But because of this, we have this aspect of inertia. And you may recall Newton's first law of motion, which is that objects are always in motion. There's never a point where things just stop. Because the objects are always in motion, they will want to randomly distribute themselves. And that is essential to appreciate a concentration gradient. If you have an area that's more concentrated, they're too tightly packed, they're bouncing off each other, they want to get a little space. And so if they can, they will diffuse or they will move into an area randomly, chaotically, erratically, that allows them to be at a lower concentration.
- 2.7** The concentration gradient is the difference in the number of particles in a particular space. And because of the randomness of movement, these things want to be equally distributed, or in other words, they always want to reach an equilibrium. So let's go back to the two compartments or the two containers of water that I had. We have water in this container, water in this container, and I start adding salt into these two containers. I'm concentrating them, make them into a salty water solution. I put, let's say, 10 units of salt in this container, and I only put five units of salt in this container.
- 2.8** And then I take a membrane that is capable of only letting one or the other pass. And in this case, because we're talking about diffusion, it's going to let the molecules pass. It's not going to worry about the water. It's going to let the salt molecules move. And so I have this, again, concentrated and less concentrated or more dilute. And I have a membrane that I'm going to then put in between them. So I lay the membrane here, boom, I connect them.
- 2.9** Now, remember, it's only the salt molecules that are going to want to move here or that are moving here. This one was much more concentrated than this side. And so the salt wants to go down its concentration gradient, so it starts moving through the membrane to this side, which was more dilute. This is diffusion. Now, within this same example, as the concentration is, as the molecules are moving down its concentration gradient, from the high concentration area to the low concentration gradient area, this is called a passive transport or a passive movement. There is no work required at all.
- 2.10** Because this is so concentrated, the molecules are just fighting with each other. They want to be randomly distributed, and they sense, oh, hey, I can move through here into an area where there's more room. It's going to move down very readily. You don't have to tell it what to do. That is passive transport or passive movement. No cellular energy required. However, let's say now, within the body, we have areas where we need things to be more concentrated. It's supposed to go up. It's supposed to be one compartment more concentrated than another.
- 2.11** So let's say we took two equally concentrated containers, put them together, but we need this side to actually be more concentrated on that side of the membrane. In other words, perhaps, to zoom it back to the body, we need the water within the cell to be

more concentrated with a particular solute than the water outside the cell. Those molecules do not want to move from an area of low concentration to high concentration. It's too crowded. Let's stay in the cell. It's too crowded in the cell.

- 2.12** Why on earth is the—let me be specific here—why does the potassium, which is a perfect example for the cell, there is more than 10 times more potassium within the cell than there is outside the cell. How can we maintain that? Why doesn't the potassium just flow out? After all, it wants to. It's so concentrated here, 10 times more potassium in the cell. Why doesn't it just start moving out of the cell? How on earth can we maintain such a difference in concentration? It's because now we are using energy to move the potassium in, which is called active transport.
- 2.13** And for something to be actively transported against its concentration gradient, against its diffusion direction, we need cellular energy to do that. And we'll talk about cellular energy later, of course. At this point, I just want you to appreciate diffusion and how molecules or solutes want to move from an area of high concentration to low concentration. And if they're able to, will do so without the use of energy. However, if we need to concentrate something and have it go against its concentration gradient, making more solute in a particular solvent or a particular compartment, then we need energy to do it, and it's active transport that is making that work. So now we're actively working.
- 2.14** An interesting example of this is glucose. When we eat a meal, our glucose levels will spike considerably in our blood, indeed, to a level that is much higher than that that is found within cells. And of course, there's some other signals here, like insulin going up, but some cells don't even need insulin, like the liver. The liver has open transporters ready to move glucose in at any moment. And then the moment glucose starts going up, well, these transporters are open and ready. And so, no surprise, they sense, if you will, hey, there's less glucose in those liver cells, and the door is wide open, so I'm just going to go in. No energy required.
- 2.15** And then the glucose is able to go down its concentration gradient, from an area where it is high, which is in the blood, in an area where it's relatively lower, namely within the cells of the liver. So the glucose diffusion after we eat a starchy, sugary meal is a perfect example of this passive transport. And indeed, it is going through a transporter, the glucose transporter that is always essentially an open door on the surface of the liver. All right, so at this point now, then you should be able to describe diffusion, the movement of solutes in a solvent, and understand now concentration gradients and how those gradients are affected or influence, in turn, active and passive transport.
- 2.16** Now, with that idea in mind, we'll move from diffusion into osmosis. Osmosis is complementary, yet distinct. Whereas diffusion is the ability of the solute to move across a membrane, osmosis is when the solvent is moving across the membrane. So this would now be water, in the case of the human body. Although there are other, of course, chemical solvents, but this would be now water's movement. So there are

some instances when it's the water that is moving for various reasons, and we'll explore some of them. So diffusion is the movement of the solute or the molecules, the salt, the electrolytes, whatever it may be, the glucose. And then the osmosis is the movement of the solvent, or the water, in the case of the human body and human cells.

2.17 Now, there are some fundamental principles here that we should just go over. As I mentioned before, we always want to move something. Something always wants to move. It has this almost inherent desire to go from an area of higher concentration to low. So if we have two compartments that are separated by a membrane that is capable of allowing things to move, it always wants to match the concentration on both sides. Whatever's on this side, a concentration of a particular molecule, it's separated by a semi-permeable membrane, we want to have the same concentration on this side. That's generally what the cell is trying to get to, unless it's forced to do otherwise, which it, of course, always is in using active transport.

2.18 So if a membrane is permeable to a solute, so if a molecule is capable of moving, then it will use diffusion. Diffusion will be what's moving around or what's responsible for balancing the concentration. So if we have these two containers—in fact, now I'm going to invoke this idea of a U-shaped pipe. Because that's going to help us, that's going to be helpful in understanding some of the differences between diffusion and osmosis. So again, with diffusion, let's imagine I have a U-shaped pipe.

2.19 And within that U-shaped pipe, I've been adding solutes within a solvent. I have water that's in this U-shaped pipe, and I'm adding solutes or molecules. And at the bottom, right separating at the crux of that U-shaped pipe, I have a membrane that is permeable only to the solute. So only the molecules are moving, not the water. And so if I start sprinkling more salt, let's say, or some molecule on this side, because this membrane is permeable to these molecules, this side is a lower concentration, which direction are things going to move? Of course, it's going to come from here to here, seeking an equilibrium. So this was temporarily, this side, much more concentrated than this side. But I give it a few minutes and sample both sides. They're both the exact same because diffusion wanted to spread that concentration out. It wanted an equilibrium.

2.20 Now, let's say then that we have to invoke the principle of osmosis. And remember, osmosis is not the movement of the molecule, of the solute, rather what we're dissolving, but of the solvent, the water. So now we have the water that can move through this membrane, but not, in this case, the salt, for example, or whatever the solute is, whatever the molecule is that we're sprinkling on both sides of this U-shaped pipe. Now, let's create this really weird scenario where I'm concentrating this side of the U-shaped pipe, but only the water can move. They have at first the exact same amount of water, but this side has more of the solute in it. It's much more

concentrated than this side. This side is much more dilute. The solute cannot move. There's no diffusion, only osmosis.

- 2.21** What's going to happen, then? If we were looking at this through a clear plastic tube, these clear plastic tubes, which direction is the water going to move? Of course, it's going to move this way. And so we would actually see, apparently defying the laws of physics, we would see that, through this water-permeable membrane, water is moving in, in order to make this side a little more concentrated by getting rid of some of its water, but giving its water to this side, which is a little too concentrated, making it a little more dilute.
- 2.22** So by moving the water or the solvent from one compartment to the next, through this water-permeable-only membrane, now we have what's called this osmotic change. And at the end, though, the equilibrium is matched. The concentration of the solute was matched. This was too concentrated; this was too dilute. So it moved water in order to make this side a little more concentrated, this side a little more diluted. So now it got matched, or reached equilibrium in concentration, but it now has different volumes of water in this case.
- 2.23** So the volume of the solvent will not be the same. That is very important as we start to understand the components of the body or the intricacies of the body. And this creates, this can actually be quantified. If we're moving water over in order to concentrate this side of the U-shaped tube, we could actually determine, all right, what is the resistance here that we could push against? How hard do we have to push in order to keep that water? Or how hard is it pushing up? That creates something called osmotic pressure.
- 2.24** So the osmotic pressure is the ability to be pulling in water, more and more water, in order to match a concentration. So osmotic pressure can be quantified. It is a real pressure that starts to push against its surroundings. And an obvious example of this is evidenced in the human called edema. Edema is that you all, whether you know the word or not, you've seen it, which is when a limb gets swollen. So I've been standing for a period of time now. Before I actually started lecturing and standing for the last hour or so, my feet and lower legs were actually smaller volume than they are now. As I have been standing, I've been creating this pressure in my lower legs that is resulting in them retaining a little more water.
- 2.25** There has been in this osmotic change, but it has had to elicit a pressure or a pushing because it has to overcome the natural elasticity of my skin. My skin wants to keep things nice and tight—of course, less tight every year I get older, but it wants to keep things nice and tight. But as the water is moving from my blood into the space around my blood vessels, and it's just the water moving out, just like the water is moving through this U-shaped pipe, it's creating a pressure that's slowly going up, and so it starts to press out against my skin, making my skin start to appear a little more puffy. So edema is a consequence of the movement of water, or in other words, osmosis.

- 2.26** You, of course, are familiar with diabetes. In type 1 diabetes, especially, where insulin is essentially zero, there is a very reduced ability of the body to control blood sugar levels. And all of this is somewhat background to getting to the osmotic principle in just a moment. At the level of the kidney, and I'm acting out here the base unit of the kidney, which is something called the glomerulus—at the glomerulus, we have the blood vessels wrapped in this thing, this capsule. And at the glomerulus, we have glucose that is being filtered. The glomerulus of the kidney is filtering all kinds of things. Glucose is just one of them.
- 2.27** So the glucose is getting from the blood into what is called the renal tubule or the tubes of the kidney. Now, glucose doesn't belong in the tubes of the kidney. So when someone has relatively normal blood glucose levels, even if you spike it temporarily by eating or drinking something sugary or starchy, there's only so much glucose coming out into the renal tubule that all of it gets reabsorbed back into the blood. There are little blood vessels surrounding the renal tubule, basically just waiting to pull things back and say, hey, you got my glucose. I didn't mean to give you that glucose. Can you please give it back?
- 2.28** Interestingly, as the molecule is moving, it starts to pull water with it. So water is moving. We have this osmotic movement where the water is thinking, oh, hold on, glucose is getting more concentrated in the blood vessels again. So I need to follow the blood glucose, lest it get too concentrated. However, back to type 1 diabetes, with poorly controlled diabetes, when the person has blood sugar levels that are maybe 10 times higher than they should be, the amount of glucose getting filtered into the renal tubule is so high that the kidneys can't reabsorb at all. And so there is a load of glucose that is staying in the tubule.
- 2.29** And in contrast to the scenario before, this actually starts pulling in water because the water is sensing in these two compartments, kidney tube versus blood vessel tube, boy, there is so much more glucose in that kidney tube. It's too much. It's too concentrated. And so water starts to move, just like in that weird dynamic of my U-shaped pipe. So the water starts to get pulled into the kidney tubule, resulting in a much higher level of urine production.
- 2.30** In fact, that is what the word diabetes means. Diabetes means excess urine. And that's why. The urine is in excess because of an osmotic pull, or this pressure to try to pull in the water. And so the water is staying in the renal tubule because the glucose levels are so high that it's forcing the water to stay there. And as more water is staying in the kidney tubule, more water comes out in the form of urine. So that is an application, this clinical application, of the principles of the osmotic pull or the osmotic pressure.
- 2.31** Now, also within the human body, we have this concept of tonicity, and tonicity refers to the osmotic differences outside and inside of a cell. Now, I'm going to bring in a couple other words because it'll be helpful for the sake of this next analogy or example. Outside the cell is the watery compartment. Outside of every cell is the watery area

known as the extracellular fluid, and that encompasses a lot of stuff. That is blood, that is lymph, which goes into our lymph vessels and our lymph nodes. Any of the water that is outside of a cell is called the extracellular fluid. And then in contrast, what is inside the cell is the intracellular fluid.

2.32 So here we have the ICF, intracellular fluid, and around it we have the ECF, the extracellular fluid. And the membrane of the cell can readily move water. You'll recall that water moves very readily. It also is capable of moving solutes in and out, but that's much more controlled and indeed much more slower. It's much more slowly managed, more carefully, whereas water can move very, very quickly.

2.33 Let's imagine then that we had a scenario where here's a cell, and then the ECF, the fluid around the cell, got too, let's just say, salty because it's such an easy example. It got too salty or it's too concentrated. Well, the cell, the water in the cell, abhors this difference, and so too does the ECF. They're both saying, hey, we used to be matched in concentration of salt; we need to match back up. The ECF is saying, I'm too concentrated, give me your water. And the cell is saying, You're too concentrated. I need to match your concentration, so I'm going to give up some water. Very much like that U-shaped pipe.

2.34 So again, the concentration of the ECF is too high. The concentration of the ICF is relatively too low, and so it wants to start moving water. So same water, let's just say, and then this is more concentrated than this. What direction is the water going to go? If here's the cell that's very dilute, here's the ECF outside the cell that's very concentrated, and only water is moving so we're talking about osmosis, the water will move from within the cell to outside the cell. This will result not only in the concentrations getting back to normal, or rather, not normal. They're both a little more concentrated now. But the ECF was able to give up some of its concentration by taking some of the water.

2.35 So they're both relatively more concentrated than they were before, but now there is a distinct difference in volume. The ECF volume has gone up. It took some of the water from the cell. And as a result, the cell is actually more shrunken. And you can see this on a microscope if you look at red blood cells. So one of the instances of determining whether someone has too much of some concentration, it doesn't just need to be salt. It could be any number of things. Salt is just an easy example. You would actually take a blood sample, and then the pathologist or the clinical team would look at that under a light microscope, and it would see that the cells were shrunken and shriveled, rather than being nice and round like they should be.

2.36 When there's an equal or matched concentration, no need for osmosis going in or out, the cell is normal and round. If things are matched, that would be what's called an iso-, iso meaning equal, tonic solution—when the outside of the cell, the ECF, is the same concentration of general molecules and total molecules or solutes as the inside of the

cell. So if the ECF is too concentrated, or what's called hypertonic, that would result in the cell shrinking.

2.37 Now, in contrast, let's imagine a scenario where the person was drinking a lot of pure water, and this can happen, and it becomes lethal. If the person is drinking pure water, maybe they're doing a water challenge. This actually happened on a radio show one time, and someone died because of it. It was who could drink the most water and win some little prize. This person kept drinking, kept drinking, kept drinking, and died by diluting their ECF too much. So let's just talk about that. If the ECF gets too diluted, then the concentration—so I'm saying diluted, but in reality, its volume is going up. The concentration is going down as a result of this.

2.38 Now, the cell is thinking, I'm way too concentrated. And the ECF is saying, I'm way too dilute. Remember, water is going to be the first thing that moves, so osmosis is going to try to correct this. Now, the cell is saying to the ECF, give me some of your water. That will help you get more concentrated, and it will help me get more diluted, and our concentration will match. But as a result, what happens to the size of the cell? As we're trying to dilute it to match the diluted ECF, the cell gets bigger and bigger and bigger and swells. And so if you look at that under a light microscope, you see that the normal structure of the red blood cell is now big and balloony. Indeed, it can even pop. That's a process called hydropic degeneration. So the concentration differential can be so substantial that osmosis is going too much and the cell bursts, which, of course, is very problematic.

2.39 Now, this can also happen at a clinical level. If, let's say, we had a medical intern who is coming to give a patient an IV, and you've seen the patient lying there. They have all these tubes hooking into their arm, and then they have these bags of solution that are dripping things down into the IV line and getting it into their blood. Let's just say that this clinical or medical intern made a mistake and picked the wrong bag of saline or a salty solution to infuse into the patient, and it was too dilute. It was, rather than being isotonic or matched to the tonicity of the human body, it was at a lower level. It was what's called hypotonic.

2.40 So they start infusing a hypotonic solution. First of all, what's going to happen to the volume of the ECF? Because any time you're infusing anything, it's not going into a cell. You're not pricking a needle that's going into a cell. That's impossible. You're infusing it into the blood, which is part of the ECF. So if you're infusing a solution, what's happening to the volume of the ECF? Of course, the ECF volume is getting bigger. But because you're infusing a diluted concentration, the ECF is getting more diluted. So while its volume is going up, its concentration is going down. That's a problem, right? The ICF and the ECF want to be matched. Now they're not. The ECF concentration has gone down. It's been diluted.

2.41 And so this is very much just like the scenario we were talking about. So now what's going to happen? Osmosis is what's going to happen. So, as the cell is now more

concentrated, the ICF is more concentrated than the ECF. It's going to start taking in some of that water, and it's going to expand the cells. So the cells are going to swell a little, hopefully not too much, but they will swell with water, until the body can get to the point where it can start to excrete this excess. It would want to excrete the excess water and hold on to salt more, or solute more, because it got too diluted. We were infusing a diluted solution.

- 2.42** Now, the end result then is that the ECF volume went up, the ECF concentration went down, and then it exchanged some water with the cell. And so the cell took in some water. That brought the volume of the ECF down while bringing the volume of the ICF up. And at the same time, it concentrated the ECF. So the ECF's concentration was down here. By giving up some water, its concentration went up. By the ICF taking some water, its concentration went down. And so everything ended up getting matched.
- 2.43** Hopefully, it's able to get matched with a volume change that does not result in that hydropic degeneration, which is like analogous to us filling a water balloon too much. And you're telling a kid, hey, stop now, stop now, stop now. You didn't stop in time, and now it blew up, which, let's face it, every kid would be delighted to have happen.
- 2.44** All right. Now we understand the principles of osmosis. We also now understand diffusion. And a lot of this has been talking about the membrane, and we had not only earlier defined the membrane itself. So the final section here that I want to go over with you is actually discussing the nature of transportation, the movement of these molecules. Whatever the molecule is, the cell needs to not only be able to recognize it and determine whether it should move at all, but also determine how much of it to move. So at the end of this subtopic within the cell biology lecture here, I want you to be able to understand the unique and the scale of various transport systems across membranes.
- 2.45** Now, there are two basic types or two unique forms of movement across the membranes. Firstly, there is membrane-based structures. So these are structures that are embedded within the membrane itself. And then second, there is what's called vesicle-mediated transport. Vesicle-mediated transport. And that takes multiple forms, things like endocytosis or exocytosis, and we'll look into these in a little more detail.
- 2.46** So let's start with the first one, which is the membrane-based structures. I've mentioned some of these, actually, throughout this lecture in the form of some examples that I've given. So the first one is a channel. A channel is something that is closed until it's told to open. And when it's told to open, it creates simply an open passageway. But it will be specific to something. It won't let anything through. It is particularly designed to have something come and say, oh, yeah, you fit. You're able to go through. You don't fit. Get out of here. You're not supposed to come in right now.
- 2.47** But again, it's not always open. The channel needs to be told when to open, to allow a molecule to go in or out. And that signal usually comes in a few different forms. In the

Fundamentals of Physiology lecture, we talked about many of these, but one of them would be what's called a chemical gated channel. So this would be, for example, calcium. Calcium has the ability. Earlier, I mentioned that some cells, like muscle cells, will have a specific type of endoplasmic reticulum that is loaded with calcium. So it's basically like these big calcium sacs. Well, the calcium can be eliminated from that and then affect the cell membrane, telling a channel to open. So you can have something like a calcium-gated channel. That would be one stimulus to open the channel.

2.48 At the same time, you could have different molecules moving called electrolytes, and then you can have an electrical-gated channel. This would be when a nerve is sending a signal. And what we call the electrical signal is, in fact, just actually these ions, these positively and negatively charged molecules, moving. That is, of course, what an electrical charge is. It is this imbalance of positive and electrical charges. So some of these channels can be electrically gated, and some can even be mechanically gated.

2.49 It would be, like me, in fact, being able to feel my finger pressing against my arm is because of a mechanically gated channel. I'm pressing, and this physical pressure is causing some of these channels to open, which is then allowing a nerve signal to come back to my brain to actually sense this contact. Those are various mechanisms of opening channels. And then again, the channel is going to be specific to some particular molecules.

2.50 Similar but distinct are the types of membrane-bound or membrane-locked molecules called transporters. Now, a transporter is not the same as a channel. Whereas the channel just opens up and allows things to just rush in as rapidly as they want, the transporter is a specific carrier that's going to have a conformational change. It basically can look something like this, where it will have a molecule from the outside come and bind it, and then it will confirmationally change and say, OK, now you're here, I'm full. I'm going to shift, and now let you come into the cell. But in so doing, regulate anything else that would be attempting to sneak in. And then it will revert back to where it was.

2.51 So it's facing outward, and then this is the transporter. It binds it and then lets it in. Opens it up, binds it, and lets it in. This is responsible for the movement of a variety of things, including blood sugar, so sugars, as well as amino acids and a handful of other molecules. So again, they're moving through transporters. Now, within the family of transporters, it gets a little more complicated. There are what's called uniporters. So this is a uniporter transporter. All of these are transporters. And then there are symporters and antiporters. Let's define each of them briefly. Let's start with the last one that I just mentioned, just to mix it up.

2.52 So the antiporter is when it's confirmationally changing, and rather than only allowing things to go one direction, it's so efficient. It says, hey, you are on the outside of the cell. You can come in. I'm ready. You combine, and then I flip backwards, and now you can come out. But hey, while I'm here, I may as well pick up something from the inside and send it out. And so it's shaped accordingly. Something from the inside comes and

binds and says, all right, passengers are ready. Let's flip back out. And it lets that thing out.

- 2.53** So this is, in its most famous example, is something called the sodium-potassium, well, antiporter. It's taking sodium from inside the cell and dumping it out, and then it's taking potassium from outside the cell and pulling it in. But you'll recall earlier when I mentioned potassium, the cell has a lot of potassium in it already. And so, no surprise that this requires energy, this particular antiporter. It requires cellular energy because that is active transport. It is moving something against its concentration gradient. So that's an antiporter, where it opens from one side, moves something in, and then, having changed it, says, I'm here, I'll take something out at the same time to get ready to pull something back in.
- 2.54** A symporter is when it's moving two distinct different molecules in the same direction. So this is when the transporter is facing outwards and it says, OK, one molecule is here and bound and ready, but it says, I can't move yet. I need to wait for the other molecule to come. And so when the other molecule comes, then it will move. That is a symporter. Again, one molecule won't be enough.
- 2.55** And then the uniporter is when it's just moving one molecule of glucose at a time, or one molecule of whatever it is. Sorry, I say glucose, and that's going to be an example we cite in just a moment. So it's open, ready to go, binds one molecule, flips, and allows it to go back, to come in. Then it just simply switches right back out. So that's the uniporter. So we had the uniporter moving one single molecule at a time. We have the symporter, which is allowing the movement of two distinct molecules, not the same. And then lastly, we have the antiporter, which is moving two distinct, not the same molecules, but in opposite directions.
- 2.56** Now, as I have done a few times, let's use glucose as an interesting example. When you eat or drink something that is starchy or sugary, through the process of digestion, it's going to get converted into these base little individual molecules of glucose. Now, on the surface of the intestine, when we get down to the small intestine, that's where we actually can start moving things in. We have all of these molecules of what we just ate or drank. Now we need to move them into the blood.
- 2.57** Along the surface of the intestine are these things called sodium-glucose symports, or symporters. And these are the specific transporters that are called—the abbreviation is SGLT. So along the lining of the intestine are the SGLTs. They will not work to move glucose alone. They need sodium as well. And so any sodium that is in the guts, they will actually bind and tell the symporter that it's time to go. So there could be lots of glucose, and the glucose is bound and ready. It is waiting, but not until a sodium comes will it actually move it in.
- 2.58** So at that first level of the movement of glucose, we have an SGLT or the sodium-glucose symport transporter. So that is the transport mechanism of moving the glucose

from the guts into the first level of the cell. On that side, it's what's called the apical, or the intestine-facing, side. So on the apical side, or the front end of the epithelial cell, the lining of the intestine, we have the SGLT. That's allowing the glucose to come in. And again, it's because of the sodium. And the sodium is moving because of a concentration gradient. There's higher sodium in the gut than there is in the cell. And so sodium is ready to move in, and it's pushing its way in, opening the door, and glucose sneaks in with it.

- 2.59** And this is actually the mechanism of glucose movement, not only in the guts, but also in the kidneys. Earlier when I was mentioning the example of the glomerulus, which is filtering out glucose, and the glucose comes back into the blood, that is a type of transporter called SGLT2, which is easy to do because there's so much sodium here, and the glucose says, hey, I'm going to hitch a ride with you and come back into the blood. So this is SGLT2. In the guts is something called SGLT1. They're, of course, very related, but they are distinct.
- 2.60** An interesting application of this in the clinic is a very widely used class of drugs for diabetes called the SGLT inhibitors. If you've ever heard of a drug with the suffix gliflozin or flozin, that is going to fit within the class of an SGLT inhibitor. Some of these SGLT inhibitors will inhibit SGLT1. In other words, you eat the starchy or drink the starchy, sugary meal. There's lots of glucose getting digested, but you can't move it into the blood. It stays in the guts because you've blocked SGLT1. And by staying in the guts, it actually creates an osmotic gradient.
- 2.61** So while the patient is able to eat something that would normally spike their blood glucose, but not have it spike their blood glucose, because of blocking the absorption, the unfortunate consequence is that all of that glucose stays in the intestines, creating an osmotic gradient, pulling in water, resulting in horrific diarrhea. So that is one of the unfortunate consequences. In fact, it gets so bad that SGLT1 inhibitors are very rarely used because it's just not worth it. The patient has such gastrointestinal distress and such potentially socially awkward moments of bowel movements that it's not worth it.
- 2.62** Now, not quite as socially catastrophic, but still related, would be the patient using an SGLT2 inhibitor. So now, we have the glucose here, but by blocking SGLT2, the glucose is forced to stay in the kidney. But that results in an osmotic pull this way. And so not only do we have a lot more urine being produced, so they have a lot of urine. They're always thirsty, constantly worrying about dehydration because they just can't stop peeing. But there's a consequence to putting that much glucose in the urinary tract, because the urinary tract is sometimes a bit of a dirty place. There's bacteria always trying to work its way up.
- 2.63** It's like the song, The Itsy Bitsy Spider, that may be sung in little Sunday School classes. The Itsy Bitsy Spider went up the water spout, but down came the rain and washed the spider out. Normally, you're urinating enough that you're keeping—and there's such a composition of the urine that you're washing any of the bacteria out that are

attempting to invade and infect the body. But when you are pushing all that glucose into your urine, bacteria eat primarily one fuel, glucose. And so by forcing all of that glucose to stay in the urine by blocking SGLT2, now you're feeding the bacteria.

- 2.64** It's almost as if the bacteria are running a marathon up your urethra, and you're creating these little aid stations on their marathon saying, hey, you look like you're getting tired. Take a Gatorade. Take an energy drink. Here's an energy drink for you, to make sure you just don't stop till you get all the way to my bladder. So no surprise that, when people take SGLT2 inhibitors, blocking the reabsorption of the glucose, they end up getting substantially more frequent and more serious UTIs. Alternatively, of course, the diabetic could simply eat less glucose, but that requires too much self-discipline in some instances. It's easier to take a drug.
- 2.65** Now, the glucose journey is not over yet. We used SGLT1 to move the glucose through a symport. So we had the symport movement of glucose into the epithelium from the top side of the cell of the gut. But now it has to still get out of the cell of the gut into the blood. So there's a back side, what's called the basal side, of the cells of the gut. Now there's no SGLT here. There's what's called glucose transporters. So now we have a glucose transport. It's still fits within that transport family. On the top was a symport, now it's a uniport. And the uniport will allow one molecule of glucose to go out. It can come out. It says, hey, glucose, you're ready? OK, you go out. All right, now you're ready and you go out.
- 2.66** And it's the glucose transporters that are abundant throughout the rest of the body. So when you look at every single cell of the body, starting from that exact moment, any other movement of glucose in or out of a cell is going to be through this family of glucose transporters. Now, within that family of glucose transporters, one of them requires insulin, and that is the glucose transporter that is the most abundant on our muscle cells or our fat cells. Those are glucose transporters that are tucked in the cell.
- 2.67** Then insulin will come to its insulin receptor, which is, it's membrane-spanning, and it will knock on the door, and then there will be a response. Part of this response will be that these glucose transporters open up. And again, that's only found on cells like muscle and fat. So muscle and fat have insulin-dependent glucose transporters.
- 2.68** But then there are 12 other glucose transporters that are abundant throughout the rest of the body that are always ready to go. The moment glucose goes up, they don't need to be told what to do, like the muscle and the fat does. They can only pull in the glucose when the insulin gets elevated. But like the cells of the brain, the neurons, they don't wait. If the glucose goes up, they pull it in. The cells of the kidney, they don't wait. If glucose starts to go up, these transporters, these uniporters, are ready to go, and they pull the glucose in.
- 2.69** Now, muscle—one last point on this is unique because it is capable of a fantastic shift in its metabolic demands. Muscle can have a relatively low metabolic rate, never very

low. In fat tissue, and this is something we measure in my lab, actually, actively, fat tissue has a very low metabolic rate. It doesn't do a lot of work. Muscle, even at rest, just has a higher metabolic rate. Now, just as a point of interest, nothing even comes close to being as high as the kidney. The kidney has the highest metabolic rate by 10 times. It is so metabolically active, but somewhere below it is the muscle.

2.70 What's unique about the muscle is that it can increase its metabolic demands substantially. For example, watching this, listening to this, you're very sedentary at the moment. You're sitting, I assume. And your metabolic rate of your muscle is very low because the metabolic demand placed on the muscle is very low. But if you were to get up and start walking, the metabolic demand would skyrocket very quickly. And of course, it would skyrocket again if you started running or lifting weights.

2.71 And in that instance, the metabolic demand is so much, or in other words, the muscle gets so hungry, that it doesn't have time to wait for insulin. You'll recall that, just a moment ago, I said that the glucose transporters, the symport on the muscle cell, usually needs insulin to come and tell it to open. But when the muscle starts to work, it starts to contract; it has its own way of saying, I'm ready, and it has this back door; and it can open up the glucose transporters itself, without the need to wait for insulin. But if the muscle stops and things settle down, then it comes back to waiting for insulin. So the muscle has the back door based entirely on need.

2.72 Now, let's just wrap all of this back up as we finish up with the transporters and then get on to the vesicle-mediated transport as a final point in this big lecture. We started with the glucose in the intestine. And along the intestine is this intestine epithelial cell. Epithelium, meaning it's lining a tube or a part of the body. On the inside of the intestine, we have, in this case, the only two molecules we care about is sodium and glucose. And along this epithelium is the sodium-glucose symport.

2.73 Now, remember, that takes advantage of sodium's concentration gradient. So the first thing I want to bring your attention to is on this basal side, on the bottom side of the epithelial cell, the cells lining the intestines, there is this sodium transporter, and it is an active transport. It is pumping the sodium from the cell, where it's relatively low concentration, out into the blood, where it's already fairly high. There's way more sodium, about 10 times in the opposite direction of potassium. So there's tons of sodium outside that epithelial cell of the gut than there is inside the cell, and yet it's pumping it out. So it's pumping sodium out, and to do so requires an active transport.

2.74 So we have an energy-based transport, and that allows the cell to have a very low amount of sodium, which is good, because then if we go back to the top of it and look in the guts itself, we need to have the sodium want to move in. Remember, we want the sodium to come in. And because there's so little sodium here, because we pumped it all out, it is able to move in passively. So now there is this passive movement, but glucose now is able to slip in with it. So glucose slips in now through this symport based

on sodium's concentration gradient that, again, was caused because of the active sodium pumping at the bottom of this epithelium.

- 2.75** And then at the same time, when we've moved the sodium in to the epithelium or the lining cells of the gut, we need to get it out. And to do so, we had the glucose transporter, which is a uniport. So just within this one single cell, we have an antiporter that's actively moving sodium and potassium in order to keep the sodium very low within the cell. Then on the top part, we have this active transport of a symport moving sodium along, moving glucose along sodium's concentration gradient. And then ultimately, to move the sodium out of the epithelial cell and into the body or into the blood, we had an individual uniporter that was capable of just allowing the glucose to move. So one single cell encompassing three different parts, the three aspects of transport mechanism.
- 2.76** All right. Now, as I said, let's finish it off by talking about vesicle-mediated transport. Now, the vesicles are interesting because they are structures within the cell that have their own phospholipid bilayer. So you'll recall that the structure of the cell membrane was a phospholipid bilayer, these hydrophilic heads and hydrophobic tails with the tails facing in to protect them from the water which they so despise. This is basically now, these vesicles, are small little cell shape. They're not individual cells. They're little pockets, but they use the same physics of the phospholipid bilayer. So a vesicle is a phospholipid bilayer that is enclosing a portion of the solution, like something from outside the cell to pull it in, or something from inside to push it out.
- 2.77** Those are two different processes that we'll cover briefly as we finish this first lecture in the Basics of Biology—the first being endocytosis, which is the process whereby the cell takes something into itself using a vesicle, and then exocytosis, the process whereby a cell is giving up something that it has produced from within itself.
- 2.78** All right, so the first one, the first part of endocytosis that I want you to be familiar with, is something called phagocytosis. Phagocytosis, it comes from the word phage, which is eating. That's when the cell has encompassed some particle and taken it in. Now, eating makes it sound like it's eating some nutrients, which is not what's happening here. It's engulfing. It actually looks like, if you look at a phagocyte, which is a cell that's capable of phagocytosis, so pulling in or endocytosing something. Here could be the molecule of interest or the particle, and then the cell will actually form around it and close around it. So it looks like it's eating it. That's where the term phagocytosis, or cell eating, comes into play. And then this particle that's been enclosed will then be brought into the cell to go to some other compartment. So that's phagocytosis.
- 2.79** A different but very related phenomenon is something called pinocytosis, pinocytosis being a term for cell drinking. Now, that makes it sound like the cell is taking in water, like it's taking a gulp from a drinking fountain, which is not the case. This is a system of the cell absorbing certain molecules, like certain B vitamins, or vitamin C will actually

come into a cell through pinocytosis. They will come to the cell, edge of the cell membrane. The cell membrane will close around it, and here's the rest of the cell here, and then the membrane pinches it off, and then this molecule will move in and then disperse its contents from within the cell. Just very passively and very readily. If there's some B vitamins or vitamin C, for example, boom, it pulls it in.

2.80 Related again to this is the final form of endocytosis, which is receptor-mediated. This is when the cell surface has some receptors, something to sense a particular molecule. And then once these receptors are filled, it will enclose around them and pull it in. An example of this is actually LDL. You've heard of LDL because you've been told that it's the bad cholesterol. Well, LDL itself isn't cholesterol at all. It is simply a big molecule of fats and proteins in its technical definition.

2.81 LDL will come to a cell, bind the LDL receptor, and then the cell will say, oh, I know what to do with you. Thanks for knocking on my door. I'm going to wrap around you and pull you in, so I can get access to all of the fats or all of the cholesterol or the proteins that you're carrying, because after all, cholesterol is a molecule for life. We must have it. And now we have endocytosis because a transporter was activated because of this receptor. So we had the receptor-mediated transport in the eventual endocytosis.

2.82 And then lastly, as I mentioned, we have the phenomenon of pushing things in reverse, where let's say a cell has made a product. In fact, let's take the beta cell, which produces insulin, a cell that I'm very familiar with as a diabetes scientist. So the beta cell is saying, hey, look, I have a really important job, in that I'm the only cell in the entire body that is going to produce insulin, so I'd better do so right now. And so it produces insulin, and it's storing it in these little vesicles. And the moment glucose levels start to go up, the beta cell will sense this and say, now I know what to do with these. And so this little vesicle, which has its own phospholipid bilayer, and it is filled with insulin, will move to the membrane.

2.83 And because like attracts like, the membrane, the phospholipid membrane itself, has this phospholipid membrane vesicle bump into it, and they just merge seamlessly, and then releases its insulin into the cell. This is found in many, many instances throughout the body. Indeed, every signal from a nerve to either another nerve or to an end cell—and please watch my Fundamentals in Physiology lecture, we talk all about nerve signaling—every time a nerve is sending a signal to a cell next to it, it's using this exocytosis. It'll have little vesicles, like my little freckles here, filled with some molecules, and it will bump them out and release them across this little cleft, telling the other cell some signal, telling it what to do.

2.84 Then the final part of all of this cytosysis is one I didn't even mention earlier, but it's called transcytosis. And let's go back to the cells of the gut, because this is where it's mostly relevant. Part of the absorption of amino acids or peptides, when you eat protein, it'll get broken down into its little component parts, these little amino acid

chains. When we want to absorb some of these little, small little peptides or these links of amino acids, it will come to the top of the cell of the gut, the epithelial cell, and it will enclose around it, making a little vesicle. And that little vesicle will simply move through the epithelial cell and then be dumped at the back end. So it pulls it in through endocytosis and then dumps it out through exocytosis, never really interacting with the cell in this case. And so we just call it transcytosis. It's just passing through.

2.85 So with this subtopic completed, you should be able to describe some of the unique transport systems that abound across a cell, enabling the cell to take things in or take things out across its semi-permeable membrane. And in conclusion, we've covered a lot of totally relevant material. We've discussed, just to remind you, the three aspects of the cell theory, including defining what is life, what is alive, how to determine it. We talked about the families of cells, from within the prokaryotes against the eukaryotes, and we mentioned the rare form of prokaryotes being the archaea and the bacteria. Those are the prokaryotes. And then the eukaryotic cells being the cells that make all animal and plant life.

2.86 We also described in detail the utility of the various cellular components and the organelles, some in more depth than others. Then we moved on to the principles of osmosis and diffusion and concentration gradients, and how important those all are as aspects of a cell knowing when to transport something in or out, and exactly how it does so. So this is the cell, and the next lecture will be the central dogma.

Lecture 3: Biomolecules as Building Blocks

At a glance — Biomolecules as building blocks.

- 3.1** With a firm understanding of the functions of the cell, we're actually now going to move on to a topic called central dogma. That name implies one of the most important aspects of a cell, but we'll get to that in a moment. The general learning outcomes with this particular lecture is, I want you to understand the differences and the definitions of the four major biomolecules, so the four essential organic molecules in every cell. I also want you to understand the function of these biomolecules, including their role in enzymes, and the role of enzymes in mediating their production.
- 3.2** And then lastly is the central dogma itself. So all of those topics are building up to the final pièce de résistance, the central dogma, which is the flow of genetic information that ends up defining who we are or identifying who we are within our each respective and unique bodies. So let's start by discriminating or learning the differences between the four major biomolecules. Now, you'll recall in the previous lecture, I referred to every cell as a busy little factory that is following instructions and has all of these machineries, all this machinery, in order to produce whatever it needs to create.
- 3.3** Among these things, within that list of functions and aspects, we have a key few so-called biomolecules. They are ones you've heard of. Proteins, fats or lipids, carbohydrates, and one that is a little unique and less commonly described as nucleic acids. So again, proteins, lipids, nucleic acids, and carbohydrates. All of these are referred to as biomolecules. In other words, they are what's called organic molecules that are found in living things. Now, that does not mean that the molecule itself is alive. It does not meet the criteria that we identified previously, but they are necessary for every living thing, for every cell to function well.
- 3.4** Within the variety of all of these biomolecules, the one thing they all have in common that allows us to consider them components of organic molecules is that they all have the element carbon. All of these are built on carbon, and carbon is a very unique element. Because of its unique chemical properties, it's actually capable of forming very strong bonds with many other elements, including itself. Carbon can bind carbon. It can bind oxygen. It can bind hydrogen. It can bind and can combine very strongly with a myriad of other elements.
- 3.5** And because of this, carbon can be at the heart of a vast array of diverse molecular structures, including the ones we mentioned earlier. But that allows them, also the variety of carbon allows itself to be structured not only in chain structures, but also in ring structures. And it's this that ends up creating the biomolecules that I mentioned earlier—the proteins, the nucleic acids, the carbohydrates, and the lipids or fats. So carbon is essential to the chemistry of life.

- 3.6** And interestingly, in the upcoming lecture, cellular energy, we'll actually explore how that carbon actually gets into the biomolecule for us to use in the first place by nature of the energy from the sun. But until then, we will just appreciate the essentiality of carbon within every organic molecule. Indeed, the presence of carbon giving rise to defining the biomolecule in the first place as organic. Let's go through each of these biomolecules. After all, that's the first and most important part of this early part of the lecture.
- 3.7** So carbohydrates, the name itself ends up providing some understanding of what that means. The carbo, referring to the carbon, and then hydrate being the oxygen and the water. Those are the three essential components of every carbohydrate. In fact, they're found in every one of the biomolecules. The carbohydrate is exclusively those three things. It is carbohydrate with hydrogen and oxygen. And the hydrogen-oxygen, of course, together create water. That is not to say that within the carbohydrate is water per se, like we would think of running, flowing, drinkable water, but still the molecules are there.
- 3.8** Because of the unique chemical structure of carbohydrate, and its hydrogens and oxygens in particular, carbohydrates are very polar molecules, so it has a charge to it. Because of that charge, earlier I mentioned, the charge actually makes it unsuitable for slipping through the cell membrane, which is why we have to have specific transporters for carbohydrate. But at the same time, the polarity of the carbohydrate does make it very hydrophilic. So it allows it to dissolve and play very well in water. So carbohydrates and water, both being very polar molecules, end up playing together quite well. So that's an essential part of carbohydrates in our consumption of them and digestion of them.
- 3.9** Carbohydrates, of course, are most famously known in their context as an energy source. Now, that's selling carbohydrates a little short because they are involved in far more than just giving the body energy, but that is their most famous effect, and we'll get to that in more detail in just a moment. But a third and final component to carbohydrates, before talking about their energy source role in a moment, is their role in cell identity. Earlier, when we were talking about the cell membrane or the phospholipid lipid bilayer, I was mentioning some of the components that are stuck in to the phospholipid bilayer, and we talked about this in the context of autoimmune diseases.
- 3.10** Autoimmune diseases happen when the immune cells don't recognize other cells as being part of the same great whole. That recognition is actually due to these unique types of carbohydrates that are flagging or sticking up out of the cell. Now, I don't mean carbohydrates that can be burned for energy. Carbohydrates is a pretty broad family of molecules itself. But suffice it to say, some carbohydrates act as little signals, little flags, popping up out of a cell to identify their loyalty to the rest of the body, hopefully recognized appropriately and not attacked or destroyed.

- 3.11** Now, let's just revisit the role of carbohydrates as an energy source. Carbohydrates come in three general different classifications based on the number of molecules that we're talking about. A carbohydrate can be a monosaccharide, a disaccharide, or a polysaccharide. Now, of course, that prefix ends up really revealing the key here. Within the monosaccharides, we end up getting to the most basic form of the carbohydrate. Indeed, the form of carbohydrate that is necessary for the body to even pull it in.
- 3.12** So through the course of digestion, if we're consuming a bigger, more complicated carbohydrate, which we'll get to in a moment, it needs to be snipped apart, until we get down to the the singular version, or the monosaccharide. These are things as commonly discussed as glucose or fructose or less commonly known galactose. These are all the base individual carbohydrates that are now small enough to be absorbed through the intestine, through the processes that we mentioned earlier, with the symporting with sodium and then the transporting on the back end. And we went into all of that in the previous lecture.
- 3.13** Now, again, because of the hydroxyl groups, which is an -OH at the bottom, flagging off the bottom of the carbohydrate ring, that makes it polar, which allows it to be dissolved in water very well, but also means it doesn't go through membranes particularly well.
- 3.14** Now, if we go one step up in complexity, we are capable in nature of finding and consuming disaccharides. That's going to be the smallest of what's commonly produced in nature. So it's going to come together. Even the most basic food for animals, for mammals, at least, which is going to be milk, milk is lactose. Lactose is a disaccharide. We have a glucose and we have a galactose linked together. Now, that's going to be too big for the body to absorb, so we'll need to snip it apart.
- 3.15** But another example is sucrose or sugar, the sugar that we are adding to our drinks or that we're drinking or sprinkling onto our food. Sucrose is a disaccharide that consists of a bounded version of glucose and fructose. They're linked together with a little carbohydrate bond. Well, that is something we can eat and consume, but it needs to be snipped apart or digested before we can absorb it. So it will eventually, the disaccharide, be broken into its two component parts, whatever it may be. So this is a very common source of nutrition and energy for the body.
- 3.16** Now, the last one is the most complex of all of them, which is the polysaccharides. This is where you have a bunch of carbohydrates linked together. And the most common example is a bunch of glucose molecules. You'll recall glucose is one of the monosaccharides. Well, you can link glucose and glucose and glucose together in a long, what's called polymer. And I'm going to use that word more frequently throughout this lecture. A polymer is a chain of bound molecules.
- 3.17** If you take a polymer of glucose, you end up with these big starches, or more technically, the polysaccharides. But this is the type of carbohydrate you're getting

from very commonly consumed foods, like the carbohydrates in wheat, so eating bread or crackers or cereal, the carbohydrates in potatoes and other vegetables. Those are all polysaccharides, these big long chains of glucose that, when they're consumed, need to be snipped apart. And thankfully, we have the digestive enzymes, which we'll get to later, these enzymes that are capable of making that happen.

3.18 All right, let's leave carbohydrates behind and talk about lipids. Lipids are fats. Now, when you think of fat, when we think of fat, we're thinking of what we can pinch and jiggle. But there is an enormous variety of fats. I remember vividly when I was doing my postdoctoral work, I was studying a particular type of fat, which we'll talk about. And then someone at this meeting said, "Are you going to become a lipid biologist?" And I said, "No, I don't think so." And he said, "Thank your lucky stars. It's the most complicated realm of chemistry." And indeed, I am grateful. I don't do that, because it enables me to study things that are much more interesting than just fats and all of their variety. But there is an enormous variety, and they are tremendously important.

3.19 So in the carbohydrate, there is a roughly, not equal, but a pretty balanced amount of carbohydrates, hydrogen, and oxygen. In the case of fat, there's very little oxygen to it. There may only be one or two little oxygen molecules or atoms attached to this, but there's going to be a lot of carbon and even more hydrogen. That's something we talked about in the cell lecture earlier when we were talking about fats and the lipids and the phospholipid bilayer of the cell membrane. Saturated fats, monounsaturated fats, polyunsaturated fats, they're all a link of carbons, and they are surrounded by hydrogen. So lots of carbons, lots of hydrogens, and relatively little oxygen.

3.20 They're not polar. They're hydrophobic, so they don't like water, which is something we talked about earlier. They're also energy dense. Because the sun has worked so hard to link these carbons together, there's an inherent energy within those carbon bonds. And because there are so many carbon bonds in a fat, we have a lot of potential chemical energy that we can get from fats. That's why fats are said to have more calories, which is itself just a unit of energy, than something like carbohydrate.

3.21 All right, now, there are a few different classes of lipids that I just wanted to go through with you for just a moment, so that we can appreciate some of the variety and different roles of fats within the cell. Glycerolipids, phospholipids, sphingolipids, and then steroids, which is its own family. So firstly, glycerolipids is actually among the most common in the body, depending on how fat we are. Glycerolipids, the most famous form, is the triglyceride. Triglycerides are what constitute the fat that we can pinch and jiggle. That is fat cells that are loaded with a big bubble of triglycerides. So triglyceride is the main form of stored fat that can be used for energy. Not all fats are going to be used for energy. Indeed, none of the rest we're going to talk about are used for energy.

3.22 The glycerolipids are used for energy. And the glycerol- part refers to the structure. So on the structure of a triglyceride, we have a glycerol backbone. That's the glycerol- part of glycerolipid. There's a little glycerol backbone, a few little carbons linked together.

Each one of those has a fatty acid coming off it. You'll recall that the phospholipid has this phosphate head and two lipids dangling off it. The glycerolipid of the triglyceride has a glycerol backbone, and it has three fatty acids.

- 3.23** Well, it ends up being amphipathic, because of this. You'll recall amphipathic is the term referring to a molecule that has both a water-friendly part, in this case the glycerol backbone, and a water-afraid or phobic part, which are the fats themselves. And so this is— we'll revisit this, the triglycerides, and their energy role in the cellular energy lecture at the end of the Basics of Biology. But for now, you'll just appreciate that glycerolipids are among the most abundant of lipids, and they're the type of lipid that is used when we want to get energy from our fat.
- 3.24** Phospholipids are something we've already talked about, and they are primarily responsible for helping create membranes on cells or on some of the organelles. Now, interestingly, there is a dynamic or an interaction between glycerolipids, in the form of triglycerides, and phospholipids. Earlier, I had mentioned that, within every fat cell, everything we're storing to be used for energy later, we have a big bubble of triglycerides, which is a glycerolipid.
- 3.25** The reason that's not just burned all the time is that it's actually surrounded. Within the cell, it's surrounded by phospholipids, and the phospholipids are protecting the glycerolipid. It's basically telling the fat cell, no, you can't take any of this fat yet. And in fact, in order to get access to the triglycerides to break them down for energy, we have to have a series of events, which we'll talk about in the cellular energy lecture, that allows the phospholipid to move out of the way. And now the phospholipids stop protecting the triglycerides, the glycerolipids, which can then be used for energy.
- 3.26** Now, earlier, I had mentioned that as a postdoctoral fellow doing just research before I was a professor, I was studying a type of fat that is relevant in cellular processes. That's the next type of fat called the sphingolipids. The sphingolipids are named after the enigmatic sphinx, because for so many decades, we did not know what they did. And initially, they were thought, once there was a role discovered, we thought that they had no more relevance to the cell than telling the cell when to live or die, which is admittedly pretty relevant. But they're also important in signaling and telling the cell to do other things, including responding to insulin. Well, sphingolipids play a part in that.
- 3.27** Now, the last type of lipid I'd mentioned that is very relevant for the biology of the cell are the steroids. I've already alluded to this in the previous lecture, but steroids are fats that are built on the cholesterol molecule. Anything that goes on to become a steroid was first a cholesterol. Then the cholesterol gets changed. And again, the most famous form of this is the sex hormones, like the estrogens and the androgens, like testosterone, but also other things like hormones involved, like cortisol. Cortisol is based, the stress hormone cortisol is based on cholesterol. A salt-regulating hormone called aldosterone is based on this.

- 3.28** But even other components that are not hormone at all, like, as we talk about later in the cell energy lecture, we'll talk about the electron transport system. So the fundamental business part of the mitochondria requires a molecule that's a tweaked version of cholesterol in order to work. All right, now let's leave the lipids and move on to the proteins.
- 3.29** The proteins are the main machines in a cell. When we think about a cell doing something or some work getting done, it's usually a protein that's doing it. This takes the form—the proteins have so many different roles. They end up creating enzymes, which are molecules that get something done. It will change one molecule into another molecule. They are receptors that are on the cell or sometimes within the cell. For example, earlier, I had mentioned that the steroid hormones are able to move right through the cell membrane. They don't respond to a membrane receptor. They go in. But even if it goes in, there still is going to be a protein that binds to it, that receives it, that receives it. So even the intracellular receptor, let alone the surface receptor, for all signals, for hormones, are going to be protein-based.
- 3.30** Channels are protein-based; transporters are protein-based—the same channels and transporters we talked about earlier. Structural workings, so it's a molecule is actually providing structure to the cell itself, like the components of the cytoskeleton. That involves proteins as well. So you can see that there are countless parts for proteins to play within the cell to help the cell work well.
- 3.31** Now, what is a protein? A protein is a polymer, or a long bound chain, of amino acids. You've heard of amino acids before. People want to take amino acids as a supplement. But regardless, when you eat a protein, the protein itself is too big. In order to absorb it, what we have to do is break it down to its most basic unit, which is going to be breaking it down generally to its individual amino acids. So the amino acids are the building blocks that come together in a polymer to form a protein. And this is necessary to appreciate, because when we start talking about central dogma at the end of this lecture, that's going to be the end result. We want to get from amino acids to protein. So that is the last part of the central dogma theory of the flow of genetic information.
- 3.32** And with all of this, there are just 20 different kinds of amino acids, but can be mixed in countless ways and folded in countless ways in order to create a substantially greater number of proteins. The base structure of the amino acid is unique because it has what's called an amino group. It has a carbon-based group with a couple of carbons linked together, but it has an amine group, or amino group, that has a nitrogen. So that makes it a little unique within the structures that we've covered so far.
- 3.33** Now, some of the most famous amino acids, just as a funny little tangent, are the branched-chain amino acids. If you're familiar with any weightlifting fellow, guaranteed somewhere within his kitchen, he has a little scoop in a bin of branched-chain amino acids. That is some of the amino acids that constitute the muscle. And they do have a unique structure. Interestingly, the branched-chain amino acids is a threesome of

amino acids. Two of them are totally irrelevant to muscle-building, and only one, which is the amino acid leucine, has been shown to have any direct effect.

- 3.34** Now, some of these amino acids are used to build other things that are very unique. For example, thyroid hormone, one of the most important hormones in the body, is built on a backbone of the amino acid, tyrosine. Similarly, the hormone melatonin, the so-called sleep hormone, is built on the backbone of an amino acid called tryptophan. So some of these amino acids are directly used, and almost individually used in some instances, in order to build another molecule like some of these hormones.
- 3.35** Now, within the polymers of the amino acid that end up creating the protein, they are bound together through what is called peptide bonds, and the body has digestive enzymes to snip those apart. But the peptides are—the amino acids are individual amino acids, of course. A peptide would just be one amino acid bound to one amino acid, but also it can get a little longer. If you have a polypeptide, it can get up to 20 amino acids long. And then proteins, to use the term protein, different from amino acid, that's something going to be greater than 50 amino acids linked together. So a protein is going to be the biggest. A polypeptide is going to be intermediate. An individual peptide is just going to be two. And then we have the amino acid at the most basic size.
- 3.36** Now, as we start stacking the amino acids up, getting a long molecule of amino acids, a polymer, there are four different levels of protein structure that need to go through, at least some of them. Some of this needs to happen in order to turn that long string of the polymer of amino acids into a functioning protein. The first level of structure is called the primary structure. In this state, it's unusable. So you can have all the amino acids linked together as a long polymer or a long protein, but the primary structure is, in fact, just that simple string, that little string of pearls, or the polymer of amino acids, in what would be a straight line. That is the primary structure.
- 3.37** Again, you have all the parts required to make the protein, but it is not the protein yet that we may need. There must be some follow-up alteration of that long string. So from a primary structure, it starts to bend in some interesting ways that give rise to what's called the secondary protein structure. And there are two very famous parts of the secondary protein structure. This is when the amino acids start to wrap in a helical chain called an alpha helix. So a little wrapping loop, like the spirals on a candy cane—that is one part of the secondary structure.
- 3.38** And then the next one is these wrinkled little straight—well, it's not straight. It's a planar edged shape. And that is called a beta sheet. So the alpha helices and the beta sheets end up creating the beta pleated sheet. It's like pleated pants. It ends up creating the so-called secondary structure. Each of those is a secondary structure. And then as we start to connect those, wrap them around each other, alpha helices with beta pleated sheets, that creates the tertiary structure, which is now a functional protein.

- 3.39** So if we get to that tertiary structure, we now have something that can work. And in fact, the most famous example, by my estimation, would be something like insulin. Insulin is a tertiary structure of this amino acid polymer. It has wrapped into helices and pleated sheets, and those are closed in on each other. And then we have a functioning protein, in this case, insulin.
- 3.40** Now, the last level is the quaternary structure, or the fourth level. The quaternary structure is when we take multiple tertiary structures and link them together. And the most obvious example, actually, is hemoglobin or the functioning part of the red blood cell. If we were to look into the red blood cell and peel it back, we would see these distinct tertiary structures, these multiple proteins linked together. That's a quaternary structure. Not all proteins will get that far or are needed to go that far, but it's got to get to that tertiary structure for it to be really useful.
- 3.41** Now, interestingly, if we step out of the biology of the individual cell for a moment and look at the relevance of this within the body, let's just stick with insulin and diabetes. You'll know that when a diabetic has to take their insulin, they must do so intravenously or inject it. Why not just take a pill? Wouldn't it be so much easier if they could just take a pill of insulin and just gulp it down? That'd be so much nicer than having to carry around a syringe and poke themselves with a needle multiple times a day. It would be simpler, but it would be totally ineffective because those peptide bonds, when they get to the stomach and the other enzymes of the small intestine, it starts to snip them apart.
- 3.42** And so if you took in the insulin, even though you're taking it in its perfect tertiary structure, that won't last for long because in the process of digestion, it's going to snip those wraps and start to spread it out and then start cutting the polymer apart. And so all you'd end up getting is its individual amino acids, which the body would rebuild however it wanted to. Most certainly, it would not be insulin because the patient doesn't have any beta cells to make the insulin. So if there's ever a hormone that is a peptide-based or protein-based hormone, it must be injected.
- 3.43** In fact, in this day and age, the most famous are what's called the GLP-1 agonists. These are some of the most commonly used weight-loss drugs that are available on the market right now. They must be injected because it's a peptide, and if you were to swallow it, it would get broken apart.
- 3.44** All right, now let's move on to the last one, and this becomes a good segue for the central dogma topic in just a bit, namely the nucleic acids. Nucleic acids are itself a broad term, including energetic molecules like ATP, which we'll talk about later. ATP being the most important molecule for allowing a cell to have energy and do work. But within that family of nucleic acids, we have the nucleotides. And the nucleotides are essential because they store our genetic information. The nucleotides are DNA and RNA. And let's start with DNA.

- 3.45** So DNA, or let's just define the nucleotide structure first, and then we'll distinguish between DNA and RNA. The nucleotide structure has three key parts. It has a phosphate group, which, as I'm acting it out, we'll just say is my thumb. So it has a phosphate group. Then it has this five-carbon carbohydrate in the shape of a ring. Now, I only have four fingers here, but it's actually not too bad because there are four main carbons, and then there's one sticking up to bind onto the phosphate. So we'll say that that's my thumb here. So we have a carbohydrate ring, sometimes referred to as a sugar, or just, well, ribose or deoxyribose, depending on whether it's RNA or DNA, respectively.
- 3.46** So we have the phosphate, we have the sugar ring or the carbohydrate ring, and then we have a base, sometimes referred to as a nitrogenous base or a nitrogen-containing base. And that base ends up identifying the nucleotide. The phosphate can be in common, the ribose can have in common with others. But to make it a unique nucleotide, it's going to have one of four different nitrogenous bases, which we'll get to in a moment. And the DNA will have one of four, and then the RNA will have one of four, and they have three in common, and one is unique to each of them.
- 3.47** And so within that, back to this structure, the phosphate and the carbon structure, it's the carbon structure, that little carbohydrate, that defines or identifies it as DNA or RNA. There's going to be one little difference on one of those little carbons, whether it's bound to an O, an oxygen and a hydrogen or hydroxyl, or whether it's just bound to a hydrogen. That's the only difference between them. So the deoxy part of the DNA, that's going to have just a hydrogen dangling down, and then the RNA is going to have a hydroxyl group dangling down from it, instead of just the hydrogen. So that's the only difference. It's literally one single hydrogen that distinguishes the difference between DNA versus RNA.
- 3.48** All right, now let's get into that in a little more detail. So nucleic acids are going to be a polymer, in this case, of nucleotides. And the DNA stores the genetic information in the nucleus. That tells the cell who we are. And then it will be turned into RNA. So DNA stores the genetic code and RNA expresses it. And that becomes part of the flow of genetic information known as the central dogma, which we'll get into. So let's just talk about DNA for a moment and then RNA for a moment, just to provide a little more detail, because they're so important to the central dogma and the cell being able to replicate itself and have the body be what it is.
- 3.49** So the DNA has the carbohydrate in the form of the deoxyribose. That's that little five-carbon ring. Then it has the bases. Remember, I said it has a nitrogenous base. There are four bases that are abundant on the DNA. They are split up into two different categories, purines and pyrimidines. And the purines are a little longer. The purines actually end up having two little rings attached to it in its little nitrogenous structure, so two rings versus the pyrimidines having only one ring.

- 3.50** And the two-ring structures of the purines include adenine and guanine, which when you're looking at it as a code, genetically, it's just going to be an A or a G. Those are a little bigger. They have two rings attached to it. So that base has two rings. So it ends up being a nucleotide that is actually a little longer. So if the DNA has an adenine or a guanine or a purine, it's going to be a little longer. It's going to stick out a little more than if it's a pyrimidine, which is cytosine and thymine. So we have an A and a G as the purines, and we have a C and a T, cytosine or thymine, as the pyrimidines, slightly shorter.
- 3.51** And that's nice, one being a little longer, one being a little shorter, because they're going to—a long is going to connect with a short, when we start to bring two strands of DNA together. And indeed, that is an essential part of DNA. DNA is a double-stranded molecule in its functional shape within the cell, specifically within the nucleus. And it also is wrapped around in a helix because of its connection with each other. So there's a connection binding them from one strand to another. But then there's a connection with itself where it wants to start to coil in.
- 3.52** And you've seen this before, where you could have something that's been twisted, but if you let it go, it wants to wrap up. That's exactly what happens with the helix, why it forms a helix with DNA. It starts to get attracted to each other. Within its own line, it starts to want to pull in on other molecules that are above or below it. And so it starts to twist, giving rise to the famous double helix that the scientists Watson and Crick identified as the central unit of genetic information, and indeed, won the Nobel Prize for it.
- 3.53** Now, there is within this a law of complementary base pairing, where you have an adenine, or an A, on one side; as a purine, being bigger, it wants to match and pull. It will couple up with a thiamine on the opposite side. So where there is an A on one DNA strand, it will match with a T on the other in DNA. Similarly, if there is a cytosine on one side, it will want to match with a guanine on the other.
- 3.54** In fact, maybe to make it a little more accurate, I should shorten one up because we have the pyrimidine side, which is going to be a little shorter. The pyrimidine being cytosine and thiamine, but then coupling with a purine, which is longer, the adenine or the guanine. Now, you'll notice that I'm actually putting my thumbs intentionally in opposite directions because that's an essential part of this complementary base pairing, is that as one strand is running in one direction, the other is actually running in the opposite direction. So they are what's called anti-parallel. They're not parallel, they're anti-parallel.
- 3.55** So the actual orientation of these molecules is flipped. And it's that orientation, depending on the flipping of that ribose, that sugar, is it flipping one way with its three carbon down? And then that's usually the way—well, we're going to talk about how it reads the molecule when we talk about DNA replication and the central dogma. But the orientation of it determines the direction that the genetic code is being read. And so

suffice to say, it's going in opposite directions. So the DNA double strand is referred to as anti-parallel. The structures are flipped in opposite directions, but coupling in the middle, meeting up in the middle.

- 3.56** So now DNA has this double helix structure, and they are matched with purine binding across to a pyrimidine, but they do form a double helix. That's the main difference. With RNA, there's a few differences. First and foremost, of course, is that that sugar molecule has an extra oxygen on it. So there's a hydroxyl group off of one of the carbons, which is why we call it RNA and not deoxyribonucleic acid or DNA. So the RNA just has a slightly different sugar structure or the carbohydrate structure.
- 3.57** But then also another difference is, it doesn't have a thiamine. And so an A, when the adenine is wanting to bind with its partner, it doesn't. It binds to a uracil. So uracil is an important part of the RNA that is replacing the T or the thiamine in DNA. As we start to take the DNA and want to express it in the protein, it has to turn into, we have to copy it into an RNA form. In every length of RNA, it can have Cs, it can have Gs, it can have As, but it will not have Ts for thiamine; it will have uracil instead. And where there would have been a thiamine, or a T, opposite of an A, it's going to be a uracil. So there's a slightly different, there's a slightly different code here when it comes to RNA.
- 3.58** Now, RNA itself actually comes in a variety of versions. The one that I've been describing is known as messenger RNA. So when we're talking about taking the genetic information of a DNA double helix and turning it into a single strand, but maybe helical still, a single strand of RNA, the RNA will not be double helix. It's a single strand. This is messenger mRNA. It's the message coming from the genetic code. But ultimately, as we end up fully expressing the gene as a protein, we're going to have to go through or use two other types of RNA, one called transfer RNA or t-RNA, then the last one called ribosomal RNA, sometimes just refer to as the ribosomes. And indeed, that's how I will refer to them when we get to the central dogma in just a bit.
- 3.59** So with all this having been said, you are much more familiar with the four major biomolecules—proteins, lipids, carbohydrates, and now nucleic acids, which again are so important in the form of the nucleotide polymers to store genetic information. But before we get to that genetic information, you need to appreciate how the cell is going to allow us to express that genetic information and, indeed, do much more. So let's just take a few minutes and talk about enzymes.
- 3.60** An enzyme is a form—it's one of the things that proteins can become. So over the next few minutes, I want you to get familiar with the nature of enzymes and how they can facilitate basic chemical reactions. Even the most simple of cells, like a red blood cell, is still performing hundreds of thousands of reactions. There are so many different things the cell needs to do in order to survive, let alone to play its proper function within the greater whole of the body or the organism. But the cell, among these hundreds of thousands of reactions, in the 200 different types of cells, in the 30 trillion cells in total,

you can see that that is a lot of work that needs to be done. And the enzymes help us do that work.

- 3.61** Now, before we actually describe the function of the enzyme, let's just talk about why we would even need it in the first place. Overwhelmingly, enzymes are needed in order to break a bond or to form a bond. Well, that requires energy in either direction, depending on the nature of the bond. And so there are two main types of bonds that are going to be relevant within biology, covalent bonds and hydrogen bonds. A covalent bond is the type of bond that we get with, actually, carbon, the main molecule that defines an organic molecule. Carbon will have a covalent bond with any of the other elements.
- 3.62** And to do so, it actually is a strong bond because it shares its electrons. It is strongly bound together. All of these elements have electrons swinging around them, and it starts to share the electrons, and it comes together very, very strongly. So a covalent bond is a very strong physical chemical bond, where one element is sharing its electrons with another, and boy, they are tightly bound. It's going to be very difficult to pull them apart. But a cell may need to, and so we have an enzyme to help.
- 3.63** A weaker bond is a type of bond called a hydrogen bond. In this case, it is not covalent. It is not sharing electrons, but it is sharing an electrostatic attraction. So this is when you simply have the phenomenon of a negative signal attracting a positive signal or vice versa. Different molecules will have a different charge or different parts of it will have a different charge, positive or negative. The fact that they want to come together is a hydrogen bond.
- 3.64** So water ends up becoming or representing this uniquely polar molecule with positive and negative charges. So the hydrogen bond wants to bring something together. But again, it's just an electrostatic. So it's bound, but it's somewhat weak, and can be very easily overcome either through enzyme or even just something as simple as heat. For example, you can have a pot of water. You're not adding an enzyme into there to break the water molecules apart. They're attracted to each other, which is why they're generally in a liquid form. But all you need to do is turn up the heat, literally, and you're breaking those bonds. And now you're getting individual hydrogen atoms that are just evaporating up.
- 3.65** So a hydrogen bond is real. It is very, very functional within the cell, but it's also quite weak, and it can be relatively easy to overcome. Even still, whether it's hydrogen bonds, because, of course, you're not boiling a cell, so you can't rely on temperature to do it, to get something done, or whether it's covalent bonds, an enzyme is going to help break that bond or form the bond in the case of a covalent bond.
- 3.66** But an enzyme, but that's going to require energy. There's a certain amount of energy that needs to be put into this reaction. Some of these energies, if we left the molecule alone for years, it may just happen over time, but we don't need it to happen

eventually. We want it to happen now, whatever the reaction may be. We want to take that carbohydrate and start breaking it apart so we can get energy from it, from that glucose molecule, which we'll talk about in the cell energy lecture later. So we need to overcome the resistance. We need to overcome the strength of that covalent bond, and that's what an enzyme can do.

3.67 An enzyme essentially takes what would be an insurmountable amount of energy in order to break and brings the energy demand down. It says, hey, you want to break that? I can help you. And so I'm going to come in and help you pull this apart. And now, hey, look, we did it. That's the role of the enzyme, essentially. Now, there are certain properties in an enzyme that help determine its function. The first, and just some general rules of enzyme, one is that the enzyme is going to be very specific for one set of reactions. So one enzyme is generally going to have one reaction.

3.68 Similarly, the enzyme itself is not changed in the process. It comes in, and in this earlier example, helps you break something apart. Well, that something that broke apart is now very different. But the enzyme isn't. It remains unchanged throughout the reaction, ready to do it again. And similarly, the enzyme is not consumed. It's not broken apart itself. It's not changed, nor is it consumed. And usually, just as a term of nomenclature, if you want to wonder whether something is an enzyme or not, it is always going to have the suffix "ase" tagged onto the end of it, or -ase, -ase.

3.69 Earlier, I had mentioned ATP as the main molecule of energy. Well, ATPase is a molecule that is going to deal with that ATP. It actually is a term to refer to building it up or breaking it down, depending on the nature of the ATPase. Look, for example, sucrase. Earlier, we talked about the disaccharide sucrose, which is sugar, table sugar. There's a glucose and there is a fructose, bound together through what's called the glycosidic bond. Sucrase is going to take that bond and say, you are sucrose. I am here for you. I'm cutting you apart. Boom. And now we're left with the monosaccharides, the glucose and the fructose, which can then be absorbed. So if you hear a word and it ends with -ase, it is an enzyme always. That is the main method of identifying it within the realm of chemistry.

3.70 So let's just go through an example of how an enzyme might work. And let's take the first enzyme in a pathway called glycolysis. Glycolysis is a term you're going to be very familiar with by the end of this whole series. Glycolysis is the word for describing the breakdown or the metabolism of glucose, taking a glucose molecule and pulling it apart in all its varieties in order to get energy. That's what we want the glucose to do. We want to get energy from it. The first step in what is called the rate-limiting step—so this is the step that will be the slowest, and the rate at which it works will determine the rate at which all the other enzymes work in this long, multi-step process of glycolysis, which you will learn all about by the end of the series.

3.71 The first step is an enzyme called hexokinase or glucokinase. You can hear the -ase on that, but it takes the glucose molecule as it comes into the cell and then immediately

locks on a phosphate to it. And I use the word lock because it actually locks the molecule into the cell. Otherwise, if it just stayed as glucose, you remember the transporter is opened up and the glucose could just slip right back out. The moment the glucose comes in, this glucokinase acts, also known as hexokinase, it will immediately act on it and lock in on a phosphate, turning the glucose into a molecule called glucose 6-phosphate, because on that sixth carbon, we now have a phosphate added onto it.

3.72 In so doing, we essentially commit this molecule to the glycolytic or the glucose-burning pathway. So that first enzyme is going to take—we have the enzyme itself, the glucokinase. It is built to bind on to glucose. So it takes on the glucose. Now, there are some instances where an enzyme can take in its, what's called substrate or base-building working molecule, and just let it go. That can happen, but it's not as common. Usually, once the enzyme binds it, again, glucokinase binding glucose, it will do what it wants to do.

3.73 And in this case, it wants to take a circulating phosphate molecule and say, glucose, I need you to stay here and I need to burn you. So I'm going to charge you up with this phosphate molecule. And so it took in a glucose, it took in a phosphate, and released the new molecule, glucose 6-phosphate. And it created that covalent bond. So it's bound those two molecules together. So that's a strong connection.

3.74 Now, having done so, it releases the glucose 6-phosphate, and then the glucokinase is ready to do it again. Because, like a true enzyme, it has not been consumed, it has not been altered in any way. It is ready to continue to phosphorylate glucose and enable the process of metabolizing glucose through what we commonly call glycolysis, an essential process for, indeed, every cell. Every cell of the body readily uses glucose as a fuel.

3.75 Now, within the enzyme understanding, there are a couple other points or a few other points I want to mention. The first one is that there is a specificity that usually an enzyme is only going to work with one particular substrate or one particular molecule. So if there is a bunch of molecules, like, let's stick with glucokinase, the first enzyme involved in glycolysis or the burning of glucose, there are tons, of course, thousands of different molecules within that cell. But glucokinase is only going to recognize glucose, for the most part. And so it just pulls in the glucose, whereas all these other molecules are surrounding it, it doesn't have any eyes for those molecules. It's only paying attention to the glucose. That is its purpose. So it's specific. So there is a substrate or molecular specificity to every enzyme.

3.76 There's also the relevance of the concentration. If you have more and more substrate, the enzyme is going to become more and more active. It's going to be much, much busier. It's pulling in one molecule and changing it. And then there's another one right there, and it grabs it and changes it, grabs it and changes it. So as the concentration of the substrate starts to go up, the reaction rate will also go up. So if the building blocks are more abundant, the enzyme is much more active.

- 3.77** And then lastly, there is an affinity. And this is part of the substrate specificity, but it's a little unique because there's an additional aspect to it that I'll introduce. But affinity suggests that the size, or rather the strength, of the binding between the substrate and the enzyme has to be strong enough to want to pull it in in the first place. Not only is it specifically designed to pull it in, but it has to have an attraction or an affinity.
- 3.78** Now, this affinity isn't always going to be exclusive. There can be this concept of competition. So there can be instances, and indeed there are, depending on the enzyme, where the enzyme has a little bit of affinity for another molecule, even if that molecule isn't a suitable substrate for the enzyme. So let's talk about glucokinase again, the first enzyme involved in glycolysis or the metabolism of glucose. Its natural substrate, or building molecule that it wants to take in, is the glucose.
- 3.79** Interestingly, it also has a slight affinity for its own product. So it pulls in the glucose and the phosphate and releases glucose 6-phosphate. Well, if there becomes to be a lot of glucose 6-phosphate in the cell, it actually can start to lock on that spot. It has enough of an affinity for its own product that the product can come and occupy the binding site on the glucokinase, thereby inhibiting the enzyme's ability to continue to phosphorylate glucose. I hope that makes sense. So the affinity, if an enzyme has an affinity for more than one molecule, we've inherently brought in the idea of competition, where there may be some other molecule that's competing with the molecule that the enzyme is supposed to be utilizing. So that's affinity and competition.
- 3.80** All right, now, to this point, you should be familiar with the nature of enzymes and how they facilitate basic chemical reactions, by basically lowering the amount of work that the molecules would have to do on their own. The enzyme accelerates and facilitates a chemical process or a chemical event. And remember, there are a lot of chemical events because even plain simple cells have hundreds of thousands of chemical reactions they must engage in. If it weren't for enzymes, they largely wouldn't happen at all.

Lecture 4: The Central Dogma

At a glance — The central dogma of molecular biology.

- 4.1 Now that we have the understanding of the biomolecules in mind and we have enzymes appreciated and generally understood, let's move into the meat of the matter, which is the central dogma. Now, by the end of this part of the lecture, within this subtopic, my hope is that you'll be able to understand the overall principles that constitute the central dogma of biology and why it is so important.
- 4.2 So firstly, let's just start with defining the central dogma of biology, often just referred to as the central dogma. I've used this term already or this expression, but it is essentially the flow of genetic information from DNA as it is transcribed into RNA, which is then translated into protein. So to say all of this in a less technical way, the central dogma helps us understand how our genes determine our body. And it is everything here.
- 4.3 Now, just as a point of interest, this was first proposed by the same individual who first identified the structure of DNA, which is Francis Crick in the UK. So in 1958, Francis Crick proposed the central dogma of biology. Having discovered the overall structure of DNA, it then became the scientist's delight to try to understand how it fit within the broader context of biology.
- 4.4 But again, to say all of this in other way, the central dogma is going from DNA, your base genetic code, to who you are. Because who we are—yes, we are made up of cells, but the function of these cells is really determined by proteins. So as DNA is transcribed into RNA and then translated into proteins, that is what determines who we are. Height is determined by the proteins that are expressed, our general body size, our body shape, the color of our skin, the color of our hair and our eyes, all of that is a function of the proteins that are being expressed. And the proteins that are being expressed are a function of the RNA, the expression of the genetic code, which itself, of course, is a function of the genetic code.
- 4.5 Now, there are, of course, environmental factors that play into some of this, but the environmental factors are only going to have a very modest influence, very modest. It is the DNA to the RNA to the eventual expression of the protein itself that determines the overall makeup of our bodies. Size, shape, color, and everything else that plays into the variety, the beautiful variety, of the human body.
- 4.6 Now, to use a different view of looking at this, the genotype is the genetic code. That's basically like the blueprints for building a home. And then the phenotype is the physical manifestation of that blueprint. So that would be the home itself. And so, of course, the biological corollary is the genetics are the blueprint, then the protein is the structure, the end result. But again, back to this example, it's the blueprint for the home and then

the actual home itself, the genotype or genetic code, or the phenotype and the expression of the protein and the end result.

- 4.7 So when we look at what makes up the final product, what makes up the composition of our bodies or the structure of the home, we, of course, are made up of cells. And then the cells are influenced by all of the proteins that determine the functionality of what the cells are doing, giving me my freckles because of the strange expression of melanin, or giving me, in my case, my blue eyes. That would be a function of the proteins that my cells are expressing.
- 4.8 And just as a reminder, a protein is a sequence or a polymer of amino acids. You'll recall we talked about the polymer of amino acids, how there's the single-level structure, and then we have the secondary structure, then we have a tertiary structure. That all plays into the protein. But before that, we have the DNA. And the DNA as the genetic code, it is a sequence of specific amino acids, just like a computer code. Imagine an old ticker tape has a series of information on it, so too does DNA. It is a series of codes, or it is a code in general, broken up into smaller little genetic codes or genomes or genes.
- 4.9 Now, what is the gene? The gene is a DNA sequence that encodes for a particular product, whatever that product is. If we look at the length, if we unwound the DNA molecule from a chromosome, and we'll talk more about that in just a few minutes, within that would be distinct regions where we would say, OK, nothing, nothing, nonsense. OK, here's a gene that expresses something. So it's relevant. It's a gene because it can eventually turn into a protein product. Nothing, nothing. OK, here's another one. Here's another one.
- 4.10 So not all of DNA, not all of a chromosome, which is a length of wrapped-up DNA, is going to have genes on it. It's not all gene from front to end. A lot of it is just wasted space, as we'll talk about. The genes are in selected areas, and that little gene will be recognized and turned into RNA, an expression of the genetic code. And that RNA will be turned into a version of some protein, some amino acid chain.
- 4.11 Now, we talked about the genetic or the DNA sequence earlier, which we could call it the genetic sequence, where we have the nucleotides, the DNA molecules themselves, the individual nucleotides, broken up into the purines and the pyrimidines. The purines being adenine and guanine, A and G, and the pyrimidines being cytosine and thymine, C and T. That's the DNA structure. And as a reminder, they run anti-parallel. That was on that DNA molecule, we have the sugar molecule with the phosphate and then its nitrogenous bases coming off this side.
- 4.12 I'd mentioned earlier, not only the anti-parallel nature of it, but it's very now important to understand that there is what's called a three-prime (3') end and a five-prime (5') end. One strand of DNA just starting at the bottom as we're orienting it here, one strand is starting with the 3' facing down. And the 3' is going to be one of the sugars that is on that base, ribose or deoxyribose, molecule. On that sugar little carbohydrate

molecule, the 3' is going to be one of those bottom ones, whereas the 5' is going to be what my thumb is awkwardly doing here, where there's a phosphate bound onto it.

4.13 So on one strand, 3' is at the bottom, 5' is at the top, as we are stacking these molecules up. All of them are in that same orientation on this side. 3' at the bottom, 5' at the top. On the other side, it's anti-parallel. In this case, starting at the bottom, it's 5'. And as it's winding up in the double helix, 3' is at the top. So again, anti-parallel, stacked against each other, 3' to 5', or 5' to 3'.

4.14 Now, I know you're wondering, "Ben, Dr. Bikman, why on earth are you talking about this?" That becomes important when we start to peel things apart to read the genetic code. The directionality becomes important in how we express the genetic information. So again, within the double helix, we have this anti-parallel structure of moving on different ends from 3' to 5', which is something we will revisit.

4.15 So at this point, just to have the general overview of the central dogma, we're starting within the nucleus, and we have DNA, a long genetic code string. That DNA will, through processes we'll get to in a moment, be transcribed into RNA. RNA will then have the ability to leave the nucleus out into the cytosol, the cytoplasm. That's the water and everything else that's there. And then it will interact with other molecules to undergo what's called translation, ultimately giving us the protein. Then the protein will have a variety of fates, maybe staying in the cell to do something or maybe being released in order to send some signal somewhere else in the body.

4.16 So at this point, you should be able to understand and describe the overall principles of the central dogma, again, being the flow of genetic information that takes our genes and turns it into who we are at the ultimate end. So the first part of the central dogma is DNA replication. And then we'll go on. So let's discuss DNA replication for a few minutes. Then we'll go on to talk about transcription, which is when the DNA is changing into mRNA, the messenger form, the genetic form of RNA.

4.17 And then lastly, we'll talk about translation, which is when we convert that genetic expression through RNA into the protein itself. And then again, just to say it for the 10th time at this point probably, it's the protein that determines the overall appearance and functions, or what the cells are doing, and then the overall appearance of the body makes us who we are.

4.18 So firstly, with DNA replication, in the next upcoming lecture, we're going to talk about mitosis and meiosis, the process of the cell actually dividing. If it weren't for that, of course, we would have no life. We would die. A cell is not immortal, and if that cell died, we'd be dead. So thankfully, we can live longer than the life of an individual cell because the cells can divide, and they can pass their genetic information not only to their own dividing cells, keeping me me, so that I don't start becoming someone else, but there's also the ability to create germ cells or gametes, which allows the human, the animal, to reproduce. Those are both the two processes.

- 4.19** But again, before we have that happen, we have to have the ability of taking the genetic code in a DNA and replicating it so that when we're about to pull a cell apart through mitosis, which we'll get to in the next lecture, we have two copies of all of the genetic information so that they stay identical. I don't want my liver cells to start spontaneously changing into someone else's cells or a different kind of cell entirely. I don't want my liver cell turning into my kidney cells.
- 4.20** And so the fidelity and integrity of DNA replication means one cell can become two cells, and they are identical, allowing me to stay me and every human to stay whoever they are. So DNA replication is the first part of this, which is a necessary event in cell division and genetic heritability, passing on information to offspring.
- 4.21** Now, over the decades, there were actually several proposed mechanisms. There was tremendous scientific debate in order to understand, how does the double helix of DNA end up sharing its information or get replicated? How does it turn into more copies of itself? There were multiple different theories, and I'm only going to spend time talking about the one that has been the most validated, and I would say, proven, which is referred to as the semi-conservative model of DNA replication.
- 4.22** With semi-conservative, it means that there is always a piece of the original. So we have these two copies of DNA together. And then, as we're going to go through in just a moment, the DNA will pull itself apart and start to get copied, and new strands of DNA will be produced, an exact replica of one existing strand. And then eventually, when we start to divide the cells, the two original strands of DNA will bind on to two of its identically made copies.
- 4.23** And so in the first cell, we had 100% original. In the first round of replication, as this cell wants to divide into two separate new cells, one original strand will be in this one, one original strand will be in that one, coupled with a newly-produced identical strand. So they each have 50% of what existed in the first 100% of the original cell. Now, when these cells will eventually divide, if we are only tracking the original DNA that came from this cell, now, these two cells have each one original copy, one new copy, one original copy, one new copy. When they divide, now the original copy will stay in one.
- 4.24** So one of those cells will have 50-50. But then there will be one new daughter cell, it's called, that doesn't have any of the original line of DNA. Now, it's a perfect copy. So for all intents and purposes, it's the same. It is functionally the same, but technically, it is not the exact same strand of DNA. That has been diluted and worked out into other cells.
- 4.25** So that's the semi-conservative model, where as we continue to replicate, we get more and more cells that have none of the original strand, but it will always exist. The original strand will always be there, getting passed down through these so-called daughter cells. When one cell divides, it has daughter cells. And we actually use the term daughter or sister when we start referring to the DNA strands themselves in just a bit.

- 4.26 All right, so the highest level, when it comes to DNA replication, let's have in your mind the idea of the double helix, the two DNA strands linked together, wrapping around each other in that helical fashion. The first thing we need to do is get to the individual code or the individual nucleotides. Remember the A, Gs, the Cs, and the Ts, the individual nucleotides that are creating that line of genetic code as they're all bound together. In order to do that, we have to unwind. We have to pull this double helix apart. It's pulled in too tightly, we can't get to the information. We have to open it up.
- 4.27 So as we go through this, we're going to describe the process of opening up or uncoiling the DNA. So it's all tightly coiled. We start to open it and expand it. Now we have enzymes that can get on there and replicate the DNA. Now, why am I going like this? It's because the directionality matters, as we'll get to. You'll recall that there's an anti-parallel structure of the double helix. Because of that, we end up reading it and, if you will, opposite directions, although it ends up being a little segmented in one instance.
- 4.28 Once we have expressed this new line of DNA, we have this big expanded line that can then start to close back in or separate and join onto the new strands of DNA. But we have what's called this replication fork at either end. So we have the tightly wound DNA as it has opened up. We have a replication fork here and a replication fork here. But that's basically just where the DNA starts to coil back into each other. So we can have these two ends of what has opened into this area of replication in order to copy the DNA.
- 4.29 Now, let's talk about the individual components that allow this to happen, that take the tightly coiled DNA, start to open it, and then start to copy it. Among the proteins that are relevant within DNA replication, the first one is helicase. And helicase is what actually comes into the tightly coiled double helix and opens it up. So helicase comes in and starts to spread it apart. Once it's spread apart, it basically anchors it open by the insertion of some single-stranded binding proteins. So helicase comes and opens it, but then it basically uses these molecules. It's kind of like kicking them in place if you want to try to get something stacked open.
- 4.30 If you've ever locked your keys in your car, one method of getting your keys out of your car is to slowly pull open the door and then put a book in. Then you tug it a little more and put another book. You tug it a little more and put another book. So the tugging is the helicase opening up something that's very tight. Then the books are the single-stranded binding proteins keeping the thing open. So helicase starts to run along the length of the DNA molecule, opening it, opening it, opening it up. But why doesn't it just collapse back in on itself? Because of the single-stranded binding proteins. They're keeping it open.
- 4.31 Now, once it's opened up, we have another enzyme called primase that's going to become very important, to establish this little area and say, OK, this is where we're starting the replication, right here. So the primase primes the copying. Then the DNA

polymerase, polymerase being that it's making a polymer of nucleic acids. It's making this new link by copying. And then lastly, there will be something called the ligase that comes and just sinks or connects everything together.

- 4.32** Now, as another reminder, the DNA polymerase or the enzyme that's actually copying, making a new polymer of the DNA information, only goes in one direction. This is where the directionality is so important. It reads the template in a 3' to 5' direction. So as the polymerase comes and binds one strand of DNA, it's starting on the 3' end, and then starts laying down a new copy on top of it, making a new DNA strand. So this opposite strand, as it's being created, of course, is being created in the 5' to 3' direction. I hope that makes sense.
- 4.33** So the copying molecule, the DNA polymerase, will come in and start on a 3' end, which is, in this case, my elbow. Here's the 3' end. You'll recall the anti-parallel nature of the DNA. The polymerase comes in, binds, and starts, or it reads the template or the original strand from the 3' end moving down towards the 5' end. But because it's creating an opposite, its own new anti-parallel DNA line, it's actually creating the DNA from the 5' end down to its own 3' end. So again, simply anti-parallel.
- 4.34** So we took an existing anti-parallel double helix, pulled it apart. But what we are now creating is a new anti-parallel double helix. So it's reading from the 3' to the 5', but creating from the 5' to the 3'. Lots of primes, potential confusion, but that's the way it goes. Now, as we start to open up the DNA, one of these DNAs is going to be called the leading strand because of its directionality, 3' to 5'. The other strand is, of course, going in the opposite direction, but the polymerase can still only move in one direction.
- 4.35** So if I open up my DNA here, here's 3' to 5', the polymerase has no problem. It will bind right here and just continuously go unbroken because it wants to go this direction, especially if we're opening it up in this direction. We're opening, the helicase is opening, and we're going to go into a little more detail. I'm just whetting your appetite because I can tell you're so excited about it. So we are opening this molecule and the polymerase just continues to go continuously, unbroken, uninterrupted.
- 4.36** But remember, the polymerase can only read the strand from 3' to 5'. So on this other strand, it's called the lagging strand. And if the polymerase is still going this direction, sorry, it has to go in the 3' to 5', but it continues to open up in this direction. So now on this lagging strand, polymerase will come and bind going from 3' to 5'. But as it goes from one section here, it's already opened up more this direction, which I hope makes some sense.
- 4.37** Now we have this polymerase, which has to disengage. It's coded this direction, getting to a closed replication fork. And then it says, oh, wait, hold on, back up this way, back up closer to my 3' end, we have more of an opening because the helicase has continued to open the DNA strand. And so this molecule will jump forward ahead and then come

this way until it bumps into the existing one. Then it will jump further ahead until it will bump into this existing one. And so we keep creating these small little segments.

- 4.38** I'm getting a little ahead of myself, so let's just go back to the leading strand. We have the tightly coiled DNA. It has a helicase that will come and pry it open. Once it's pried open, in place, we'll skip in these single-stranded binding proteins, which will keep it open. Then the helicase continues to move along, opening the molecule up more and more. And now the DNA polymerase, the molecule that's going to make a new polymer of DNA or make new nucleotides and bind them together, it has to bind on to these templates, both of them.
- 4.39** And depending on the direction, remember, it can only create a new peptide or a new nucleotide sequence in the 5' to 3' direction. That's what it is creating. It's starting with its own new 5', moving to its own eventual 3'. It's continually adding on. 5' is where it starts. Now it's building that on more, more, more, more, building it in the 5' to 3', but to do so, of course, it's reading the original template from 3' to 5'.
- 4.40** So, as I'd mentioned, we have a leading strand, and the leading strand is going to be whichever strand, as this is getting unwound, has on its one end of the replication fork, the 3' end, because the polymerase wants to bind onto that 3' end, and now it can just continue to go. As long as we continue to open this molecule up, this DNA polymerase just can continue to go and create a copy strand of DNA, continuously, without any problem. That is the leading strand.
- 4.41** As long as we have the molecule continue to stay open by nature of the single-stranded binding proteins in the helicase continuing to open up and unzip the DNA, on that leading strand, the polymerase just can continue to read from the 3' end on and on and on until the very end of where we get to the 5' and termination sequence. And then it's done its job. At the same time, though, we have what's called the lagging strand. And the lagging strand is a strand that is going—its directionality is 5' to 3'.
- 4.42** So as the DNA is opening up, we have on this side, so in my awkward acting it out, this is the leading strand, 3' to 5', and then the DNA polymerase just continues to go because it wants to read 3' to 5'. On the lagging strand, as it's opening up, it will get access, as it continues to uncoil and open and open. The DNA polymerase will say, OK, here's the starting sequence. I'm going to bind right here and go and go and go. Oops, I got to the replication fork. I'm done.
- 4.43** But then again, as the molecules continue to open, there's a new point work in access. And so it will come in further down. The DNA polymerase will say, I can start at this 3' end and go and go and go. Oops, I bumped into the existing fragment. And so it continues to skip ahead, making new little sections. But now there are all of these distinct little DNA sections that are all called Okazaki fragments. They are little DNA fragments that have been produced because of the polymerase skipping ahead, skipping ahead, skipping ahead. Remember, only on the lagging strand of DNA.

- 4.44 But those are not bound together. And so eventually, then, that's when we have the final enzyme that I alluded to earlier, which is ligase. At the very end of it all, along that lagging strand, where the DNA polymerase has continually skip ahead. Here's one section, it read it. Now it comes off to the earlier section and reads it. Then it comes off to the earlier section and reads it. They're all disconnected and fragmented. So then the enzyme ligase comes in and seals them up, seals them up, seals them up.
- 4.45 So now we have an identical, but in opposite fashion, another DNA strand that was capable of being produced from the lagging strand. So that allows us this overall process of leading strand in its DNA polymerase, lagging strand and its DNA polymerase, were able to continue to obey this sequence of reading from 3' to 5', but doing so in opposite directions because of the anti-parallel nature of the DNA, ultimately giving us now exact replicas of the original DNA strand, maintaining the genetic information, and allowing the cell to divide, because all of its genetic information is now copied or duplicated.
- 4.46 Thankfully, DNA replication is very, very accurate. This ensures that as we are copying our genetic information, we don't inadvertently introduce mutations or changes so that we have different types of cells and we stop becoming who we are. The three mechanisms that ensure accuracy—the first one is hydrogen bonding, where when we have the law of complementary base pairing, the A being coupled to a T, in the case of DNA, or the G being coupled to a C, the purines and the pyrimidines linking across each other, they have a stronger bond than they would with anything else.
- 4.47 So if you, for example, had an A and accidentally coupled it with a C rather than the T, that's such a weak bond that there will be a weird little disrupted shape within the DNA helix, and then it gets recognized and fixed. So the bond isn't as strong if it's a mismatched combination, which allows itself to be more easily recognized. Whereas if it is a coupled bond, A to T, G to C, the hydrogen bond is stronger and we have a tighter DNA coil.
- 4.48 Also, the active site of the DNA polymerase makes it so that it doesn't mismatch very readily. So DNA polymerase has its own site recognition software. It knows where to start and when to end. But also with DNA polymerase, it has the ability to proofread. It can look back and determine what it just put down and recognize an error before it even lasts more than just a fraction of a second. So DNA polymerase, in addition to writing the new code, can edit or proofread its own work, where the DNA polymerase can literally back up, break down one section, and rebuild it, and then continue to go.
- 4.49 So DNA polymerase is a pretty multi-talented enzyme. Whatever we want to call it, it is certainly totally essential to the replication of DNA. And remember, the replication of DNA is the first and perhaps most complicated part of the central dogma of biology. And with that perhaps most complicated part behind us, let's move on to the next part, which is the transcription. So my hope in this little mini section is that you can understand and then describe the details of transcription.

- 4.50 So as a reminder, the central dogma of biology is taking the raw genetic code of DNA, transcribing it, which is the topic now, into the genetic expression, which is the messenger RNA, or we can just call it RNA, but it's messenger RNA. And then eventually we'll get to, in just a few minutes, the translation, the conversion of that genetic expression into the protein itself. So DNA coding for RNA is transcription. mRNA coding for the protein is translation. And remember, it's the protein that determines the function of the cell.
- 4.51 Earlier, I mentioned this, and now it's an appropriate time to revisit it because they're all about to become very important, which is the three different types of RNA. The one that I'm going to talk about mostly now, when we talk about transcription, is messenger RNA. That is the template that has been produced, the genetic code copy, from the DNA. Remember, DNA stays in the nucleus. It does not come out. But protein synthesis happens in the nucleus. So how do we get that signal from the nucleus without damaging it into the cytosol? That's the messenger RNA. It's sending the genetic message out from the nucleus into the other part of the cell, into the cytosol of the cell.
- 4.52 Once it gets out there—and that is the process of transcription—once it gets there, it will interact with the other types of RNA, transfer RNA, which is recognizing a specific sequence on the mRNA, and then ribosomes, the ribosomal RNA, which allows the transfer RNA to do its job, basically. Those two types of RNA work together on the messenger RNA in order to give rise to the translation or the production of the protein itself.
- 4.53 Each of these has its own RNA polymerase. So just as there are three different types of RNA, there are three different types of RNA polymerase. Earlier, for DNA replication, we talked about DNA polymerase, making a new polymer of the DNA nucleotides. Now we have an RNA polymerase, RNA polymerase one, two, and three that is responsible for the ribosomes or the messenger RNA, or ultimately, the transfer RNA.
- 4.54 Now, thankfully, the process of transcription is much simpler than the process of DNA replication. So you may be thinking, gosh, Dr. Bikman, why didn't you start with that? Well, because it doesn't start with RNA replication, unfortunately, or RNA message production, the transcription. But thankfully, having gone through that awfully complicated system of replicating DNA, this ends up becoming a bit of a walk in the park. The first thing that is different here is that, as we start to copy the DNA, now rather than DNA becoming a new DNA, just to make sure we're locked in, this is a DNA code that's been separated out, and now it's getting copied into an RNA strand.
- 4.55 So it's not a perfect copy, because you'll recall that one of the differences between DNA and RNA is the presence or absence of thiamine. What is thiamine, or a T, in the DNA is actually uracil in the RNA. And there are a handful of suspected theories of why that is, and for the sake of time, we won't bother. But suffice it to say, it's not the exact same, but functionally it is. It is this looser, more easily created copy, but that is RNA. But

instead of all those enzymes that I mentioned earlier, the helicase, the single-stranded binding proteins, the ligase, the polymerase, RNA transcription only has one, which is just RNA polymerase. But it is doing the function of all of those other things.

4.56 So we have the tightly coiled DNA. We need to open it, and we need to copy it from the 3' to the 5' direction. RNA polymerase does all of it. So it does pay that we took the time earlier because now it's just that simple. Uncoil the DNA, RNA polymerase comes and binds the 3' end, and starts reading from 3' to 5', creating now not a DNA replica, but an RNA replica. This is transcription, the middle part of the central dogma of biology. So it is creating the RNA strand from its 5' end to its eventual 3' end.

4.57 So that is this process that involves three distinct parts. Firstly, the RNA polymerase must recognize a start signal. So there is the initiation, which includes the RNA polymerase weaving its way in, edging its way in, uncoiling the DNA, and then reading, reading, and saying, oh, you're a start signal. So there's the initiation. Then there is the elongation as it's running. It's actually creating this long RNA tail that's just coming off it. That's the elongation. And then, eventually, it gets to a stop sequence. It will recognize a stop sequence and say, OK, I've read a particular length of gene or genes. I'm done. I'm ready to move on.

4.58 Just as a point of interest, the start sequence is a very specific sequence, just like the end sequence, the terminal sequence, is. So the RNA polymerase knows what to look for, as it's starting to run down the 3' end moving, moving. I'm starting now. Reading, reading, reading, or transcribing, transcribing, transcribing, to be more precise, creating a sequence and saying, oh, you're the end of the gene. I'm done. That's one individual message ready to send out. And then if it wants to, it can keep going and identifying other sections to encode for new genes. But that is the simple part of transcription. Thankfully, simple now because we took the time to dive in to replication with DNA replication.

4.59 However, when we have produced the mRNA strand, it exists in an immature form. We need to mature this. One problem is that it's too big for the pore. You'll recall way back when we talked about the nucleus, when we discussed the organelles in the first cell biology lecture, we talked about how, yes, the nucleus contains the genetic code in the form of DNA and now mRNA, but we need to get that out. And there are microscopic, of course, teeny, but there are pores that are lining this nucleus. And the mRNA in this native form is too big. Thankfully, it is also immature.

4.60 There are, along its length, sections that need to be snipped out. And then having snipped it out, we now are left with a mature RNA that not only just contains the essential genetic information, but is also conveniently small enough to slip out of the pore. Exiting now for the first time in the life of this cell, perhaps, exiting the nucleus, getting into the cytosol, where it can then undergo its next and final step, namely translation.

- 4.61 However, before we mention that, along the immature length of mRNA, there are distinct sections known as introns and exons. The exons are the actual areas of coded genetic information, whereas the introns, all the in-between, is non-coding genetic information. In other words, it doesn't account for a gene that can produce a protein. It's just wasted space. In fact, it's a matter of speculation to wonder why it even has that at all. Why have all these long sections of introns that do not code for any genetic information?
- 4.62 Well, we don't need them at this point, whatever role they may play. And so we have to get rid of them. And we get rid of them through this process of splicing. But importantly, the end result is an mRNA strand that ends up being far less than half. It can be a third or a quarter of the length of the original size, of the original length. As we start to snip out the pieces, then piece it back together, by splicing, snipping, and rejoining, snipping, rejoining, we end up with a molecule that is substantially smaller than the original. And again, thankfully now small enough to conveniently slip out of the nuclear pores.
- 4.63 The process of splicing involves something called the spliceosome. And whenever you hear that -ome ending in biology, it is referring to a complex of something. The proteome is the study of all the proteins; the lipidome, the study of all the lipids; the spliceosome, the spliceosome, is all of the aspects that are involved in splicing or snipping apart an mRNA length and joining them together. So as the spliceosome, all these various little proteins, they will recognize a sequence and say, you're an intron. I'm going to snip you here at the 3' end. I'm going to snip you here at the 5' end, and I'm going to join you back together in my wake.
- 4.64 Let's keep going. You're another intron. I'm going to cut you on this end. I'm going to cut you on that end, and then merge the remaining chain back together. So at the end of it all, after the spliceosome has done its job, we've removed all of the non-coding, or we could say, wasted space. All the non-coding sequence is gone. No more wasted space. The only thing we're left with is the essential coded information that will then go on to make the protein that the original DNA strand was coded for, with mRNA simply being the intermediate step, the step of transcription, and then eventually translation.
- 4.65 So at the end of the splicing and the actions of the spliceosome, we are left with the mature mRNA. Still mRNA, that's still the only RNA we've been talking about so far, even though I've introduced the other two. Now that we've gotten to the mature mRNA that is now small enough and mature enough to leave home, it's big enough to leave the nucleus and go into the cytosol, now we can take the final step. So translation is the final aspect of the central dogma. This is the process of taking the RNA and its genetic copy of the DNA and turning it into the protein that ends up defining the function and the appearance of the human body.
- 4.66 At this point, we now need all of the parts of RNA. We need mRNA, we need the transfer RNA, tRNA, and we need the ribosomal RNA, or I'm going to refer to as the

ribosomes. And all of this is going to occur in the cytosol or the cytoplasm, we'll say, because it's involving different components, so outside of the nucleus. Now, let's just take one little step back and look at a big picture for just a moment, which is the genetic code that is coming out itself. When we are copying the DNA onto the RNA, now we're reading the RNA. That's about what we're going to do. We're going to translate the RNA strand, the mRNA strand, into this sequence or a polymer of amino acids.

4.67 Interestingly, there is this magic rule of three, where every three amino acids on the RNA is what's called a codon. Every single codon will encode for a specific amino acid. So as we start to go through this process of translation, which we'll talk about, just keep in mind, please, that it's always happening in little segments of three. Every three nucleotides on the mRNA will constitute one individual amino acid. Then the next three will constitute a new amino acid. And so, as we have these little codons, sections of three being read, one after the other, it ends up giving rise to one individual amino acid after the other being coupled together. And we end up with the polymer of amino acids, which becomes, of course, a polypeptide and then a protein as it gets folded.

4.68 Now, interestingly, while all of the codons encode for a specific amino acid, individual amino acids can actually be created through distinct codons. So in other words, you could have an amino acid like leucine, the one I mentioned earlier. Leucine is the main branched-chain amino acid. It is the main amino acid of building muscle. That is the most important part of every protein we eat, if the focus or the purpose of that protein is to build muscle. Well, there are six different codons that can be coded for leucine, that will give rise to that amino acid signal. So there are six different versions of this codon.

4.69 This one, it's not the same as this one, but coincidentally, they both actually are leucine-based. So not each amino acid has only one individual codon. In fact, that doesn't happen at all. Every codon or every amino acid will have multiple different codons that can code for it. All right, so we have the codon, which are these little triplets of amino acids, and now we're ready for translation. Before we do that, I need to describe to you the ribosomes or the rRNA. The ribosomes are these molecular machines that are complex molecular machines that can measure or read and follow along the RNA.

4.70 There are two different parts to the ribosome. There's a small fragment and a large fragment that end up coupling together in order to read along the RNA and create and actually find a place. They end up giving a structure for the transfer RNA, the tRNA, to come and bind. And then so the ribosome is reading the RNA and providing a scaffolding for the transfer RNA, then the transfer RNA is what actually comes in and transfers the amino acid.

4.71 So let's just talk about that process very briefly to wrap up the central dogma of biology. So we now have the mRNA strand that's in the cytosol. It will bump into, first

of all, the small fragment. So the small fragment of the ribosome will come along and recognize a start codon. There's going to be a specific—just like on DNA, there was a start sequence— there is a specific codon, a triplet, a specific codon that the small RNA, the small ribosome, will sense and say, this is the starting point. Now, the large fragment ribosome will come and bind, and then the transfer RNA will come.

4.72 Now, every time I'm acting out the transfer RNA, I'm showing this unit of three because it has a specific sequence of its own, its own sequence that will recognize only its exact opposite. So now we have what's called an anticodon that the transfer RNA has. So within the cytosol are all of these different types of transfer RNAs, one corresponding, well, each with different amino acids at the top of it. So we have the two ribosomal units that have come in since the start codon. And then we will have a transfer RNA that represents the exact opposite, but anti-parallel signal, called an anticodon. And on its top, we'll have an amino acid.

4.73 And so it will come on and recognize it and say, I was here, I recognized it. Now I'm off. And then transfer RNA leaves, but the large ribosomal unit holds onto that amino acid. Then as it continues to move, it gets to the next codon, and it says, oh, hey, you, you transfer RNA. It's turn. It comes and binds, the anticodon, recognizing the sequence, leaving its amino acid on the large ribosomal unit, and then leaving. The transfer RNA leaves, but it leaves behind its original or its bound amino acid.

4.74 And this continues where the ribosomal subunits are moving along the length of the mRNA, and every time it gets to a new triplet or codon, it will grab the relevant transfer RNA, which is going to be anticodon or the matching opposite sequence, and then it will come and sense it, leave its amino acid, and then move on. And so, as the ribosome is moving along the length of the mRNA, the transfer RNA is coming and binding when relevant, then getting kicked off.

4.75 What's left is this growing little sequence. It's kind of dangling off the top, as we have the large ribosomal unit getting all its transfer RNAs and only leaving behind the amino acid. The amino acid chain starts to get longer and longer and longer and longer until we get to a stop codon, and that's the termination sequence. Then the ribosomal units say, oh, we're done. Everyone take a break. They break apart, and now we're left with this long amino acid chain, which is, again, this peptide or this polymer of amino acids that now can become something practical.

4.76 Now, one thing to note here is that, at the front end of it, near the very beginning of this sequence, it recognized the start codon, started with that, and then it created something called the leader sequence. And the leader sequence will then determine where this amino acid chain goes. Is it going to go back into the nucleus and do something? Maybe it's a protein that has to get changed and go into the nucleus. Maybe it's going to go to the mitochondria. Maybe it's going to go to some other organelle, or maybe it's even going to go out.

- 4.77 But regardless of where it goes, it has to undergo what's called post-translational modification. And this really is the final point to make in the central dogma, that once we have that long string of amino acids, you'll recall that that is not a functional thing yet. It needs to be wrapped into its secondary structures and then wrapped again into at least its tertiary structures, if not then being coupled with more to create a quaternary structure. So that's going to happen through the process of post-translational modification.
- 4.78 At this point, there's no more genetics to it. There's no more transcription or translation or replication. That is all done. Now it just has other proteins that come and start to fold it, and linking it in, creating the functional protein. And then again, where that protein goes is going to be determined by the leader sequence. And then if it's going to go out, it'll move through some of the other organelles we talked about earlier, like the Golgi apparatus. If it's going to excrete that protein, it gets packaged up into a little vesicle like we talked about earlier. Then we have exocytosis, and the protein is being released from the cell to then go and do something else in the body somewhere far distant.
- 4.79 So in the time that we have had together discussing the central dogma, before we could talk about central dogma, we needed to discuss the four major biomolecules—carbohydrates, lipids, proteins, and nucleic acids. The most important of those nucleic acids, arguably, for the sake of this lecture, being DNA and RNA. We also described some of the function of those biomolecules, including the relevance of enzymes, enabling bonds to be broken or formed.
- 4.80 And then lastly, we actually went through the formal steps of the central dogma, being DNA replication, allowing the body to survive through the process of cell division, including transcription, taking the protected nucleus-bound genetic information of the DNA and transcribing it into a mobile message, a genetic message, through the messenger RNA. That getting matured through splicing, moving from the nucleus through the small nuclear pores through which it can fit now, into the cytoplasm, where it then undergoes the final step, namely translation, actually converting the genetic information or translating it into a functioning string of amino acids, which finally undergo post-translation modification, to fold up and become a functional protein to live its best life and actually do what the original genetic information intended to have happen. Thank you.
- 4.81 Thanks for all the information. How we can maintain the cell healthy or in good function?
- 4.82 How to maintain the cell in good function? That's a great question with a lot of different answers. I will pick one part of that to answer, and that would be to make sure that you are consuming the right stuff. So for example, we have a hard time, our body has a hard time metabolizing an abundance of polyunsaturated fatty acids. PUFA, P-U-F-A, polyunsaturated fatty acids. That's a problem because polyunsaturated fatty

acids have now become the single most commonly consumed fat in the human diet globally.

- 4.83** So humans eat, now, we eat more of our fat from these refined seed oils than any other fat source, which is totally new. A hundred years ago, 120 years ago, we didn't eat any. There was literally no presence of refined seed oils in our diet. It was all saturated and monounsaturated fats, largely from animal sources. However, with the fear of those fats, came the birth in the 1950s or so of these refined seed oils, which are very heavy in polyunsaturated fats, to a point that we would say we're poorly suited to handle that much of these fats.
- 4.84** And when we eat such an abundance, these fats become part of our membranes. And so if we have cell membranes, like we talked about earlier, that are heavily enriched with these polyunsaturated fatty acids, they can readily undergo a process called peroxidation. You've heard of oxidative stress. Those types of fats can undergo oxidative stress very, very readily. And in so doing, they become very damaging molecules that can start to rip the cell membrane apart. But then if they get into the nucleus, they can actually mutate DNA as well.
- 4.85** So, among the many ways I could answer that question, one of them, I would say, is make sure you are maintaining a good integrity of your cell membranes by eating real fats, not these very industrialized fats from refined seed oils. How do I define a real fat? A real fat would be a fat that we, as humans, have been eating since time immemorial. That absolutely includes all animal fats we've been eating since the very beginning, and fruit fats. Fruit fats are when our ancestors would have only needed to scoop out the flesh of a fruit and press it or step on it, somehow squish it, and we would get oil.
- 4.86** You cannot do that with seeds. You do that with soybean, you do that with corn, you do that with any of the other seed oil. Canola, you just get a pulpy mash. That doesn't work. But if you take a coconut or olive or avocado and squish it, you will have a pure oil dripping out of it. We've been eating those since, again, the beginning of time, as far as we know, and so we're very well adapted to them. So saturated fats and monounsaturated fats in abundance will help maintain a very healthy cell membrane.
- 4.87** Thank you so much. Between the exons and the introns, I was interested in knowing—it was hard to believe that the introns really have no purpose, and maybe we just haven't figured it out yet. But is it possible in the process of evolution that they're there ready to be activated with a new gene in case we need it or something in terms to that? Also, because I was listening to your response to Naomi's question about having, polyunsaturated fats can even change the genes?
- 4.88** Yes. So the first question, well, let's start with the second part. So polyunsaturated fats are so readily—again, the process is called peroxidized, that they become highly radical. So anytime, you've heard the term oxidative stress before. Oxidative stress starts with what's called an oxygen radical. Radical because it's unstable with this— it's too

electrically negative, and it's not balanced, it's not neutral. That has the potential to bump into and pass that radical oxidative capacity onto the lipid.

4.89 Now, the problem with this lipid is that lipids move very readily. As I said before, all the membranes are made up of fat, and like attracts like; fat is OK with more fat. And so this means that this highly reactive lipid peroxide now, which is almost always built on a polyunsaturated fatty acid, it can move very freely into a cell membrane, into the mitochondria, and damage the parts of the electron transport system, giving rise to new reactive oxygen species, new oxygen radicals. But also it can slip through the nuclear envelope, which is again a fatty envelope, a lipid membrane, and slip in and start bumping in and passing this oxidative stress onto genetic information, onto DNA, just making DNA breaks.

4.90 So that's one of the reasons why I'm such an advocate of avoiding refined seed oils and why I'm an advocate of avoiding these so-called lab-grown or these fake meats. Because the fat that they put in them is these fats from refined seed oils because they're the cheapest fat. And they're naive enough, or corrupt enough, that they just say, oh, fat is fat. No, it's not. There's natural fats, ancestral, that we've been eating since time immemorial. Then there are new fandangled manmade fats because they're more cost effective or whatever.

4.91 The first part of your question was the introns then. I very much agree with what you just said, actually, and that's why I tried to frame it the way I did. We don't yet know what their purpose is. I do believe things exist for a reason. I see the world in a very rational way. All of creation I look at through the lens of an intelligent design. Unapologetically so. I just cannot believe how anyone could see it differently, but I respect anyone who does. But to me, there's an intelligent design behind the functioning of biology.

4.92 Now, that's not to say there isn't chaos and disorder because things happen. It's the randomness of biology. But at that level of the cell, I certainly think if something's there, it's probably there by design. So I can only speculate. And one speculation that comes to mind is that maybe there is something to be said for having some space to just buffer. Maybe if a reactive polyunsaturated fat comes in—because so much of the length is a non-coding, it's just more statistically likely it will bump into a non-coding region, and maintain, leave the coding regions intact. And then that little separation can be just spliced back together and have no influence on the genetics of it whatsoever.

4.93 I'm purely speculating, but that's a fun part of science. When I'm teaching my science classes, I'm always quick to tell my students, as a scientist, we can only answer how something happens or what is happening. We can never answer why. That is strictly within the realm of the gods. That is a divine question that can be asked. Why is the system working the way it is? We can only speculate, and that's the realm of philosophy and religion, not the realm of the basic science. As much as I enjoy dabbling

in that, it's fun. We can never really know. It really is just trying to wonder at the intelligence behind the system. As a scientist, I have to just focus on what I can focus on and find answers to, which is the how and the what.

- 4.94 Parting from her question, her question, can you tell us a little bit about epigenetics? How what we do, our environment, what we eat, how that affects our genes?
- 4.95 Yeah, wonderful question. In fact, I very much debated on whether to include that material in this lecture. So when I talked about post-translational modification at the very, very end, that would have been the moment when I insert epigenetics. So epigenetics is when you take a normal code, or whether it's post-transcriptional or post-translational, actually, at any one of those points, you can introduce some bit of messiness, like methylation or carboxylate, carbonylation. You can have something that will get in the way of reading or translating the code and resulting in, even though the original raw genetics are all there, you've had some modification happen. Again, methylation is one of the most common, but it also is affected by things like metabolic status.
- 4.96 One of the most interesting studies ever done was studies done on people in Holland. It's the so-called Dutch famine study. So I think in the wake of World War II, there was a period of time where there was severe rampant starvation. And you would have, within one generation, a substantial departure from the metabolic health of the initial generation. So let's say, Mom was surviving, maybe pregnant during this Dutch famine, this period of, I think, months, maybe even over a year. So it would have been getting pregnant sometime during that.
- 4.97 She passed on her genetics to her child from her healthy husband, who also went through this famine. Their genetics are perfectly expressed in that child. And yet at a population level, once the famine ended, these offspring were much more likely to become overweight and much more likely to become diabetic than the parents were. Within one generation, that's, of course, not a mutation or an adaptation or evolution in genetics. It is entirely accounted for by epigenetic changes based on the metabolic, in this case, the metabolic environment. Basically, telling that little fetus, hey, nutrition is scarce, so be very thrifty. Anything you get, try to store it. Try to store it. Don't be wasteful with it, because we never know when that's going to happen again.
- 4.98 But similarly, you have other more dramatic instances, in fact, more common nowadays with things like radiation exposure. Radiation exposure can cause these other epigenetic changes, resulting in a differential expression. The gene is there, but when you go from genetic gene in the DNA to full-on protein, something has broken it, changing the expression, giving maybe too much of that protein or too little of that protein.
- 4.99 Good questions, guys. Thanks so much. That was awesome. Very, very good questions.

Lecture 5: The Cell Cycle

At a glance — The cell cycle.

- 5.1** At this point in our journey through the cell, in these lessons of the basics of biology, we have gone through the cell itself to understand what it's made of and what it does. And then we discussed the central dogma, which is the flow of genetic information that ends up determining how our genes become our bodies. Today, we're talking about the life of the cell cycle. This is the cycle that a cell will go through when it is time to split or divide. But there's some nuance here, and there's different types of division that we're going to explore.
- 5.2** And then we're also going to discuss how sometimes things go a little wrong in these divisions, where it's a miracle that it ever goes right and that it goes right as often as it does. All right, so let's get started in discussing the cell cycle and then transitioning into this process called mitosis, which is the end result, usually the most common end result, of the cell cycle in most of the cells of our bodies. So at the end of this lesson, I want you to be familiar with what the cell cycle is, as well as the steps and the purpose, the steps involved and the purpose of mitosis.
- 5.3** Now, before we get too far, we need to take a moment and make sure we have a common understanding of a very important word, and that word is chromosome. Throughout this lecture, I'm going to be referring to the word chromosome. I may say chromosomes, but within, and we'll define some of the numbering and the accounting of this, throughout some of these processes we're going to discuss, I may be referring to a chromosome that may actually have a matched or sister pair. Even though that is technically plural, and this will start to make sense soon, I'm going to still refer to it, that duality or that pair, as a single chromosome.
- 5.4** And that's going to be very relevant when we start talking about the cell division through the process of meiosis, which will come later. And that's the part of the cell division or the cell cycle that's involved in reproduction, of course, an essential part of life of every organism. So the chromosome is going to be a singular thing, even though sometimes it may actually have a coupled pair or a twin.
- 5.5** Now, within the human body, you've heard this before, we have 23 matched or homologous chromosomes. So in other words, there are 23 general families of chromosomes, each of which has two chromosomes. So amongst this 23 of matched chromosomes, we actually have 46. So human cells have 46 chromosomes. We are what's referred to as diploid organisms. So for every general chromosome, we actually have two of them, one from each parent. We inherit one chromosome from father, one chromosome from mother. And so every one of our cells in our body has 46 chromosomes.

- 5.6 Now, each of these chromosomes that we've gotten, one from mother, one from father, has a matching— they're part of a matching set, but we're not calling them one single chromosome. Even though they are sort of a matched couple, they are distinct. So we still have 46 chromosomes, but in general, they're in 23 different themes. And you can imagine, you might have seen this before, there's something known as a karyotype. A karyotype is something that a clinical team can do in a laboratory. Take a cell and then break open the nucleus to get to all of the chromosome information, all the chromosomes.
- 5.7 And when the chromosomes are nice and tightly packed, they actually form these little lines. And then that clinical team can start to match them up. They're not actually connected. They're not even actually in the same space necessarily. They're all in the nucleus. But the nucleus can be this chaotic thread of DNA. And then it's only when the DNA winds up tight do we look at it and say, oh, that's a chromosome now. It's linked up nice and coiled all tight. And thus we have the chromosome that we can literally image and view under a microscope.
- 5.8 But they're not all balanced beside each other. But in a karyotype, when it's presented in conventional biology teaching, it almost makes you think as if they are just stacked right next to each other. They're not. There could be chromosome one over here, chromosome one over here, but the clinical team will cut the image and move them together, when they want to look and just, in order to determine whether there is some genetic or chromosomal anomaly or some problem. And we'll talk about some of those problems later.
- 5.9 So within the human body, we have cells that are diploid cells. That is the majority of all of the cells of our body. In fact, using the word body is appropriate because we call these cells somatic cells. Soma, the word for body. So somatic cells are the majority, the vast majority, of all the cells in our bodies. They are somatic. They're the body's cells. And we have that double chromosome arrangement. So 46 total chromosomes. Now, however, within the human body, we have some cells within the gonads. Within men, that, of course, means the testes, within the testes. Within women, that means that there are cells within the ovaries that are capable of producing what's called haploid cells or germ cells or gametes. Each has a specific name depending on the sex. Men will produce sperm, women will produce eggs.
- 5.10 Now, there are other words, and I'll use some of them later. But now the difference in those cells is that they have half the number of chromosomes. They're not diploid, they're haploid. They have half the number that we had originally. So we have 46 chromosomes in the somatic cells of all of the body. And now in these particular germ cells or cells of reproduction, now we have, or these seeding cells, we have half of that. So now we have only 23 chromosomes. They no longer have a matching pair or a matched set. It is only one. And we're going to talk about how that happens.

- 5.11** When we think about growth with all of this genetic information and cell division in mind, when we think about an animal or the human body growing, that's a combination of two things. A modest part of that growth happens because each cell may be capable of some growth itself. Now, that's very modest. Some cells can't grow at all. Some cells can grow substantially. But overall, when an organism is aging and growing, it's a result of the cells dividing. That's a process called hyperplasia or just multiplication of cells.
- 5.12** So for something to grow, the cells must divide. That includes skin cells, muscle cells, bone cells, you name it. Now, some cells won't divide once they're big and mature, like a neuron. That's not going to start dividing, and it's going to be very difficult to replace it. One cell type, unfortunately, that is capable of not only substantial growth, but also substantial hyperplasia or multiplication, are the fat cells, unfortunately. Let me just talk about that for just a second because it provides such a good example.
- 5.13** As a cell biologist, scientist, the fat cells are one of the main cells that I study in my lab. Throughout the life of a person who's gaining weight or getting fatter and fatter, that fat growth will be a result of both of those processes, both the hypertrophy of each individual fat cell, as well as the potential for the fat cells to multiply. Now, during the life of an individual, both of those are happening in the early years of an individual's life. Indeed, from birth all the way through adolescence or puberty, both of those are happening. Fat cells are growing a little bit, but they're multiplying as well.
- 5.14** And then once the person starts coming out of puberty, generally, the number of fat cells they have is set for the rest of their life. So the fat cell number isn't generally going to change anymore at that point. Thus, any future fat gain, post-puberty or post-adolescence, is going to be a result purely of the expansion of the fat cell itself. Unfortunately, while that is the most common way of getting fat, it's also the most problematic. But I won't get into that now. That was the topic of my "why we get sick" series that I've already done. So I refer you all to that.
- 5.15** Nevertheless, here we have an example of a tissue with its cells that are capable of both dividing and growing in size. But it's the division that we care more about. So let's come back to the cell cycle itself. The cell cycle is a programmed series of events that is necessary for the overall proliferation or multiplication. There's a very specific activation to this. There's a very specific timing to this. But the genetic information, the DNA, wound tight into chromosomes, must be passed as the cell is getting ready to—it's growing, growing, and it's ready to start splitting. As it splits, we must have that genetic information get copied and moved into these two cells. That is the essential part of the cell cycle.
- 5.16** And then when we actually have the division, that can take the form of either mitosis, which we'll talk about now, or meiosis, which we'll talk about just after. So with that in mind, let's actually move right into mitosis now. There is a helpful acronym that every student of biology has learned in order to think through the cell cycle, and that is

IPMAT. IPMAT. That stands for interphase, prophase, metaphase, anaphase, and lastly, telophase. That is the IPMAT acronym. Again, every student of biology memorizes at some point in order to do well on the exam, to think through the cell cycle.

- 5.17** Let's start with the interphase. There's actually a part of the interphase that you're already familiar with. Interphase is the part where we deal with DNA replication. And we've talked about DNA replication now in very thorough detail as I was acting out this odd uncoiling of the DNA. And then you'll recall the DNA getting copied and replicated from the 3' end to the 5' end, on the leading and the lagging strands. All of that is what's going on in the cell cycle first phase of the interphase. Now, the interphase itself, once it has occurred, so this is important to keep in mind—once we get through with the interphase, now the cell will choose which direction to go based on its inherent capability.
- 5.18** Does it go through mitosis, splitting and making an exact duplicate of itself, or does it go down the path of meiosis? Is it now splitting and having its chromosomal information, creating the gamete or the germ cell, or the sperm or the egg? Now, remember, there are only a few cells in the body strictly within the gonads, testes and ovaries, that are capable of meiosis at all. Most cells of the body, every other cell, indeed, cannot go down that path. It can only go through the path of mitosis. But nevertheless, when I get done explaining interphase and its three distinct phases, just keep in mind that that will be common as the preceding event for both mitosis or meiosis.
- 5.19** Now, what are those three phases? I mentioned that there are three phases to the interphase. And remember, that's the first. That's the I, the first letter in the IPMAT acronym. The phases are G1, or a Gap 1 phase; S phase, the synthesis phase; and then the G2, or Gap 2 phase. And then after that, interphase has ended, and then we move into the prophase of either mitosis or meiosis. And we'll get into both of those in just a moment.
- 5.20** So in the G1 phase, the Gap 1, it could also—that G does stand for gap, which is meant to imply that there is a gap in the, there's a pause in the timing of it all, through this. So this is a state where the cell wants to assess the overall health and fitness of the milieu or the body in that moment and decide, am I OK to commit to dividing? Is this a good time for me to do this? So there's this temporary gap in the process just to check. But the G could also stand for growth because there is a lot of growth. The cell is undergoing hypertrophy as it begins duplicating and copying all of its organelles.
- 5.21** So the number of mitochondria is going up. The endoplasmic reticulum, the peroxisome, the lysosome, all of those things we spoke about in the cell biology first lecture, all of those things are replicating in order to get ready to pull apart. But at this phase, still, the cell has 46 chromosomes, one from mom, one from dad, that has been, of course, copied over the generations of the cells in that person's body. So it still has

46 chromosomes, and it's just assessing. Am I ready to do, is everything OK? Can I actually start growing and copying myself? So that's the G1 phase.

5.22 If that goes well, now the cell has moved into the synthesis phase or the synthesis or the replication of DNA. This is when DNA replication occurs, as we described it. But we would say that it's synthesizing copies of the chromosomes. Now, during the synthesis phase, the chromosomes are all duplicated. So we had 46 chromosomes, and now each one of them has been copied, and that's the synthesis phase. Now we move into the G2 phase.

5.23 So it's another opportunity, and especially at this point, the cell really does a thorough examination of itself. Am I healthy? Is everything going all right? Am I really prepared to divide? Assuming that's a yes, that everything has gone well, that the replication of the DNA has gone without any mutations or problems, then the cell commits to the next phase, the prophase, and interphase has ended. And again, now it either moves into mitosis or meiosis.

5.24 Now, there's some important structures. Let's pause for just a moment and then talk about some of what's present within the cell at this moment, having gone through G1, S, and then G2, all in the process of growing the cell and copying everything, even down to the DNA, the chromosomes themselves within the nucleus. Now, a couple important things. First of all is the centromere. Normally, within the nucleus, the chromosome is just a long stringy thread of chaos. All of them are. Now, they're still distinct. It's these long lines of DNA, and they are really long, but it's a cloudy thread of just chaos within the nucleus.

5.25 Now, during this phase, what's starting to happen is the DNA is starting to wrap nice and tight and forming a distinct cylindrical chromosome. Along this chromosome, we have something right in the middle called the centromere. All chromosomes, when they wrap up tight during the cell cycle, will have a centromere, a central section, and bound to that centromere or the center of the chromosome is something called a kinetochore.

5.26 Kinetochore, you hear in that word, the word for movement, kinetics. There's a kinetochore, and that becomes important, because on that kinetochore, which is right at the centromere or the center of the chromosome, there are areas where these things called spindle fibers to come and hold, to grab. That's going to be the site of movement or the kinetic pulling away from each other that's going to happen in a moment.

5.27 But those spindle fibers are coming from these other organelles that we talked about in the first lecture called centrosomes. The centrosome are these organelles that dictate the movement of these little, these spindle fibers, they're called. But we could just refer to them as little threads that are reaching out and extending through the cell. So the centrosome is anchoring and dictating the movement of the threads. And these threads

are going to come and bind to the kinetochore, which is bound to the chromosome at its center. And then it's going to, as we go through in a moment, it's going to arrange them, organize them, and then pull them apart.

5.28 But we'll get into that in a moment. But it's necessary to just keep those three terms in mind. The centromere, which is the center of the chromosome, the kinetochore, which is this little protein that's binding to that center, and then the centrosome, which are the organelles that has been duplicated now, that is dictating the movement of these little threads or these spindle fibers that are going to come and bind to the kinetochore and start pulling it apart.

5.29 All right, so let's move into those next phases. With the IPMAT acronym in mind, we've gone through interphase. The most important part of that is the synthesis or the replication of the DNA, duplicating it, which we have in this moment, temporally, through time, that's where we're at right now. Then we move into the prophase, metaphase, anaphase, and telophase. And thankfully, none of these are overly complicated. In the prophase, the chromatin, which is the word for the loose, thready DNA, condenses into the chromosome like I was describing, where if you were looking at it under a high-powered microscope, you would now see these distinct little clumps of things. They're not all stacked up nice and neat, unfortunately. It would just be a chaos of these teeny little rod-like structures.

5.30 Then each chromosome now, because of its duplication earlier, is coupled with its exact sister. Now it has twinned itself. That was part of the DNA replication of the S phase earlier. So of all these 46 chromosomes, each is now bound to its exact copy. So we have 92 chromatids, or individual little chromosome stacks. But we still say that there's only 46 chromosomes, to be technical. But each chromosome is literally bound, pressed together, with these kinetochores at their centromeres. Each is bound to its exact copy that it made earlier through the S phase.

5.31 Then the mitotic spindles, these little fibers, start to extend from those centrosomes as they start to move through the cell. So the centrosomes have copied themselves. That happened during the G1 phase earlier, like all the organelles have been copying themselves. And these little directed structures that are going to direct the movement, they're starting to extend their little spindle fibers, their little threads, while they're moving to polar opposite ends. That's all the prophase.

5.32 Now we move into the metaphase, and that is all about the centrosomes. These little organelles that are spreading out these little fibers, the spindle fibers, that are now lining, literally lining, all of these newly paired or coupled or twinned chromosomes along this equatorial line, they literally get pushed. The spindle fibers are directing all of them to just get lined up exactly down the equator of the cell, right down the middle. That is the metaphase. So they're all lined up with the spindle fibers from the centrosomes binding to the center kinetochore, or the center little proteins that are locked right on the middle of each of the twinned chromosomes.

- 5.33** So in my little fingers here, these matched chromosomes that are literally pressed together, there's a kinetochore protein on this middle section, the center section, and there's one on this center section. And the spindle fibers are coming and locking onto all of them, lining them all up. And you can imagine what's about to happen next.
- 5.34** Next is the anaphase. Ana, from the Greek word ana of back. So the centrosomes are backed up to the far polar ends of the cells, the opposite ends of the cells. And now they literally pull those twins apart. So we had 46 chromosomes, each duplicated with its twin. But we still say that, as this one chromosome was duplicated during the S phase of interphase, we still call it one chromosome. But there are two identical copies now. Now, they get pulled apart during anaphase. They get pulled back to the opposite ends of the cell. And so the chromatids have now separated from each other, getting pulled to the opposite ends.
- 5.35** And now we go to the last phase, which is the telophase. During this, earlier on, although I failed to mention it, during the prophase, the nuclear envelope dissolved. So that allowed the centrosomes to actually bind on and start moving the chromosomes and lining them up. So very early on in the prophase, the nucleus dissolved, allowing them to get pulled apart. Now that nucleus starts to reform around these exact copies, even though the cell is not technically separated yet; it's not pinched itself apart. It's pulled itself apart so that you could imagine there's these two round cells, and they're connected by a small little space. They're still connected, but the nuclear envelope, or the nucleus, starts to reform around the chromosomes as they have been pulled into these opposite directions.
- 5.36** And then the final step is a step called—it's distinct from the telophase. It's called cytokinesis, which is the cell movement. Now, earlier, the very first day, one of the organelles we talked about was microfilaments. And I described how it's involved in cell division, but it's also part of what's involved in muscle contraction or movement. Anytime a cell is moving, there are microfilaments that are directing that movement.
- 5.37** Now there's a small ring that begins to form as you think of the cell being pulled apart, like two bubbles pulling apart, and the connection between them starts to get smaller and smaller. This little ring of microfilaments forms and actually contracts. It flexes and pulls them and finishes off that pinching apart. Now we're left with two distinct cells. And that last step, again, was cytokinesis, all the result of this muscle-like contraction of a little ring.
- 5.38** Now, this is what's referred to as mitosis, and it is also referred to as an equational division. Equational because they have equated in creating two exact copies of each other. With the exact same number, equation referring to the number of chromosomes. So it had 46 chromosomes originally, and then during the interphase, it copied, still with 46 chromosomes, but now an exact copy of each one, then those exact copies were split apart now. One parent cell, we could say, gave birth to a

daughter cell, or another way of saying it is we have two daughter cells now that came from one parent cell.

- 5.39** 46 chromosomes in each. This is a fundamental process of tissue growth and the replication of cells as they're getting old. They survive, they persist, and the body persists similarly because of the cell's ability to divide. That is mitosis. So at this point, as we wrap up mitosis, you should be generally familiar with the cell cycle, keeping in mind the IPMAT acronym. And now you're familiar generally with mitosis. And of course, it all was based on the DNA replication that we spoke about before, making copies and then going through the cell cycle and pulling itself apart into two now-identical cells.
- 5.40** Let's move to the other cell process here, the other part of the cell cycle, which is meiosis. So throughout the next few minutes, I want you to be familiar with the steps of meiosis, and it will sound very familiar because it's just a slightly different version of the same process of mitosis. But with that in mind, I want you to appreciate the differences between the two. Now, again, as a reminder, meiosis is a unique type of cell division that only occurs in the germ cells. When I say germ, think about a seed, germinating and growing and reproducing. So this is why that term is used.
- 5.41** And of course, that's happening only in the gonads, the testes and the ovaries, of males and females, respectively. And even within those tissues, there's only few cells that are actually still capable of doing this. It is the spermatogonia, which give rise to, which undergo meiosis, to give rise to the spermatocytes or sperm in men, in the testes; or in the ovaries of women, these are cells that are called the oogonia, which give rise, through meiosis or dividing, creating the oocytes or the egg. So that is when this is relevant. There is no discussion of meiosis in any other cell type.
- 5.42** Now, earlier, I had mentioned these terms diploid or haploid. The somatic cells, the cells of the body, are always diploid. They have 46 chromosomes, one inherited from mother, one inherited from father, as you'll recall. Throughout this process, now, we give rise to the cells that have half of that number or the haploid cells. But keeping in mind this term that I used earlier called homologous chromosomes, where there is a matched pair, or I hate to use the word pair because I've used it earlier in a slightly different context. But there is the one from mother and the one from father that is still here.
- 5.43** And so, let's say it's chromosome number 15 of the 23 total chromosomes that are available. It's chromosome 15 from mother and 15 from father. So the cell still has that. And those are called homologous chromosomes, whether it's chromosome 1 or all the way through 23. And 23, of course, what we refer to as the 23rd chromosome becomes very important in this discussion now, because that's the chromosome that determines sex, whether it is XY, determining the male, or whether it is XX, determining usually female. And we'll explore that in a little more detail.

- 5.44** With these homologous chromosomes now, we have to split them up. And this is the part of meiosis. This is the point of meiosis. This is, in fact, an essential part of reproduction. We have to take these diploid cells, each with its matching homologous chromosome, two of each type, and then split them apart to give rise to the germ cells or the reproductive cells, the sperm and the eggs. With meiosis, unfortunately, it's slightly more complicated than mitosis, because with meiosis, there are two steps. There is this first step, which is called reductional. That's meiosis I, and I'll define all of this in a moment. And then there is the second phase or process, this mini cycle, which is meiosis II, which is again referred to as an equational division.
- 5.45** So the first one being reductional, we have 46 chromosomes within this parent cell, and now we're going to pull these homologous pairs apart. So the homologous pairs separate. And remember, we're starting this right at the end of the interphase. Everything we talked about earlier with mitosis, you have to treat it as an entirely separate thing. We have to then imagine that that new cell has similarly gone through that interphase again. So before we start meiosis, just as a reminder, you have to imagine that we just went through the interphase, which is the replication.
- 5.46** Each chromosome, now again, we've gone back in time, and we're going to take the other path. Each chromosome has its matching sister chromatid or another chromosome, but we're not saying that it's a separate chromosome. They are literally bound together. Earlier during mitosis, we spoke about that process of them just getting pulled apart, giving rise to two cells, each with 46 chromosomes. We have to go right back to that moment now as we begin discussing meiosis I. So there are 46 chromosomes, and we're about to start splitting all of them up.
- 5.47** Now, one essential event that is happening in meiosis I during this prophase, the first of the PMAT letters that are coming, in the IPMAT acronym, an essential event that is totally unique to meiosis is what's called a crossing over. So there's going to be this brief moment before they get pulled apart, where we have this unique event to ensure that the offspring is going to have unique genetics compared to the parent. If this didn't happen, then any parent set, a mother and a father, every single child would be identical to each other, every single time. The fact that there is variety between every child means that there has been some mixing.
- 5.48** And indeed, it is this event, called this crossing over, where the chromosomes are going to come together and swap some information before they end up splitting apart. So that is key. So let's talk about this. And this is within meiosis I, we're now going to use that term I, that numeric, to identify the prophase, metaphase, anaphase, telophase. So this is going to be called prophase I. We have this pairing of homologous chromosomes. So remember, each chromosome has been copied. Let's just go with chromosome 10, let's just say. But it could be happening with any of the chromosomes, of all of the 23 different types.

- 5.49** So chromosome 10, each one is still bound to its sister that it gave rise to, that was created during the interphase before. Now, there is something where these two here, of chromosome 10, which have been copied and are literally bound together, pressed together by the kinetochores at their centromere, and these two of chromosome 10, each being one chromosome, even though, yes, there's two things here, they will come together and literally sync up and touch each other in what's called a synthesis.
- 5.50** So they bind together—or a synapse. This is a synapse. A synapse is a touch point. They come together and bind, and now they literally swap sections of the chromosome, or in other words, swap sections of DNA. Now, it will only be a section that has to do with the same thing. Now, most of these sections won't swap at all. Like, for example, why do humans have 10 fingers, 5 on each hand? That's not swapping. There's no mixing there. But some do. The items, the genes that are involved in some of what gives us our character, our height, our body shape, our coloring, complexion in every way, that is some of what's starting to mix up.
- 5.51** And again, that is what gives some uniqueness and variability to the offspring of same couples, which is, of course, an essential part of reproduction. You want genetic variability. It helps the species. So in this day and age, when we're all so up in arms about differences across people, we should be thanking god for these differences, just for the sake of the species. We don't want everyone to be the same. We don't want everyone to be as susceptible to the same problems. This variability not only gives some spice to the planet, but it also ensures the survival of the species.
- 5.52** So again, that's an essential event in prophase I. There is this pairing up of the chromosomes, and then a swapping over, an exchange of information, before they come apart again. Now, also happening during prophase I, that I alluded to before, like with mitosis, same thing, the nuclear envelope starts to dissolve. And the little filaments that I was talking about earlier, the little threads, they start to spread out of the centrosomes. So the same thing is happening. And indeed, as we go through metaphase, anaphase, and telophase, and then even cytokinesis, it's all the same.
- 5.53** So we have the breaking down of the nuclear envelope, the centrosome starting to come to the opposite sides during the anaphase, and the metaphase first, then anaphase, and then the spindles, lining them up and then pulling them apart. Now, remember, this is a reductional step. This is a step that requires or that takes the 46 chromosomes that we had earlier and splits them up.
- 5.54** Now, how does it split them up? You have to remember that it still has kept them as pairs. So it lines them all up during the prophase or the metaphase. They're all lining up along that equator. But remember, there's two copies still. There are still 46 chromosomes, each coupled with its sister that it was made during the interphase, and then they get pulled apart. So what was a cell with 46 chromosomes is getting bigger, bigger. The centrosomes start binding them with their spindles and pull them apart. Now we went from one cell with 46 chromosomes to two cells, each with 23.

- 5.55** But we had some exchange of information. So now they're really unique. None of these things are like the original now. And so we have gone from one cell, through meiosis I, we have two cells, and we had a reductional or a halving, a division, of the chromosomes, from 46 to 23, but each still with its sister chromatid that was made during the original interphase. But with that, we are ready to move into meiosis II. And meiosis II is the exact replication of mitosis, essentially. There is no longer any crossing over. That crossover already happened, and it is done during meiosis I in that prophase.
- 5.56** During prophase II, metaphase II, anaphase II, and telophase II, it is simply now, again, the nucleus dissolving, and then the cells lining up again because of the centrosomes, pushing them into that equatorial middle line. And then during anaphase, pulling them apart. And during telophase, reforming the nuclear envelope, and then cytokinesis is the pinching-off of the two cells. So we went from one diploid cell, splitting them up into haploid cells, two. But each still with a twin. Now we went from these two cells that had 23 chromosomes, but with a matched twin, splitting up the matched twins into four new cells, each with 23 chromosomes, but not a matched twin anymore.
- 5.57** So I'm going to act that out one more time with my fingers as awkwardly as I'm doing it. So we had the original cell that duplicated its DNA to have a matched sister chromatid with every single chromosome, 46 chromosomes. During meiosis I, the chromosomes get split up, and after they have exchanged some information, making them all totally unique, blending the genetics from both parents. That's split into two cells during meiosis I.
- 5.58** Then meiosis II was these twins getting pulled apart into four cells, none of which has the same genetic information as the other, and none of which has the same genetic information, exact genetic information, as its predecessors. Now, during this process, well, first and foremost, that's the end of it. That is the end of the cell cycle. That's the end of our understanding of mitosis. And then finally, the two different phases of meiosis, meiosis I and meiosis II.
- 5.59** But in the time we have, I want to go through some instances when things go wrong. And so let's talk about this. During the process, usually it's at some phase during meiosis, the crossing over, that's a tricky, delicate time when we're snipping and snipping and then transferring. There can be these chromosomal alterations that occur. And there are four in particular that I want to highlight—duplications, inversions, translocations, and deletions.
- 5.60** And very briefly, with a duplication, the name is pretty apparent what happens. You can have a chromosome that, during the course of exchanging, accidentally swaps and copies that same gene section. Whatever that section was on the chromosome that was swapped across homologous chromosomes, it accidentally copied itself. So we have a duplication.

- 5.61** A common example of this is the mutation in a gene and in a protein called MYC. MYC. And when MYC has mutated, it's a very, very common cause of breast cancer. So some breast cancers are a result of a duplication of this gene, which is arising during meiosis. Similarly, we can have an instance where, rather than a section getting duplicated, as it's swapping across, it can get inverted. And so now the genetic code, the DNA, is going in the wrong direction as it binds over. They were supposed to be this way oriented, and as they move over, one of them gets inverted.
- 5.62** Inversion is a common cause. If it happens at chromosome 16, when they're pairing up, these homologous chromosomes and exchanging, that can give rise to a type of leukemia, acute myelogenous leukemia, or AML. Some instances of AML is a result of this inversion at chromosome 16.
- 5.63** Even further, the next one that I mentioned is translocation. A translocation is when we've actually swapped information, but it was the accident being that it wasn't at homologous chromosomes. So normally you'd have chromosome 14 come to chromosome 14, and they change their information. But one of the most common translocations happens when 14 and 21 accidentally start exchanging information, because the tail on one and the head on the other is similar in size. So that can give rise to this aberrant or incorrect translocation. And that is the most common, this exchange across chromosome 14 or chromosomes 14 and chromosome 21.
- 5.64** That's in fact so relatively common, within the uncommon realm of translocations, that it's called a Robertsonian translocation. And that usually will give rise to a different type of leukemia called chronic myelogenous leukemia, or CML, whereas the other one was an acute form. And then the last one that I'll mention is the deletions. That's just when, in the course of exchanging, one of them just doesn't quite make the connection. And so as it's moving information across, one makes it, but the other doesn't. And it just goes missing within the cell and gets just eaten up through some other process.
- 5.65** So, a somewhat common version of this—and again, all of these are uncommon, just as a testament to how amazing it is. So many things can go wrong. It really is a miracle that they go right as often as they do. This relatively uncommon problem happens relatively more commonly at chromosome 5. And if you have that deletion on chromosome 5, it gives rise to a birth defect condition called cri du chat. Cri du chat, named after the French term cry of the cat. The baby, when the baby is born, has a very unique cry that sounds like a moaning of a cat. That's where that name comes from.
- 5.66** But it's associated with generally a very low birth weight and a very underdeveloped brain, which, of course, gives rise to mental dysfunction, but also a condition called microcephaly or very small head or small skull. So that's the cri du chat problem, or mutation because of a deletion of chromosome 5. Now, we're not quite done, even still, talking about chromosomal aberrations. What I just got done mentioning was, when there is a problem in that exchange step, giving rise to the duplication, the inversion, the translocation, or the deletion, but at the same time discussing meiosis,

it's a pretty good opportunity to talk about a potential genetic problem called aneuploidy.

- 5.67** Aneuploidy is a term for when a cell does not have the correct number of chromosomes. Of course, a normal cell should have 46 chromosomes, but you can have it, some instances, when one of those chromosomes doesn't have its matching set—one of them is gone—or in some instances, there's an extra one. And so within the realm of aneuploidy, you can have two general problems. One, and relatively more common, is trisomy. When, if you were looking at all of the chromosomes, you end up seeing that rather than having a matched set, one from mother, one from father, somehow mother or father gave two. And so at one of the chromosomes, there is three. Again, that is called a trisomy. There's three where there should only be two.
- 5.68** In contrast, there can be an instance, although much less common because it's usually embryonic lethal. In other words, the fetus simply will not be viable, and it would be a miscarriage. It can happen where rather than supposed to have two, you end up only having one. That's called monosomy. So monosomy is uncommon because usually the baby wouldn't even be born. Trisomy, still uncommon, but it can happen. And that's when you have too many. And that happens. What gives rise to the trisomy is when, during the course of, as we were acting out meiosis, one of these divisions doesn't occur. So it's called a nondisjunction event, or to say that another way, you have a failure of the chromosomes to separate appropriately.
- 5.69** And that can happen either during meiosis I or meiosis II. Let's just act through it if it's at meiosis I. So if it's at meiosis I, when normally the cell splits, pulling both chromosomes apart, each with its matched twin, it ends up going into the same one. So now this daughter cell has both sets. This daughter cell over here is empty of that one particular chromosome. And then as this one divides now, now it has split into two final germ cells, a sperm or an egg. But there is this duplicate copy of the chromosome. And then when it matches with the gamete or the germ cell of the partner, mother or father, it now takes on that one chromosome there. And so it was two from one parent, gets matched with one from the other parent who had proper meiosis in this instance. And now that one particular chromosome has three copies.
- 5.70** But now let's follow the journey of this other cell over here that didn't get any. It is still a cell that has all the other chromosomes, just not this one particular chromosome of interest, whatever one it may be. It then still merges with the cell from the other parent and is now only left with one copy. So it would just depend on which cell this it happened in, and again, in which phase. So that was if it happened at meiosis phase I. But then you can see that if it had meiosis I occur fine, you could still have a problem with meiosis II, where—let's just go over here—rather than getting split where the twin gets pulled apart, the twin stays together, and you're left with generally the same scenario as before.

- 5.71** So whether the nondisjunction occurs at meiosis I or meiosis II, in both instances, you're left with the potentiality of having a trisomy or a monosomy, both of which are in that general family of chromosome aberrations called aneuploidy. Now, among these uncommon problems, the three most common trisomies occur at trisomy 13, trisomy 18, or trisomy 21. If it's at trisomy 13, that's called Patau syndrome. And the child will be born with severe mental deficiencies, but also polydactyly is a fairly common outcome of trisomy 13, Patau syndrome. And that is when the child has extra digits on their adorable little cherubic fingers. Those chubby fingers will have an extra one or two.
- 5.72** So polydactyly can be a result, as well as microcephaly, the smaller head, because of the significantly diminished brain size. That's Patau syndrome, trisomy 13. Trisomy 18 is Edwards syndrome. In addition to, once again, some mental deficiencies over the life of the individual, the kidney and all of the visceral organs are malformed, including to the point where the infant might have developed, where the intestines are actually protruding out of the stomach. Even though the skin has formed around it, but the baby will have intestines that are literally dangling out of the tummy, and the baby will normally be very, very small.
- 5.73** And then the most common and the most viable, where the child is the most likely to survive, is trisomy 21, known as Down's syndrome. Down's syndrome is the best known example. It is the most common. Some of the key features there, in addition to some degree of an intellectual deficiency, there will be macroglossia, so the jaw will be a little too big, and certainly in relation to the upper jaw. So that's a pretty common feature. Often there will also be stunted size as well, not quite as tall as typical for the sex of the individual.
- 5.74** Now, all of those instances, trisomy 13, 18, and 21, are instances of what is called autosomal aneuploidy. Earlier, I failed to go into the detail, but among all of the chromosomes, these 23 matched sets, 22 of them are known as autosomal. They're dealing with all of the general function that determines how the body is going to be. Chromosome matched set 23 is sometimes not a matched set. That is the chromosome that determine, that usually are determining sex. I say usually because there's some instances when things go a little wrong, and we'll talk about that.
- 5.75** So chromosome 23, that is called the sex chromosome, that matched set. Of course, if it's X from mother, X from father, we normally will have a female. If it's X from mother and Y from father, because the father, of course, is the only one who can pass on the Y, then that will be male. That's the male sex or the male genotype. So you, just like we can have an autosomal aneuploidy, where we have too many, a trisomy or a monosomy in the first 22 matched pairs, the autosomal chromosomes, you can also have alterations in the sex chromosome.
- 5.76** The most common of the sex chromosome aneuploidies is a problem called trisomy X. Now I say problem, but I have to use that term a little loosely because there are women

who may have this and not even know. Its consequences are either totally irrelevant or very, very, very modest. But that is generally the most common of all of these uncommon problems. So it's the sex chromosome trisomy, where it's three Xs, where at some point, probably during mother, the egg division, there was, rather than pulling the Xs apart, the Xs stayed together. So mom created an egg with two Xs, and then baby got the one X from father, now having three Xs. Again, the female sex is already used to having a redundancy in the X chromosomes. There are two. So having one more is just one more redundancy that doesn't really end up causing any significant problem. So that's trisomy X.

5.77 Now, where things start to get a little more complicated is when we blend these Xs and Ys. So one other instance, I'm going to mention two, the first one being Klinefeltersyndrome. This is when a baby boy will be born, but with two Xs and one Y. Now that one Y packs a significant punch. Even though it's a small chromosome, indeed the smallest of all of them, it still has a significant effect, as we'll discuss more in a moment. It has a significant influence. So even though there is an extra X, the Y still determines that the baby will develop as male, with male genitalia, and we'll talk about that process soon. But the presence of that extra X does mess things up.

5.78 Not only will the boy, as he becomes a man, have likely severe infertility, but he may also have a phenotype or body that is a little more female. There can be some modest degree of breast development during puberty, not full-on mammary tissue and lactation, none of that, but also slightly wider hips, so just a bit more of a female form because of that extra X. Again, and as I mentioned, usually with severe infertility. That's Klinefelter syndrome, XXY. So three of the sex chromosomes, a trisomy.

5.79 Now, there's also an instance of a monosomy or a monosomy X, the opposite of the trisomy X, where the woman had three Xs. This is an instance of a woman having only one X, and it is called Turner syndrome. So somewhere in that development, what could have become, through the process of meiosis in a nondisjunction, what could have become a trisomy of Xs, that other egg or sperm that didn't have the right number matched with its partner chromosome may only have one. Again, that is called Turner syndrome.

5.80 This is relatively more substantial. In general, it's better for the organism to have more genetic information than less. And we see that here. Klinefelter syndrome with three sex chromosomes is very problematic, but the person can grow and survive, and even in some instances have some modest degree or some normal function. In Turner syndrome, there's really no chance of that because you've lost genetic information with just one X. So the person will usually be very, very short, will often not have any normal puberty because the ovaries are not healthy functioning ovaries. They're often so immature, if you will, that they're just referred to as gonadal streaks. They're just these small, fibrous little clumps. They're not really functioning ovaries.

- 5.81** And in addition to the very small stature, the person can have a few other physical abnormalities, like, for example, the neck. Rather than the neck being narrow below the skull, it can come out a little bit in what's called a webbing of the neck.
- 5.82** Now, earlier on in this discussion, I talked about the karyotype. The karyotype is referring to the actual chromosomes, the number and the type of chromosomes. And talking about the sex chromosomes is a nice time to bring up what is sex? What defines male and female? What is a woman? What is a man? This is something that's become a very debated topic. To a cell biologist, if it weren't so ridiculous, it would be quite amusing to look at this. However, in the end, it just ends up being sad and worrying that this is something that is debated and that people are claiming not to know what distinguishes male or female.
- 5.83** Usually it works, and often people only want to go this far to say, well, let's just look at the karyotype. If a person is XX, female, XY, male. But I just got done explaining to you how there are instances when the conditions start to get a little blurry, like Klinefelter syndrome. How would we define the sex of a person with Klinefelter syndrome? After all, this person has two Xs, which would normally define female, but they also have a Y, which would normally define male. How would we define them? If we're only relying on the karyotype or the chromosomes, then we're left with an impossible situation.
- 5.84** Similarly, we say that a person with Turner syndrome, what do we put them in? Because normally a female needs two X chromosomes to be considered female. Well, she only has one. So we do not rely on the karyotype to determine sex. We rely on the gamete type. This is why there should be no confusion with regards to the sex, and someone saying they're a different sex or becoming a different sex. We can always go back to the one simple question, and indeed, I would want you to remember this. This is the simplest, yet most precise definition of sex. What is the gamete that they create? Do they create sperm? Male. Do they create egg or oocyte? Female.
- 5.85** There has never been an instance in all of human existence where a person is creating both. It does not happen at the level of the gamete. That is the most functional, yet clear way to define the sex, because that is what matters for reproduction. That is why we can say sex is, in fact, binary. There's one or there is the other.
- 5.86** All right, at this point now, you should be not only very comfortable with the cell cycle and mitosis, the process of creating identical daughter cells, but now even the process of meiosis, which is the slightly more complicated way of taking a parent cell and creating these daughter cells that are unlike the parent cell, not only because of the exchanging of genetic information, but also because of the reductional halving of the chromosomes, taking the diploid cell with 46 chromosomes and splitting it into haploid cells with 23 chromosomes.
- 5.87** Now, let's go from the chromosomes and talk about how some of that gets applied. What ends up determining sex as we typically think of? I already defined to you that

sex needs to be determined at the level of the gamete, sperm production or oocyte or egg production. But I want to take a little time now, because this is a pretty good time to bring it up, to talk about sexual differentiation. And this, in other words, is saying if baby is going to be born with female genitalia or male genitalia, because that's going to be the next level of defining sex.

- 5.88** As we go through this, in fact, even earlier when we talked about Turner syndrome, in Turner syndrome, that is a child who is born with female genitalia. However, they do not make eggs. You need two functioning X chromosomes to do that. And so that classification of determining sex based on gamete type, what if the person isn't making a gamete? What if they are sterile, and those gonads are not functioning to the point that they're making the cells of reproduction? Then the next layer of definition comes into play, which is, what is the genitalia that the person is born with? Are they born with female male genitalia or male genitalia?
- 5.89** So let's move into that part of the discussion, because that is entirely a function of those sex chromosomes, most especially, the Y chromosome. So the purpose of this concluding part of the cell cycle, what I hope is that you're going to understand the process of gonad differentiation and genital differentiation, and some instances of genital ambiguity, when things aren't matching up the way they're supposed to. These are problems that can exist within biology. Again, as I mentioned earlier, biology is a bit of chaos, so it shouldn't surprise us that things can go wrong. What should amaze us is that, indeed, they go as right, they go right as often as they do.
- 5.90** Now, I'd mentioned the Y chromosome because it's a small little section. Indeed, a small section up at the 3' end of the Y, of that long stringy line of DNA, that can tighten up and become the Y chromosome. Again, it is small. It is the smallest of all of the chromosomes. But as I said, despite its diminutive stature, it really punches or fights in a heavier weight class. It has a significant effect.
- 5.91** So on the short arm of the Y chromosome, there is something called the SRY gene. So one small, single little coding section of genetic information on the DNA that has this SRY gene, SRY standing for the sex-determining region on the Y chromosome. Sex-determining region, SR on the Y chromosome, Y. SRY. That gene, when it goes through the process of transcription and then translation, which you're all expert in, it gives rise to a protein called TDF, testis-determining factor, and the name says it all. So the SRY gene on the Y chromosome gives rise to a protein called TDF. And this protein is going to take that primordial embryonic gonad, both of them, and it's going to say, I know what to do with you. I'm going to turn you into testes, or testicles. I'm going to turn you into the male sex gonads. Again, the testes.
- 5.92** So in the presence of TDF, those early germ cells or reproductive cells will be pushed down the path into what's called myotic spermatogonia. In other words, the diploid cells that are going to be giving rise to the sperm, the spermatocytes or sperm. In the absence of the TDF, those embryonic primordial gonads, which are up where the

ovaries would be in a woman before they get pulled down during the course of development in the male, in the male little baby fetus, in the absence of the TDF, those early germ cells or reproductive cells in those gonads will say, well, then I'm going to become these myotic oogonia, which will give rise to the oocytes or the eggs.

5.93 So again, just to restate that, if there is TDF present, those early gonads will say, I'm going to start getting ready for sperm production. If there's no TDF, no signal, then the default signal says, oh, all right, well, then I'm going to go down the path of oogonia, or oogenesis, to start making eggs. Now we're not done yet. There are other cell types within the gonads, each with a sort of male-female version. If there is TDF, then the other cells will go down—one of the other cell types will commit to the path of the Sertoli cell. And that's going to become very important for later sexual development, as I'll get to in a moment.

5.94 So again, with TDF, the early gonad says, I'm going to make Sertoli cells. In the absence of TDF, then the cell says, all right, I'm going to go down this default path and make granulosa cells, another essential cell type. Now, I'm not really going to be revisiting these cells. So the cell type itself is less important than you just knowing that they do become distinct, different cells. And then lastly, there's one more cell type to talk about. If there is TDF, the testis-determining factor, then those early embryonic gonads will say, I'm going to also make Leydig cells, which are the main cells in the male that's producing the male sex hormones, like testosterone, especially. In the absence of TDF, then those cells that would have become leading cells take the path of other least resistance, if you will, and becomes theca cells. And the theca cells are very, very prominent in the production of the estrogens for the female sex hormones.

5.95 Now, as I've mentioned, as I've gone through this, I've really elaborated how the SRY gene is necessary to create the testis-determining factor protein, the TDF protein. And then the TDF protein is essential for telling those embryonic gonads to make all of the cells of the testes or the male gonad. In the absence of TDF, all of those cells become the cells of the female gonad. Now, however, that doesn't mean that that's going to be a full functioning ovary because you do need two Xs for the normal development of oocytes. So, oogonia will not work to produce germ cells in the absence of two Xs.

5.96 So even though that ovary could have the granulosa cells and the theca cells, it will not be functioning reproductive, capable of any reproduction, like what we see in Turner syndrome that we talked about earlier. Now, that is the committing pathways of the gonads, ovaries or what becomes testes. Separate from that is the development of the internal sex organs or sex structures, as I like to call them. In the primordial little embryonic baby, in its earliest phase, it actually has both sets of internal sex structures. So in the male, this is things like the vas deferens or the epididymis or the prostate gland, these very typical male sex structures. In the female, it's things like the uterus, the fallopian tube, and the inner part of the vagina.

- 5.97** So within the baby, it can go either way. And mind you, it commits to it very, very early on, but that is separate. So the internal sex structures are separate from the gonads, and they don't necessarily go the same direction in some instances of anomalies, which we'll get to in a moment. And it depends, once again, usually it depends on what the testes have been told to do because of the presence of testosterone. If the SRY gene has properly coded for TDF, the testis-determining factor, then those early gonads have committed to the male testes, with the Leydig cells and the Sertoli cells that I'd mentioned.
- 5.98** And those are very, very important here, in part because the Leydig cells are producing a lot of testosterone, and that is necessary to make the male internal sex structures develop. So the male internal sex structures need to be told what to do. I know you can roll your eyes and think about the husband who's not loading the dishwasher like he's supposed to. So the male structures need to be told what to do. In contrast, the female structures need to be told not to develop. So the male structures need to be stimulated to develop, and testosterone provides that stimulus.
- 5.99** But at the same time that the Leydig cells are making testosterone to promote male structure development, the Sertoli cells are producing a hormone called anti-Mullerian hormone. In the Mullerian ducts are all of those female structures. So the Sertoli cells are making a hormone signal to destroy the female sex structures. And thus in the healthy functioning male baby, only the male structures survive. And that gives rise to the penis and the scrotum. And as the scrotum develops, it starts to pull those gonads down from the abdomen, down into the scrotum eventually.
- 5.100** Now, in contrast, if there was no SRY gene and no TDF, then those early gonads committed to the path of the ovary, and they would ideally have these early eggs to develop, but also they would have granulosa cells and theca cells, and there will be very little testosterone, indeed not enough testosterone to tell the male structures to develop. And if the male structures aren't told to develop, they go away and die. They get degraded into nothing. At the same time in the ovary, there are no Sertoli cells, so there's very little AMH production, the anti-Mullerian hormone, Mullerian being the term for the female structures. So in the absence of the Sertoli cell, and in the absence of this big release of AMH, the female structures stay and develop.
- 5.101** So the default setting with the internal sex structures is female sex structures, unless it is explicitly told not to because of the signals coming from the testes, the AMH telling the female structures to degrade and the testosterone telling the male structures to develop.
- 5.102** All right. Now, with all of that in mind, I want to emphasize, once again, that for functioning healthy female ovary tissue, she needs two Xs. Even if she has all of the presence of the female sex organs internally, if there's not two Xs in those ovaries, things will not work out well. Now, let's talk about some instances. Throughout all of this, there have been a few different points where things could go wrong, just within

this conversation of the sex and differentiation and the genital ambiguity that can happen.

- 5.103** There are two instances when things can go wrong that gives rise to a bit of a mismatch from the genetics, from the genotype, which is what the genetics say the person should be, compared to the phenotype, which is the physical manifestation. Just think through that process of the central dogma and that flow of genetic information. The two general things I want to talk about, the two potential causes for the sexual ambiguity, first of all is the SRY gene misbehaving and not being where it should. You'll recall that the SRY gene will encode for the testis-determining factor protein, or TDF, which will determine whether the gonads will, in that instance, it tells the gonads to commit to the path of testis formation or male sex organ.
- 5.104** But so in addition to the SRY gene misbehaving, not being where it should, you can also have an instance, which is a relatively, slightly more common called the lack of androgen receptors, or in other words, no testosterone receptors. Just as a reminder, these problems are very uncommon, and importantly, they are not getting more common. The existence and the frequency of these problems, of the genetics of sexual ambiguity, are not increasing in frequency. That's very, very important to know, because sometimes students think these disorders can explain some of what we're seeing culturally, and indeed, they do not, whatsoever. They're not getting any more common than they've ever been, and they've always been fantastically uncommon.
- 5.105** As I mentioned then, within this realm of genital ambiguity and a departure from the sexual norm, there are two general classifications—the SRY displacement, where the gene isn't where it's supposed to be, and the androgen receptor is not working well. So firstly, let's start with a person who has the genetics of XY. So their karyotype, their chromosomes, they have an X and a Y. Unfortunately, during the process of meiosis, very commonly, is when this uncommon problem would occur, the SRY gene accidentally gets deleted. You'll recall that that was one of the chromosomal aberrations I mentioned, deletions.
- 5.106** If the SRY gene gets deleted, then you have none of that series of events that I talked about earlier. In the absence of SRY, there is no TDF. In the absence of TDF, you do not have the stimulus to make sperm, or Leydig cells and all of its testosterone, or Sertoli cells and all of its anti-Mullerian hormone. And thus the person ends up following the other path, which is to be born with a female body. And so the baby would be born with a female form. Now, because of the way the gonads fail to develop properly, because there are not two Xs there. Remember, there's an X and a Y. These are not functioning ovaries. Indeed, they become very fibrous, just like I talked about with— or very scarred. Just like I talked about with Turner syndrome, they're not really functioning ovaries.
- 5.107** So not only will this person, of course, be sterile and never have any ovulation or indeed any menstrual cycle, they also will not go through puberty. And that's typically

when the problem gets diagnosed. This would be a little girl who is growing up, born as a baby girl, growing through childhood as a girl. And then when all her friends are going through puberty, she's not, and she continues not to, continues not to, even though she's getting taller, because height is not really affected here, but she will never go through any puberty. So if there is puberty that's going to happen, it will have to be medicated, medically induced. In other words, she has to be on hormones in order to make her body go through that transition of turning little girl body into woman body.

- 5.108** Now, because of the lack of the SRY gene, that not only means—so externally, the person is born female with external female genitalia. So just what's called the labia in the outer part of the vagina. Internally, think for a moment, how would the person, what sort of sex structures do they have internally? The absence, just to review to help you think through it, in the absence of the SRY gene, there was no TDF. No TDF meant there are no Sertoli cells and no Leydig cells. So there's very little testosterone being produced, and there is no AMH.
- 5.109** So there was nothing to stimulate the growth of the male sex structures, but there was no AMH, meaning there was nothing to tell the female structures to go away. So the female structures persisted. So internally, the person will have all of the sex structures of the female. Thus, even though this person will never ovulate, if they wanted, and every woman listening would wonder why, she could actually be on a hormone cycle that could induce the growth of the endometrium of the uterus and have a normal menstrual cycle. But again, not the purpose of the menstrual cycle, which would be ovulation. So why go through the pain? And I can only imagine the discomfort that's associated with that.
- 5.110** But nevertheless, that is the problem of what's called XY gonadal dysgenesis, also sometimes referred to as Swyer syndrome. So this was a person with the XY chromosome at the sex chromosomes, but the Y chromosome had a deletion where that very critical and one singular gene was accidentally snipped off.
- 5.111** All right, now, a parallel version of this is when you can have a person who has an XX genotype. And yet during the process of crossing over, that SRY gene from a Y chromosome accidentally slipped over onto an X. So it was a carryover effect, where that SRY gene no longer normally bound to the Y is accidentally found on one of the Xs. The presence of the SRY gene on the X chromosome throws everything off. So there is the SRY gene despite the two Xs, and that SRY gene successfully encodes for the TDF protein, telling those tissues that should be ovaries, and there are two X chromosomes even, but that they're actually supposed to be testes.
- 5.112** So it does, in fact, develop Leydig cells and Sertoli cells, and the Leydig cells are making a lot of testosterone, and the Sertoli cells are making a lot of AMH, anti-Müllerian hormone. And so for just a moment, we can think about how the person, how the genitalia and the internal sex structures will develop, and they will develop male. So the person inside and out, internal sex structures and external genitalia, they will have all

the male, if you will, hardware to go along with it. However, there are two X chromosomes, and so these are not functioning testes that are making sperm.

- 5.113** So this person will be totally sterile. There will be no production of the spermatocytes. And they will be testes like we think of them, located down in the scrotum, but they'll usually be much smaller and much more firm because once again, they're very fibrous or scarred. Often, in some of these instances, they're actually removed, in all of the ones I've mentioned, Turner syndrome, the Swyer syndrome, and even this one, just because of the risk of them becoming cancerous. So that is XX gonadal dysgenesis. Once again, when the SRY gene has moved over to an X when it shouldn't be, it translocated. But again, the internal structures and the external structures will all be male.
- 5.114** Now, lastly, once we've moved on from the SRY displacement problems, we get to the androgen receptor problem. And in this instance, it's only a problem in someone with an XY genotype. So they have the genetics of the male. But you'll recall that even from the earliest stages of development, let alone through puberty and future development, we needed testosterone in order to not only tell the internal male structures to develop, but also to eventually tell, well, not external, penis and scrotum, but then also to eventually go through puberty.
- 5.115** The problem here is that this person can't sense, they can't respond to the testosterone. The mutation in their bodies prevent them from sensing any of the testosterone. That means then that even though those testes went down the path of—or the gonads went down the path of testis development, and the leading cells are producing testosterone, the testosterone wasn't sufficient to stimulate any other changes internally or externally. But at the same time, those little testes do have functioning Sertoli cells. And the Sertoli cells are making the anti-Müllerian hormone or AMH. AMH telling the internal female structures to go away.
- 5.116** So externally, the baby is born female with external female genitalia labia, and then the first part of the vagina or the initial part of the vagina. But then internally, because there was not enough testosterone to tell the male structures to develop, but there was a lot of AMH telling the female structures to go away, they have no internal sex structures. And thus the vagina, rather than continuing up and connecting to the uterus, it ends up becoming what's called the blind pouch. So the vagina does not connect to anything further up.
- 5.117** Now, interestingly, when the baby is born, of course, there was no testosterone to signal male development, and thus the default became female external genitalia. But as the child ages, the child will appear to go through and will indeed have the secondary sex characteristics that define puberty, like the expansion of breast tissue and hip growth, most especially.

- 5.118** But you may be wondering, why would this person have that happen if those are testes? And they are, even though they're located up where the ovaries are and they're not making any sperm, those are still the cells of the testis, Sertoli cells and Leydig cells. And the Leydig cells are making testosterone. Remember, the problem here is that their body can't respond to the testosterone. That ends up resulting in a fascinating scenario.
- 5.119** It's a little-known fact that all estrogens in every male and in every female were originally androgens or testosterone, to be specific. Every speck of estrogen became estrogen by passing through having once been testosterone. And it is converted, testosterone is converted into the estrogens, this family of the prototypical female sex hormones, via the actions of an enzyme called aromatase. So aromatase is a very busy enzyme, much busier in females than in males, but it's taking the testosterone and converting much of it into estrogens. That's always happening.
- 5.120** Now, in this particular person, who can't sense their testosterone, there is no negative feedback. And to appreciate this, I invite any student here and in the future to watch the Fundamentals and Physiology series, where we talk a lot about negative feedback. But in a healthy individual, when testosterone levels are going up in male or female, it will start to inhibit its own production. So if the testes are producing testosterone, before they would get too high, the brain would sense, oh, hey, we reached our limit. Then it would send a signal back down to the testes saying, "OK, slow down." And so testosterone would never get too high.
- 5.121** But in the male with the XY genotype, there's this much testosterone and there's this much estrogen because some of it's getting constantly converted into estrogens via the actions of aromatase. Now, in this individual who doesn't have any androgen receptors, the brain cannot sense how much—it assumes there's none. It's not sensing any of the testosterone. And so ironically, this person looks female, but their testosterone levels are multiple times higher than they should be in any other male. They are enormously elevated because there's nothing to turn it off. There's no negative feedback. There's no sensitivity to the signal.
- 5.122** And that means we take this normal dynamic where testosterone is here and the estrogens are here, and what we end up doing is pushing it all up to the ceiling because the testosterone cannot turn off. With that much testosterone, there ends up being a massive amount of estrogens getting produced. So when this child goes through puberty, and the gonads end up producing a lot of sex hormones to turn little child into the adult version of themselves, they end up going through their version of female puberty. So little girl does look like woman. She has breast development and hip development, and it starts to have a female shape to her.
- 5.123** And again, it's because these testes are not turning off. And what ends up being this massive amount of testosterone also happens to end up being a massive amount of estrogens, making this body start looking like a female, a woman's body. But in the

absence of testosterone, there will be some things that don't happen, specifically body hair. So usually, testosterone will end up eventually limiting height. So this person ends up often being a little taller than the average woman.

5.124 And also she'll have little to no body hair. In fact, if there's no androgen receptor, she'll just have the body hair of a child. So a little bit of hair on the arms and legs, but nothing else. No axillary hair, no pubic hair, and certainly none of the coarse hair that even certainly men have, but even women have somewhat more coarse hair than a child. So slightly taller than normal and less overall body hair because of the lack of androgen signaling, because those are both determined generally by testosterone.

5.125 So here we have, at the end of this lecture of the cell cycle, an unexpected twist going into the realm of sex, which you hopefully understand better, including some instances of the genital ambiguity, when the genes and then the body, the physical form, don't end up quite matching up. Thank you.

Lecture 6: Metabolic Mechanisms

At a glance — Metabolic mechanisms, the final topic.

- 6.1** Now, we come to the final topic here. While we have been discussing all of the inner workings of the cell, we need to remember that any effort or any action requires some energy to give life to that action. Now, we come to that point. How does the cell actually get enough energy in order to do all of the many reactions we've discussed and the hundreds of thousands of reactions we've never had time to dive into? So, in this topic, we're talking about cellular metabolism.
- 6.2** And then I want you to understand through the course of this lecture what metabolism is. How is it defined? What is it made up of? And I also want you to appreciate a particular part of metabolism, namely, how the cell is breaking things down in order to get the energy that is locked within those cells, the chemical energy. How can it get access to that chemical energy in order to convert it into something that the cell can use to get some work done, some action?
- 6.3** So in that particular part of the lecture, we're going to discuss nutrient catabolism, which we'll define in a moment, as well as this citrate cycle, and then the inner workings of the mitochondria, the electron transport system, and then the final part of that, called oxidative phosphorylation. You will become expert in all of that. And then at the very end, we're going to do a brief overview of photosynthesis, the energy process that's occurring in plants, the complementary part to the energy process that we're talking about, the metabolism that we're talking about in animal cells. There is a plant parallel, which ends up really creating a beautiful symbiotic connection.
- 6.4** So firstly, speaking of the plant, we need to remember, because of its intermediary role in all of this, all energy that we experience, indeed, the energy animating our very bodies, originated in the sun. And then the sun is the source of all of the energy that we have on this planet. Now, we can't rely on solar energy, on that electromagnetic energy, the electromagnetic energy coming from the sun, but plants can. So the plant is designed to harness that energy from the sun, and in so doing, start connecting bonds together, particularly carbon bonds that we've talked about before, and we will again.
- 6.5** When these carbon bonds are connected, there are animals that can come and eat those carbon bonds and get nutrition from them. And that's where the animal, including we humans, come in. We can survive on eating the plants that have gotten their energy from the sun or other animals that, in some instances, are better designed to handle the energy in those plants. But at the end of it all, you could think of all of the energy sources that we're going to talk about, all of the nutrients, those carbons were once carbon dioxide in the atmosphere. So it's an interesting thought experiment to wonder at how we are able to survive on eating air that ends up being the fuel that provides energy to our bodies.

- 6.6** Now, earlier, I had mentioned the word metabolism, and that is a buzzword. People love that word nowadays. They don't appreciate that the term, however, metabolism, is really made up of the balance of two different processes. But metabolism, at the level of the whole body, we could define as the sum of every individual reaction, of the hundreds of thousands of reactions occurring in every of the trillions of cells. All of that would be encompassed in the term metabolism.
- 6.7** There are some reactions of these metabolic processes that are breaking things down, and there are some that are building things up. The breaking down part of metabolism, because it encompasses both, is known as catabolism. Those are catabolic reactions. And while the enzymes are breaking things down and being catabolic, they're actually releasing some of the energy that I talked about earlier. So catabolic reactions provide the cell with energy. It's basically telling the cell, hey, you need some energy to do something. So here you go. I'm giving it to you.
- 6.8** At the same time, the cell can do whatever it needs to with that energy. Some of that energy will actually be diverted into the other part of metabolism, which is known as anabolism or anabolic reactions. It starts to build things up. And you'll recall, in the first lecture, we talked about cell biology, how there are these key biomolecules. You'll recall them, I know—nucleic acids, carbohydrates, proteins, lipids. That is going to be the result of anabolic reactions. When the cell needs to build those up, and many, many more things, it is undergoing an anabolic process. But it's able to do so because of the energy that came from catabolic processes. And then the sum of all of this is metabolism.
- 6.9** Now, let's just focus again on the catabolic part, the part of breaking things down. You have, of course, heard of the term burning energy. I'm really fat-burning right now, or I'm going to eat this, drink this sugar, and I'm going to burn this sugar for energy. We use the term burn, but of course, there's no flame within the body. But that term is nice to use, at least as a teaching tool. Think about a fire. If we were gathered around a campfire and we are feeding that, that fire is a physical manifestation of energy, unlike what we can see in our bodies.
- 6.10** We can't see the energy in our bodies, or indeed, the energy all around us in its myriad forms. But that fire, that physical manifestation of energy, is burning. It is living because it's eating the carbons that are in the wood, a very good fuel source or any other source that has carbons in it. It is breaking those carbon bonds, and we're seeing the physical manifestation. Now, the wood is carbon. In fact, you'll recall that when we talked about the wall of animal cells earlier, we talked about how it is this loose fluid membrane. But the wall in plants is very rigid and tough. It is made of cellulose. It's much more rigid. It still is semi-permeable, thankfully, but it's very rigid.
- 6.11** But the wood is made of a lot of carbohydrate, and then the fire is burning it. And the result of that is a lot of heat. Now, if we look for the parallel in a living organism now, in a living cell, we are in fact—but also that flame, there's one other similarity, which is

that that flame needs to use oxygen. If we deprive the flame of oxygen, it can't burn anything. Oxygen is a necessary input. So we have a carbohydrate plus some oxygen. And the result of the wood burning is that we end up getting some carbon dioxide, some release, not necessarily in the form of smoke, but we're getting some of this breakdown of carbons, and then we're getting a lot of heat.

- 6.12** Now, an animal cell has the same exact inputs, carbohydrate and then oxygen or some carbons. It could be lipids. You could burn oil, right? We could have a little, a flame burning on oil. That works well as a fuel source. Anything that has lots of carbons, that flame can burn. We're not quite as broad in our ability to break things down. We have to narrow it down. But even still, the inputs would be the same. We need some carbon backbone, some carbon skeleton or carbon bonds, and we need some oxygen, usually. And we're going to talk about how that plays into catabolism.
- 6.13** But in animal cells, we get a little bit of heat—most animals, warm-blooded animals. We get a little bit of heat, and then we get a lot of what's called ATP. That is the big difference between the wood burning, the fire and its energy burning wood, or our cellular body is burning its own carbon source. We're burning it in order to create ATP. So that is an essential mediator of metabolism. And I'm saying the word mediator on purpose because it's kind of what's in between the catabolic and the anabolic reactions. If a cell needs to build something, it's going to need energy to build that thing. ATP will provide that energy and more—and we'll revisit that in a moment—and that ATP comes from catabolic reactions.
- 6.14** Now, ATP, very briefly, is a nucleic acid that we talked about earlier. It has this little ribose backbone, and it has even the bases. In fact, a double ring base like we talked about earlier when we talked about DNA. The difference, however, is that while DNA normally has just one little phosphate coming off of that five carbon on the little sugar molecule part of it or the little carbohydrate part, ATP is adenosine triphosphate. So it has three phosphates coming off its tail.
- 6.15** Those phosphate bonds are what's called high energy bonds, that if it breaks, you can use this—an enzyme can use that ATP to fuel itself doing something else. It's like the ATP comes around and starts animating the arms of someone who's getting some work done. So the ATP enables the enzyme to then do something else, to build something up or move something around that the cell wouldn't be able to do otherwise. So keep that in mind.
- 6.16** ATP is the essential energy currency of the cell. It is the money or the currency that the cell will use to buy something else, to get something done, whether it is anabolism and synthesizing other molecules; whether it is moving ions around and pushing something across the membrane against its concentration gradient—you'll recall that active transport; whether it's sending a nerve signal along the length of a nerve, contracting a muscle to allow us to animate and move our bodies. All of that requires some—it has to

be bought. The cell must buy that action. ATP is the currency with which it is capable of purchasing that act.

- 6.17** And thus, again, I mentioned that ATP ends up being a mediator. It sits between the catabolic reactions of metabolism and the anabolic reactions of metabolism. Now, when we think about the production of ATP—that being the end result— when people think of metabolism, they often just think of breaking something down. They only often think of the catabolic part, which, for the sake of this lecture, is going to be the predominant topic, but I am going to revisit some anabolic processes just so that you're really informed.
- 6.18** After all, you are taking this class from a guy who got his PhD in bioenergetics. An area of interest for me has long been the study of energy in living organisms, in cells. What is a cell doing with its energy, with energy in general? Is it using it to build things up or is it just breaking down and getting the energy as a result? But keep in mind that ATP is a very important thing.
- 6.19** Now, as we start getting through this lecture, I'm going to highlight three general areas, and we'll have a few little tangents or departures in the midst of these three areas. We're going to talk about nutrient catabolism or the breaking down or burning of glucose and fats. Then, we're going to talk about how those end products can come together to go into the mitochondria, into the inner part of the mitochondria, and fuel or run the citrate cycle, sometimes known as the Krebs cycle. And then, how those products can then go on to fuel the electron transport system, really the end action of the mitochondria, the so-called powerhouse of the cell.
- 6.20** Now, as much as I have been talking about ATP, when I mention now the electron transport system, when we get to that point in the lecture, and until we get to that point or through that discussion, I'm going to mention some other molecules, namely NADH and FADH₂. I'll just call it FADH. Those are not ATP, obviously. I'm not saying the words ATP at all. We're listing those letters, but they're going to be necessary when we get to the electron transport system, that end part of this lecture, that's when those will finally become relevant.
- 6.21** So, anytime I'm talking about how this reaction creates NADH or this reaction creates FADH, I want you to think, "Okay, that's going to result in some future ATP, and it's going to be a lot." So, it matters. Those are extremely important intermediate molecules that are going to, again, go to the mitochondria and say, "Hey, use all of the electron energy I have and use it to make new ATP."
- 6.22** So, as we go through this journey of energy production and metabolism, you'll hear some instances when a reaction just creates ATP, and then that is immediate energy that the cell can use right then and there. More often than that, though, is the production of NADH or FADH. And then, that's going to become very relevant later. Those have a lot of potential energy in them. They're not ATP yet, but what they have

in them is going to be used to create ATP later when we get to the electron transport system because without those molecules, the electron transport system is nothing; it doesn't work. They fuel it. They kick it into gear.

- 6.23** All right, so as I mentioned, three general things we're going to go through in order. We're going to talk about nutrient catabolism, or the breakdown of the nutrients, how those products feed into the citrate cycle in the mitochondria, and then how along the edge of the mitochondria, those products from the citrate cycle fuel the electron transport system.
- 6.24** So, first thing in step one of this three-step process: let's talk about glycolysis or glucose burning. Now, importantly, glycolysis—you'll see at the very end of it—you can imagine this as a series of events, and that's often how it's presented in biochemistry textbooks. It's a straight line down. You start with glucose, and then it is a series of reactions. Indeed, I'm going to highlight briefly some with a little more detail than others. There are nine reactions that start with glucose and end up with a final product that I won't mention yet. But when I get to that final product, it has a 'choose your own adventure.' It stands at a fork in the road, and it can go one of two ways.
- 6.25** And those one of two ways will determine whether this has ended up being what's called anaerobic glycolysis or aerobic glycolysis. There's another term for this. With anaerobic, a parallel term or synonymous term is non-oxidative, or even another term would be non-mitochondrial. In other words, that end product has the option to stay out of the mitochondria because nothing I'm about to mention happens in the mitochondria. It happens in the cytosol, in the space outside the mitochondria. So, glycolysis starts and primarily occurs in the cytosol.
- 6.26** But then, it has the choice at the end. Do I stay in the cytosol, or do I go into the mitochondria to get a lot more energy later? And that would be what's called oxidative glycolysis or aerobic. Aerobic meaning with oxygen, anaerobic meaning without oxygen. You've probably heard that term before, too. I'm doing aerobic exercise, or now he's really anaerobic. That is what this is referring to, even if the person may not be using those terms correctly, and often they are not.
- 6.27** So, there are nine reactions, and let's go through them in a little bit of detail because some of it matters as we're breaking things down or we are thinking of glycolysis in a catabolic direction, breaking it down to get energy, some cells of the body have the ability to reverse that and to go back up and to have an anabolic form to make glucose. Just as an aside, here's one of a few, although this one being spontaneous—dietary carbohydrate is not necessary for humans. It is considered a non-essential nutrient because humans are capable of making all of the glucose the body may need. And we're going to talk about that process.
- 6.28** All right, now, as you imagine glycolysis, one way of looking at the glycolytic pathway is to identify a preparatory phase, which are the first few reactions, and then the payoff

phase, which is the remaining reactions. I use those terms because some of these reactions require energy to fuel it or ATP. And so we're actually—in order to get more energy later, we're actually consuming energy on the way. There's a little bit of investment involved in the first few reactions, and that's why it's called the preparatory phase or the buy-in phase. The cell has to commit and say, "All right, I'm going to loan you some of this because I know you're going to be high yield and give me more back. I'm loaning you this energetic currency with interest, and I'm going to get more back in that payoff phase."

- 6.29** So, the very first reaction you've already heard me talk about in a previous lecture, which is glucokinase, sometimes called hexokinase, depending on the specific cell type. There's very slight differences in how those enzymes work, but the general outcome is the exact same, which is that it takes glucose and phosphorylates it. It adds a phosphate group, and it gets that phosphate from ATP. So, that is a buy-in step. That is a committed step, and it is committed because it ensures the glucose doesn't flow right back out of the cell. It's a way of locking the glucose into the cell and committing it to the glycolytic pathway. All right, so that's the very first step. Glucokinase is a buy-in. It requires a little bit of ATP to phosphorylate, to add that phosphate group onto the glucose.
- 6.30** The second reaction is just a slight change in structure, not worth really talking about. The third reaction is once again a buy-in reaction. What's interesting about reactions one and three so far is not only that they require ATP, so they're both taking in energy, they're telling the cell, "Hey, I promise this will be worth it," but they are also irreversible. They only go one direction. And you'll recall that a moment ago, I mentioned that some cells have the ability to go the opposite direction. That's going to have to be resolved when we start to turn things around. But at the moment, we don't have to worry about it. So, reactions one and three are buy-in. They require energy, and they're both irreversible. Two is reversible; that's an enzyme that can go either direction.
- 6.31** At step four, something interesting happens. Glucose is a six-carbon molecule. At step four, we actually cut it in half. Now, we're left with two three-carbon molecules. So, every reaction I'm about to go through for the remainder of this glycolysis is happening in duplicate, you can imagine. I'll mention it one time, but keep in mind that the pathway, step one, two, and three, with one and three buying in energy, requiring energy, and being irreversible— now we split.
- 6.32** So, everything hereafter is occurring in duplicate. But that is still a reversible reaction. There's no energy required. But it is also—as we split, we've reached the end of the buy-in phase or the preparatory phase. Now, we've reached the payoff phase. So, every other reaction, if there's any energy involved, it's giving energy, including reaction five. At reaction five, we now have the production of some NADH. You know what to do with that. It's not energy yet, but boy, is it going to be; it'll pay off big. It's worth having.

So, section five, remember, happening in duplicate. We got an NADH from each one of those reactions.

- 6.33** Step six, happening again in duplicate, is, once again, both mutual direction. It can go both ways, but it now creates a little ATP. There's an ATP that's being produced each time reaction six happens. So, the payoff is already happening. Indeed, the glycolytic pathway is telling the cell, "Hey, look, you already got your money back: I required two ATP at reaction one and three; I just have given you two back. You're welcome." And not to mention the NADH that is waiting to be used.
- 6.34** And then, reaction seven and eight are generally a little forgettable, just some other slight changes in the molecular structure of the molecule. No energy required, and they're multidirectional enzymes. They can go both ways. Then, we get to the last step. Step number nine is not only one way; it can only work to continue to go down; it cannot go in reverse. Once again, we get ATP from each time the reaction is happening. That final molecule is a molecule called pyruvate. That is now the end of glycolysis in a normal way.
- 6.35** We've gotten down to pyruvate, and now pyruvate is left to wonder where it should go. There are a lot of variables that are going to play into this, mind you, and I'm just going to highlight them at a fairly high level, but please keep in mind that pyruvate is standing at the edge of the mitochondria and waiting to assess the overall cellular demands to either commit into the slightly longer pathway, but with the ultimate payoff of much more energy, or just be done and turn into what's called lactate. You've heard that before.
- 6.36** Now, importantly, that then the moment pyruvate commits to turning into lactate, this is the anaerobic glycolysis. It is the end of anaerobic glycolysis. Importantly, I'm saying the word lactate very deliberately because animals do not make lactic acid. It does not exist. So, anytime you hear the term lactic acid, that would only apply to bacteria. It does not apply to the eukaryotic family of cells. We do not make lactic acid. We make lactate. More than likely, any myth you've ever heard of lactic acid is just that. Whatever you've heard from your old high school track coach or anyone similar who is blaming some pain or phenomenon on lactic acid, not only have they gotten the molecule wrong, but they've also gotten that wrong. Lactate doesn't contribute to any problem whatsoever.
- 6.37** Interestingly, when we go from pyruvate to lactate, being the final and key step of anaerobic glycolysis, we actually use an NADH. So, one of the NADHs that we made gets used up right now. We have to get what's called reducing power in order to change the pyruvate to lactate. But that is the end of anaerobic glycolysis. Now, while lactate is so heavily vilified, it is done so largely unfairly because lactate—before I talk about the consequences or the fates of lactate, rather, not consequences, but fates, I want to mention when this would happen.

- 6.38** There are some cell types where it is obligated—there is an obligatory anaerobic glycolysis, including cells that don't have mitochondria, like the red blood cells. Red blood cells do not have mitochondria. So they're a cell that the moment the red blood cell comes out of the bone, it's destined to die after just a few weeks, well, two to three months. It will not reproduce because it doesn't have a nucleus, it doesn't have mitochondria, and thus, it can only use the one energy source that comes without the mitochondria, which is anaerobic glycolysis. So, every red blood cell is only producing lactate.
- 6.39** Vigorously working muscle has such a high demand for energy that it can't take the time, and indeed, it doesn't have sufficient oxygen to fuel the aerobic version of this, which we'll get to in a moment, and we'll emphasize why oxygen is important. So, heavily working muscle will create more lactate relying on anaerobic glycolysis.
- 6.40** And then lastly, cancers. Cancer cells will rely almost exclusively on anaerobic glycolysis. Cancer abhors the use of mitochondria. Indeed, there are cancer biologists whose entire career is based on understanding that mechanism and wondering whether that can be leveraged in order to create anti-cancer therapies. But cancer cells love glucose, and this is why—they don't want to use their mitochondria; they don't really have enough. They only want to do what can happen in the cytosol, and that can only be anaerobic glycolysis.
- 6.41** Now, let's revisit the exercise quickly for just a moment. A heavily working muscle, as the muscle is working at a high intensity and reaching fatigue, it hurts. It starts to be uncomfortable. The ignorant high school coach is saying, "It's because of that lactic acid; you got to clear that lactic acid out." It has nothing to do with lactic acid. It's just a coincident event. The same overall metabolic state that is creating more lactate happens to be the same state of scenario where you're breaking down a lot of ATP.
- 6.42** As another tangent, in addition to the one before—everytime you break ATP, you create free hydrogen. And hydrogen—as hydrogen is accumulating in a particular area, it will make that particular area a little more acidic. The pH will actually begin to become a little more acidic, and that can aggravate nerve endings and tickle those nerves. But far from feeling like a tickle, it feels like a burn or a pain. So, the same scenario that is giving rise to lactate due to the energy demand happens to be the same scenario that is burning so much ATP that the ATP splitting apart actually is what's creating the product that tickles your nerves and makes it feel like a burning in the working muscle.
- 6.43** So, that is the end of anaerobic glycolysis. These are the instances when anaerobic glycolysis is needed, but, as I mentioned a moment ago, lest we conclude that lactate is a villain or perhaps slightly milder—it's useless, it is far from useless. There is something called the Cori cycle, identified by this husband-wife team, the Cories. The Cori cycle identified this process whereby a cell, like a red blood cell or a working muscle, is capable of releasing lactate into the blood, and then that lactate and those

carbons in that lactate will go to the liver, and then the liver will use lactate to undergo the process called gluconeogenesis, which we will talk about in just a moment. That's when the process starts to go backwards.

- 6.44** So, you could start at lactate, just like this anaerobic glycolysis went down all these processes, all these steps got to pyruvate, and then committed to lactate, you can run the whole thing back up. Now, there are a couple of steps that require some detours, which we'll get to—three steps—but that's possible to do, and the liver does it very readily. In fact, in a fasted state, if a person hasn't eaten in about 24 hours, all of their blood glucose has to be made from scratch now, and the liver is the organ that makes it. We'll talk about that process. The primary source of those carbons that is used to then make glucose is actually coming from lactate. The liver is so good at recycling those carbons and getting to the lactate and say, "Ah, you weren't burned in the mitochondria? I can still use you. I can rebuild and recycle those carbons and turn it right back into the glucose." It is, in fact, a beautiful instance of metabolic recycling.
- 6.45** Okay, now let's go back to where we were a few minutes ago, which is pyruvate standing outside the mitochondria. We discussed the option when the pyruvate committed to the lactate pathway, requiring one of those precious ADHs or two because it's all happening in duplicate. The alternative version of this was that the lactate or the pyruvate, rather, steps into the mitochondria. And to do so, it needs the actions of a very complicated and indeed a complex enzyme with multiple parts collectively referred to as pyruvate dehydrogenase.
- 6.46** Now, one consequence of this immediately is another ADH is being produced. Every molecule of pyruvate that comes in through the pyruvate dehydrogenase reaction—it's basically the door it has to step through, if you will—we get an NADH produced, which will pay off. We want those NADHs because they're going to give us a lot of energy later.
- 6.47** Now, when pyruvate goes through that reaction, it comes out the other side as a new molecule called acetyl-CoA. Acetyl-CoA is the branch point or the nexus molecule of all nutrient metabolism that involves the mitochondria. It can go—anything can turn into acetyl-CoA if it's using the mitochondria, and acetyl-CoA can then, in turn, be used to turn into anything else. So, it is right at the nexus of all nutrient metabolism. So, acetyl-CoA is the end result, and we get a little bit of NADH produced.
- 6.48** Now, importantly, that pyruvate dehydrogenase reaction can start to get turned off by its own products. So, let's say that there's too much acetyl-CoA, or the cell has too much ATP or has too much NADH—that's basically the cell's way of saying or the mitochondria's way of saying, "Hey, I am loaded. We got so much energy here. Don't come in anymore," and it starts to inhibit that enzyme, pyruvate dehydrogenase. Thus, even if the pyruvate were inclined to have stepped through, it is not able to, and anaerobic glycolysis is the only option.

- 6.49** All right, so we have finished the catabolic side of glucose. That catabolic story is told. But as I've alluded to a few times, some tissues, like the liver, not only have the ability to break down, or to catabolize the glucose, they also can anabolically create new glucose. And again, that is a process called gluconeogenesis.
- 6.50** Now, the cell abhors these cycles that would waste energy. So, we have to change—we have to have some points where we can regulate which direction the reactions are going. And those points are those three particular one-way enzymes that we talked about earlier. Reaction one, reaction three, and reaction nine. All the other enzymes are two-way streets. These enzymes are one-way. They only go down. So, we have to have a detour.
- 6.51** And before I mention the detour, those three points also represent the main points of regulation. For example, the energetic state of the cell will determine whether those enzymes are turning on or off. If a cell has a lot of ATP, it will start inhibiting those enzymes. It's basically saying, "Hey, don't break down anymore. We don't need any more glycolysis," so it will start to slow them down. Also, there is an endocrine or hormone signaling that plays into this, especially the hormone insulin. When insulin comes to a cell, it will turn those enzymes on, all three of them. It will say, "Hey, we got a lot of glucose in the blood. Let's burn, burn, burn." Insulin turns them on. In contrast, insulin's opposite hormone—whereas insulin's most famous effect is to reduce blood glucose, there's another hormone named glucagon, whose main job is to increase blood glucose. So, it doesn't want to be burning glucose. Thus, it's no surprise that while insulin would be, when it's elevated, stimulating those three enzymes, glucagon is inhibiting them.
- 6.52** Thankfully, you won't have a scenario where both insulin and glucagon are elevated. One will be predominating based on the load of glucose in the blood. If someone's just eaten a starchy sugary meal, insulin comes up, it will turn those enzymes on and the body's burning glucose. If glucose levels are relatively low because of, say, fasting or a carbohydrate restriction, like a low-carb or a ketogenic diet, then the glucagon will be elevated, and it will say, "Hey, we can't really afford to be burning glucose. Let's shift over to fat burning, which we'll talk about in a moment, and we need to make a little bit of glucose because some cells don't have mitochondria and can't burn fat like the red blood cells. So, some cells in the body must rely on glucose. So, it's a good thing that we have the ability to produce it.
- 6.53** Now, back to those three regulating steps of glycolysis. Because they are only one-way reactions, we need to have a detour reaction. And indeed, there are enzymes that are specifically designed that when they are turned on, they enable—but they are a point of control—they enable that particular molecule to go back while bypassing the one-way enzyme saying, "Oh, I have a detour for you. You can use me as an enzyme, and I will let you get around that one-way street. That would only be allowing you to go

down. I'm going to allow you to go up and create new glucose." That's how gluconeogenesis can happen.

- 6.54** And just as I mentioned, when insulin is high, it's stimulating the glycolytic version, the catabolic version. When insulin is high, it inhibits those detour, the anabolic version, the gluconeogenic version of those enzymes. Similarly, but in stark contrast, if you will, while glucagon is inhibiting the catabolic version of those enzymes, glucagon is activating the anabolic version of those enzymes.
- 6.55** So, we have this very differential regulation of these pathways. Insulin and glucagon will never be elevated at the same time, and thus, those pathways are never occurring at the same instance. You're never breaking down glucose and immediately building it back up, breaking it down. That would create what's called a futile cycle, and the cell abhors futile cycles. It's too wasteful. So, these two pathways are reciprocally regulated. If there's one signal that's activating one pathway, then it would be the same signal that inhibits the other pathway and vice versa. We don't want a futile cycle where a process is catabolic, anabolic, catabolic, anabolic—that's not efficient. The cell abhors the lack of efficiency.
- 6.56** All right. We talked about glycolysis. Let's move then into the other part of the main fuels, which is lipolysis or fat burning. This is still under, of course, step one because you'll recall step one of the three steps we're going through was the catabolism or the breakdown of the nutrients. So, let's go from glucose breakdown now to fat breakdown, lipid breakdown, or lipolysis.
- 6.57** Just like glucose is a carbon molecule or a carbon ring with carbons bound carbons and some hydrogens and oxygens, remember that fats or lipids are also carbons, long chains of carbons bound together with lots of hydrogens and then a little bit of oxygen. Now, the main source of those fats that we're going to burn, it can certainly come from the diet. Absolutely, we can burn the fat that we eat, but most of the fat we're burning is coming from fat we're storing that we have had stored, and most fat storage is occurring in our fat cells.
- 6.58** I've mentioned this before when we were talking about glycerolipids and phospholipids in the very first lecture, but the lipid droplet is essentially a big ball of fat within the fat cell. In fact, if you look at a fat cell under a microscope, the lipid droplet is so big you can't see anything else. The nucleus and the mitochondria, the endoplasmic reticulum, and all of these other aspects we've talked about are so pushed up against the side that you don't even see them. All you see is this big bubble. It looks like an air bubble, but it's a lipid. It's a fat bubble. That is triglyceride, which you'll recall is a glycerolipid. That's the glycerol backbone with the fatty acids coming off it.
- 6.59** Now, that glycerolipid triglyceride droplet actually is covered with phospholipids. You'll recall the phospholipids—that's what creates the phospholipid bilayer of the cell membrane. Well, the lipid droplet itself is fat, but it needs to be kept in its place. And

so the phospholipids stack up on top of it in a monolayer with its fat-loving or water-fearing, hydrophobic, fatty legs pointing downwards. And then, the watery friend, the hydrophobic head, is facing out. So, it's giving shape to the lipid droplet, but also blocking it from breaking down until it's time.

- 6.60** And it's time when an enzyme tells it to move out of the way. So then, the phospholipids will part ways, and now an enzyme called lipases—there's actually a couple of different enzymes. Lipase. Remember, you can identify enzymes by the suffix "-ase." So lipase, lipid -ase, or lipase, is capable of taking a triglyceride and starting to pull off individual pieces of it, breaking down the triglyceride into individual free fatty acids. Then, those free fatty acids are able to move through the cell and get out into the blood to be used somehow in some other way, and I'll talk about that in just a second. But again, the critical event here was the phospholipids being told to move out of the way.
- 6.61** Importantly, as we revisit in a moment, that is largely under the influence of hormones. The cell needs to be told what to do with its energy, which is thematic of all of metabolism. There's always an endocrine or a hormone component to the body using energy. When does the fat cell know it's time to break down energy? When hormones tell it to. It doesn't know.
- 6.62** The fat cell doesn't have a direct connection to the muscle cells. If the muscle cells are exercising—if we are out on a run, there's no direct nerve coming from the muscle, coming up to the fat and saying, "Oh, hey, I'm working out now. It's time to burn." The working muscle will have its own effect on changing hormones in the body and changing fuel use. Those hormone changes will then get the signal back to the fat cell, saying, "It's time to break down so that I can fuel the muscles."
- 6.63** Most especially, once again, we have a hormone called insulin that is highly relevant. That is thematic of this entire lecture. In order to understand the way a cell uses energy, you must appreciate the role of insulin. It reigns supreme. Insulin's main effect is to want a cell always to store energy. Always, always. It abhors breaking things down in this instance as well. So, if insulin is elevated, it ensures those phospholipids stay in place and guard the lipid droplet. If insulin is down, now the phospholipids say, "No one's telling me not to. I'm a lazy guard, so I'm not going to do anything. Go ahead. Break down as much as you want." Now, indeed, lipolysis occurs, and lipolysis is the word for the breakdown of lipid, taking the lipid droplet and all of its triglycerides and breaking off individual free fatty acids.
- 6.64** All right, now, when the fatty acids get into the bloodstream, they have a few different fates. Those fatty acids will get to a new cell, any cell of the body. And that cell can pull the fatty acid in and do many things with it. I'll highlight three.
- 6.65** First and foremost, it can use those individual fats to create new phospholipids. Those can be incorporated into the cell membrane or the membranes of any of the other

organelles, the mitochondria, the ER, the Golgi apparatus, all of those organelles we talked about that are membrane-bound or locked in with their own membranes. Those fatty acids can become a phospholipid.

- 6.66** Second, they can go through a process that is called re-esterification. That simply means you can take that fatty acid in that came from a triglyceride and use it to build up a new glyceride, depending on the tissue type. This is, in fact, one of the main things that happens in the liver, with fatty liver disease, the most common liver problem worldwide. It's because fat cells are leaking out their fat when they're not supposed to because they become resistant to insulin's effects. And the high insulin combined with the high free fatty acids tells the liver to pull it in and store it. So again, the fat can be pulled in and simply used to create new fat.
- 6.67** Then, the one we care about for the sake of this lecture, is that it can undergo some burning. And this is where we use the term beta-oxidation. Beta-oxidation is a famous must-know term when it comes to fat burning. This is when the burning actually starts to happen, if you will. Lipolysis, or the breakdown of fat—some people will commonly say, "I'm burning fat." No, you're just moving the fat from your fat cell. The fat burning is happening somewhere else.
- 6.68** So, that fatty acid has come into the cell, but before we can burn it, it has to get to the mitochondria. Remember, every other process of breaking down nutrients must occur in the mitochondria. So, it has to move through, and there's a checkpoint. There's a regulation point called the carnitine acyl transferase, C-A-T, carnitine acyl transferase. There's a CAT1 on the outer mitochondrial membrane and a CAT2 on the inner mitochondrial membrane. This is basically a way of the mitochondria saying, "Okay, only certain people can come in. We're a selective club. Oh, you are a fat that can come in. I'm going to link arms with you, and I'm going to escort you through these two layers of the mitochondria, through CAT1, through CAT2. Now I've escorted you in. I'm done with you. I'm going to go back out."
- 6.69** So then, that CAT molecule won't stay with the lipid. It just escorted it in. But it's a way of regulating what can come in. Once the fat is in, now we start undergoing that process I mentioned a moment ago called beta-oxidation. If you think about the fat molecule as that long string of carbons, on one end, we have the first carbon called the alpha carbon. Then next, we have that next one is the beta, and that's where we cut it right after that.
- 6.70** So, as we look at this long string of carbons, we start splitting it up into two carbon pieces. Two carbon pieces, one at a time. Those two carbon pieces become a molecule you've already heard of: acetyl-CoA. You'll recall that acetyl-CoA was the end product of pyruvate stepping into the mitochondria. So, the end product of aerobic or mitochondrial glycolysis is the same product at the end of fat burning, which is through beta-oxidation, namely acetyl-CoA. As I said earlier, acetyl-CoA is at the branch point or the nexus molecule of all nutrient metabolism. This is just one more example of it.

- 6.71** Now, with this acetyl-CoA from both glycolysis and both from beta-oxidation and lipolysis, or to put that in order, lipolysis then beta-oxidation, we are ready to engage the citrate cycle. Now, the citrate cycle goes by a lot of names, and I am happy to admit one of them being named after Sir Hans Krebs, who was the discoverer of this and indeed won the Nobel Prize.
- 6.72** It is called the Krebs, or to be more precise, the citrate cycle, because citrate is the first made molecule. It's called a cycle because it has components that will continue to be used again and again and again. Even though they're not actually structured in a circular shape, it's just chaotic enzymes bouncing around the inside of the mitochondria, it will go through a process, and then it will repeat itself, and then it will repeat itself. So, what is key here is just a few takeaways.
- 6.73** The first being that every aspect of oxidative metabolism, anything that is burning nutrients in the mitochondria, must use the citrate cycle. I've now talked about how glucose fits into it by making acetyl-CoA. Also, now with fats fitting into it through making acetyl-CoA through beta-oxidation. But you guys have likely heard of the word ketones. Ketones will turn into acetyl-CoA to be burned. Even lactate. Lactate is capable of going into a mitochondria and being turned into acetyl-CoA to be burned. So, everything that's going to engage the citrate cycle, almost everything, is doing so through having been converted to acetyl-CoA first.
- 6.74** Acetyl-CoA and its two carbons will come into molecules that are already there and ready to go, and it will feed through the first reaction. And then, once it starts moving through the reaction, and if it's capable of going through all of it, and it usually does, we will get the following products. We will get one ATP molecule. We will get one FADH₂ molecule or FADH—that's the first time you've heard that one be produced, the first and only time—and then we will also get three NADH. So, we get a lot of potential energy molecules and just a little bit of immediate.
- 6.75** Now, at the same time we're getting the energy, we also produce two molecules of CO₂. And this is why when we exhale, and we're exhaling carbon dioxide, we inhale the oxygen, and we're going to use the oxygen in the process that I'll mention in a moment through the electron transport system, but through the course of metabolism, we end up releasing CO₂. This is where the CO₂ largely comes from. It comes from the citrate cycle. The acetyl-CoA molecule with its two carbons came in, and in the process of using the full citrate cycle, two carbons came off as one molecule of CO₂ and another molecule of CO₂.
- 6.76** Now, where does that CO₂ go? We breathe it off. It works its way as a gas. It gets back into the blood. It brings it up to the lungs, and the lungs pull it in, and we exhale it. Then, when we exhale it in this beautiful circle of life, plants inhale that CO₂. And then, as we get to photosynthesis, one of the products of photosynthesis is the release of oxygen. It basically turns this whole process in reverse, although not technically. Our exhaled CO₂ is inhaled, if you will, at the plant, the plant will breathe off O₂ or oxygen,

which we then inhale to provide some basis or some energy for the electron transport system that we'll get to.

- 6.77** All right, now, that very first reaction in the citrate cycle, this becomes an opportunity to talk about catabolism versus anabolism when it comes to lipids, the breakdown of the lipid all the way, or the synthesis of new lipid. Now, that first reaction, as I said, is taking an acetyl-CoA and it turns it into citrate. That's the citrate cycle. That's the first molecule in this series of events that repeats itself again and again. If, however, the mitochondria starts to get a lot of citrate, it will start inhibiting the citrate synthase enzyme. So, the first enzyme that's taking the acetyl-CoA and turning it into citrate, as it starts to get inhibited by its own product, now the acetyl-CoA has nowhere to go, or rather, it can't continue to go into the citrate cycle, and it must go somewhere else.
- 6.78** So, where can it go? There are two potential fates for that acetyl-CoA. That acetyl-CoA can either turn into new lipid through a process called lipogenesis, the opposite of the term lipolysis, lipolysis being the breakdown of the lipid, genesis being the synthesis of new lipid, lipogenesis, or it can go through this new pathway called ketogenesis. That, of course, abounds in an instance of a ketogenic diet. It also abounds when someone's just fasting.
- 6.79** But which one does the acetyl-CoA go to? How does it know, "Should I go to lipogenesis or should I go to ketogenesis?" Remember, a cell does not inherently know what it should do. It needs to rely on other signals from the rest of the body to know the overall state of the body. And thus, it relies on hormones. And I bet by now you can imagine which hormone determines largely which pathway occurs. And that is, of course, the hormone insulin. It is all about whether insulin is high or whether insulin is low in that moment. If insulin is high, then lipogenesis wins that battle. And then, what happens is that the cell basically gives up its acetyl-CoA, essentially.
- 6.80** So, I'd mentioned that first step in the citrate cycle was turning acetyl-CoA into citrate. Then, if insulin is high, citrate will leave the mitochondria. It basically tells the rest of the citrate cycle, "Hey, it's too crowded in here. I'm going to go out of here. I'm going to go out." So, it exits the mitochondria, goes into the cytosol, and turns into acetyl-CoA in the cytosol, which, through a process of events, we basically start taking those acetyl-CoA molecules, two carbons at a time and binding them back together. The exact opposite of beta-oxidation when we were cutting them apart, now we're going through this lipogenesis, all because insulin is demanding it, and we're putting those two carbons back together, making a fat from scratch.
- 6.81** That's a process called de novo lipogenesis, making the fat from scratch, basically. And there are a handful of enzymes that are needed in that, but importantly, some of those intermediary molecules actually go back and block the fat from ever entering the mitochondria in the first place. You'll remember that I'd mentioned there's a CAT1 and a CAT2 who act as the usher, the person or the escort bringing the fat into the mitochondria. If we've started to turn on that process of lipogenesis, some of those

molecules actually will come back and block that off. It's basically the way of telling the mitochondria, "Hey, we're not breaking down anymore. We're building up. Didn't you get the message?" Insulin will make all of that happen.

- 6.82** Now, let's go back a few moments and wonder if we have a lot of acetyl-CoA, and it's wondering which way it should go: lipogenesis or ketogenesis. We already mentioned that if insulin is high, it has to commit to the lipogenic pathway. In contrast, if insulin is low, it commits to the ketogenic pathway. And so, acetyl OA can turn into ketones. And then, ketones are available as a fuel source for every cell with mitochondria. Every cell with mitochondria is going to happily pull in those ketones and burn them for fuel.
- 6.83** Now, at this point, we have this interesting divergence of metabolism, where we have the potential to be breaking down fat through lipolysis and beta-oxidation, and we have the potential to make new fat. But the one critical signal that determined which pathway it went down is insulin. As a tangent, and we've had a few now, what then happens in a body that has no insulin? What can that body do?
- 6.84** Well, in the absence of insulin, it becomes obvious, which is that there's no signal to stop fat burning. You'll recall back at the lipolysis step where the phospholipids were blocking the lipid droplet, there's no insulin at all, so they're always letting the cell have at it. So, the fat cell cannot help but be constantly breaking down its fat. At the same time, there's no ability to make new fat. So, the person can eat thousands and thousands of calories in excess, and in the absence of insulin, it is utterly, totally, and completely impossible for them to make fat on their bodies, and, indeed, to even hold on to any fat.
- 6.85** This is what happens when a type 1 diabetic, which is the person without insulin level, before they get on their insulin therapy, there are two interesting things happening. One, they have a voracious appetite. They almost can't stop eating. And two, they have no ability to gain fat. They start to waste away despite eating thousands of calories. In the absence of insulin, there is no ability of the body to hold on to its energy. The metabolic rate goes up. Ketone production goes up so high that they're just wasting ketones in their breath and in their urine. That's an outlet for the thermodynamics or all the energy. It's literally just being spent out of the body.
- 6.86** Now, once the type 1 diabetic begins insulin therapy, two interesting things happen in opposite. One, they suddenly start eating less, so calorie consumption starts to go down, and paradoxically, fat gain starts to occur. This phenomenon is so apparent to the type 1 diabetic that some refer to their insulin as their fat shots because when they first enter the hospital with this raging uncontrolled type 1 diabetes, by the time they leave it a few days later, they've already gained so much fat that they have to have a different set of clothes. This is used tragically in some instances of people with type 1 diabetes.

- 6.87** It can give rise—this fact, this phenomenon, the fattening power of insulin gives rise to an eating disorder called diabulimia, where the type 1 diabetic has learned they can eat whatever they want, and as long as they skip their insulin injections, they can be as skinny as they want. Unfortunately, their glucose levels will be through the roof. Their ketone levels will be through the roof into ketoacidosis, which is not the same as ketosis, like a fast or a ketogenic diet. They will look amazing, but they will die early. And so they'll look wonderful in their coffin, but it is a tragic scenario that you can imagine how tempting it is.
- 6.88** Whereas most other eating disorders require a degree of starvation, or they require a degree of bulimia or vomiting, in this instance, they're able to enjoy and eat the entire thing they want. And you're telling me all they have to do is not poke themselves with a needle? That would be very, very tempting. Again, there is metabolic hell to pay, but if your main outcome is to just be thin, then you might not care enough about the other consequences.
- 6.89** All right. Now, at the end of this, we've gone through steps one and two. We've talked about the catabolism and, in some instances, anabolism of the nutrients. Then, we talked about the citrate cycle and its role. You'll recall the citrate cycle has its main products of a little bit of ATP, and then NADH and FADH. Now it's time for these molecules to earn their keep, and we move in our journey to the electron transport system and oxidative phosphorylation. So, it's finally time for those energetic molecules, NADH and FADH₂, to pay off.
- 6.90** One aspect of the mitochondria that needs to be appreciated before we dive into the electron transport system in too much depth is the electronegativity of oxygen. Oxygen wants to pull in electrons. That is very, very important because the electron transport system has four reactions. And oxygen that we've inhaled is just waiting to be used. It is determined to pull on those electrons. It has a stronger pull on the electron than almost any other element within the cell or within the world. It wants those electrons very badly.
- 6.91** NADH and FADH have some electrons to give, but it all has to happen at the proper order. So, to understand the electron transport system and the role of the mitochondria here, there are two things I'm going to cover. Firstly, the electron transport system itself, sometimes referred to as the electron transport chain. I don't call it that for a reason I'll talk about in a moment, but that is the most common term, just slightly less accurate.
- 6.92** At the end of it, all of the electrochemical power that the electron transport system is generating is then finally going to be taken advantage of through this big molecular machine called ATP synthase. That is the part that is called oxidative phosphorylation. You can hear in there the phosphate—that's going to be when we make new ATP. But they are, in fact, distinct events. The electron transport system with its four complexes

are going to provide the energy, if you will, for that ATP synthase to do what it needs to do, namely create ATP.

- 6.93** When we get to the beginning of the electron transport system, the reason I call it a system and not a chain is that if you think of the word chain, you think of these linked things. Complex I of these four electron synthesis—electron transport system complexes, you may think that one leads to two, which leads to three, which leads to four, and that is not how it works, but saying 'electron transport chain' makes you think that.
- 6.94** I like to use the word 'electron transport system' because you have complex I, which is going to take the electron-giving power of NADH. So complex I is standing there ready to go, and all those NADH molecules that we've made previously now come to complex I and say, "Hey, here's my electrons." complex I says, "Okay, thanks. Got them. Awesome." At the same time, complex I is receiving the electrons from NADH, complex II is next to it but not connected to it. It is taking in the electrons from FADH. It's not getting anything from complex I. It has its own direct deposit source, which is FADH.
- 6.95** Now, both of them take their electrons and hand them to a molecule called ubiquinone or CoQ10. Then, it will take the electrons to the substrate III or complex III. One comment on ubiquinone or CoQ10. Very early on in the lecture series, I talked about cholesterol and how many roles it plays. I actually alluded to this exact moment where cholesterol will be altered and it will turn into this essential component, this electron-carrying component called ubiquinone or CoQ10.
- 6.96** In fact, cholesterol is so essential and feeds into this so readily that if a person begins taking a cholesterol-inhibiting drug in the family of statins, they often should also be taking CoQ10 as a supplement. Because if you have reduced cholesterol production, you will have reduced ubiquinone production. So, you want to replace that if you want your electron transport system to work well.
- 6.97** So, one is receiving the electrons from NADH, complex II is receiving the electrons from FADH, and then they are both handing them off to CoQ10 or ubiquinone. Then ubiquinone takes them to complex III, which then passes them to complex IV at the very end of the electron transport system. That is sufficient detail for the most part.
- 6.98** One last comment on this is that as the electrons are moving through these complexes, in order to maintain this static equilibrium with all these negative charges, the cells are also pumping out positively charged molecules in the form of protons. So protons, bound to a hydrogen atom, actually—these hydrogen protons are getting pumped out. This matters in the end. There are four protons that are getting pumped out at complex I, none getting pumped at complex II, two getting pumped out at complex III, and then finally, four getting pumped out at complex IV. So, you can see then that the number of protons that gets pumped out depends on where the journey starts.

- 6.99** If it started with NADH coming to complex I, then you had the four at I, the two at III, and the four at IV, resulting in ten protons that were pumped out. That will matter in a minute. If, however, we started this electron transport journey at complex II, receiving FADH, we skipped the proton pumping of complex I. Now we only have the two coming out at III, and then the four coming out at IV. So, we are left with only six having been pumped out in that instance. So, we had ten pumped out or six pumped out, and that matters because all of those electrical charges are starting to get irritated by being so close to each other.
- 6.100** And they're sitting in that what's called an intermembrane space. All of this what's been happening has been happening on the inner mitochondrial membrane. We're pumping out the protons, and they're getting trapped between the inner and the outer, in this inter membrane space. And again, they don't like being so close. They're like my two youngest kids. If you put them too close to each other, you're going to get some static. Something's going to happen. So, they start pushing against each other.
- 6.101** Before the cell can take advantage of this, the last and final note to mention about the electron transport system is oxygen. Oxygen has been waiting and waiting to eat an electron. It loves to eat electrons. Remember, it's very electronegative. Finally, at complex IV, when we're done with all of our proton pumping, complex IV says to the oxygen, "Okay, fine. Now you can have these electrons." But we don't want the oxygen to turn negative in its charge, so it also takes on some positively charged protons or hydrogens. And when you get hydrogen and oxygen together, you create water.
- 6.102** This is how—oxygen is called the final electron acceptor and it is able to accept those electrons without becoming a reactive oxygen species or a superoxide radical, a very damaging molecule that we've talked about in the Q&A earlier. It is able to maintain its neutrality by also taking on some hydrogens. And thus, that oxygen in accepting that electron will convert into metabolic water.
- 6.103** As an interesting aside, this is something that camels and some animals are so efficient in that they don't need to drink a lot of water. They have so much fat to burn—in the camel's case, those big fatty humps, those are filled with fat; they're not filled with water. The animal is so efficient at creating metabolic water that it literally makes takes enough water from fat burning through this exact process. Again, with oxygen being the final electron acceptor.
- 6.104** At the end of this, of course, we have the electron transport system, and waiting patiently has been the ATP synthase, sometimes referred to as complex V. I don't like to refer to it as a complex at all. It's distinct. But up until this point, it has been all the electron transport system, moving electrons through the complexes, powering, if you will, the pumping out of the protons. Now, all those positively charged protons who are getting on each other's nerves need somewhere to go. They need an outlet.

- 6.105** ATP synthase is perfectly designed to basically say, "Hey, protons, you want to come back into the matrix? Sure. Flow through me. I'm the only way for you to get back into the matrix or the inner part of the mitochondria. But for every four of you that come in,"—the ATP synthase is thinking—"I can shift myself around and take an ADP, an adenosine diphosphate, and now you're giving me the power by coming in and moving me around a little bit to take a free circulating phosphate and bind it," taking the ADP and turning it into ATP, triphosphate.
- 6.106** So, for every four hydrogen molecules that come through, it's able to shift around enough to create one new ATP. And that's where the accounting from earlier became so important, because for every one NADH molecule that came, we had four, two, and four or ten protons pumped, and that gives rise to one set of four coming through, creating one ATP, another set of four coming through, getting us another ATP, and then two more waiting, but we'll say, until we get some more, that means then for every one NADH molecule that starts at complex I, we can create the equivalent of two and a half ATP. And there is a lot of NADH that's coming through all of those processes we went through earlier.
- 6.107** And then, if there is one FADH, it having only pumped out six protons, we can only run one full cycle to create one full ATP, and then we're left with what is the equivalent to a half of an ATP. So, NADH can give two and a half ATP, FADH can give one and a half ATP.
- 6.108** If we go back to the citrate cycle then and appreciate the future ATP-producing power, every one cycle of the citrate cycle is to produce the equivalent of 10 ATP. And because every acetyl-CoA molecule can do this, you can begin to see the energy-producing or ATP-producing power of lipid. Because if you have a long set of fats, if you have an 18-carbon fat, you get nine acetyl-CoAs from that by splitting it into parts of two. Each one of those feeds the citrate cycle, and we're left with creating 90 ATP because of one single molecule, just with the citrate cycle alone.
- 6.109** Now, the final point I want to teach you to appreciate cellular metabolism is to briefly explore how the plant does it in opposite. So animal cells only have mitochondria. Mitos, the origin of this word—the Greek word for string or thread, because the mitochondria are often stringy. We look at them in a textbook and it looks like it's like a bean shape. That's actually because when you image the cell on an electron microscope, you're taking what a long string and slicing it in half, and then you're looking at it down the length. So, it looks like it's a round little kidney bean, but it is not. They're long stringy things.
- 6.110** Animal cells only have mitochondria, but plant cells, not only, actually, they cheat, they have a little bit of mitochondria, but they also have a plant exclusive parallel, which is the chloroplast. And whereas the mitochondria are long and stringy, the chloroplast are stacked like contact lenses. It's a stack of lenses, almost. The chloroplast is utterly unique because like I said, it's doing everything in reverse of the mitochondria.

- 6.111** Whereas the mitochondria is taking in carbons bound and oxygen and then creating energy, the chloroplast is taking in and releasing it— at the same time the mitochondria is creating energy and CO₂, the chloroplast is taking the CO₂ and energy from the sun to get enough power to start taking the carbons from the atmosphere, the carbon dioxide, and binding them into a carbon chain. So, it really is this opposite reaction that ends up enabling life on the planet.
- 6.112** We wouldn't want limitless plant growth. We don't want limitless animal growth. They're keeping each other somewhat in balance. One of the great paradoxes, though, of the modern moment is that as people are getting increasingly worried about CO₂ levels, it's actually resulting in this blossoming of the Earth like never before. Plants grow faster when there's more CO₂ because it is literally fuel for the plant; it becomes the building blocks of the plant.
- 6.113** So again, the chloroplast, that unique version of the mitochondria in the plant, is going to move everything in opposite because it can take advantage of the energy coming from the sun, unlike the animals.
- 6.114** At this point in all of this lecture of cellular metabolism, my hope is that you have a greater appreciation for what metabolism means, the balance of catabolic and anabolic reactions, and the necessity of the cell to be able to create energy to fuel all of the necessary work, not only to build up molecules, those important biomolecules, but also to get some work done: sending a nerve signal, contracting a muscle, whatever the job may be, the cell needs ATP to do that, and you have a better understanding now, including nutrient catabolism, citrate cycle, and then the inner workings of the mitochondria, namely the electron transport system and oxidative phosphorylation.
- 6.115** So, my final comment then is, at this point, you have gone from students to modest masters of biology in studying the interactions, the complex interactions, that make life possible on this incredible planet. And now you have a greater appreciation and understanding of life, what life is made up of, how to identify it, and also how living organisms function. Thank you so much.
- 6.116** Thank you for everything. What happens with the cell when we are taking statins?
- 6.117** Statins? Yeah. Statins are one of the most widely prescribed medications, which is largely a result of our, I would say, irrational fear of cholesterol. Statins are a medication that inhibits the body's ability to make cholesterol. Cholesterol is so essential that if we don't eat it, every cell has the ability to make it, which is one of the reasons why eating cholesterol has never been a problem. Even the most cholesterol-fearing scientists have admitted this over the decades.
- 6.118** Dietary cholesterol has no effect on blood cholesterol because if we don't eat it, we make it. But in fact, it is so important that every cell has the ability to make cholesterol. And that happens through an enzyme called HMG-CoA reductase. It's a long name of an enzyme, but it is the process of taking base fats and turning it into cholesterol. Statins

work by inhibiting HMG-CoA reductase. They stop the cell's ability to make new cholesterol.

- 6.119** That becomes particularly relevant at the liver. The liver is the soccer mom of all nutrient metabolism. It wants to give everything to the body and make sure everything has enough. The liver does that even with cholesterol. If there are some parts of the body that may need more cholesterol than it's capable of making, the liver is making it and releasing it into the blood to be carried on molecules called LDL. That's the main transporter. But it is just carrying cholesterol and other molecules.
- 6.120** So, what a clinician usually wants to see is that the LDL level has gone down, which it will if a person is making less cholesterol, then there's less need for these transport mechanisms to be carrying the cholesterol. So that's how the statins work, but there are consequences to statins. One of the most common consequences is muscle pain, a condition called rhabdomyolysis or the breakdown of muscle, but it hurts the muscle a lot. And that's likely because of the damage to the mitochondria.
- 6.121** If you reduce your cholesterol, you don't have enough ubiquinone; you do not have functioning electron transport. It can also damage the heart, potentially, creating heart failure where the muscle of the heart gets compromised. And also it can reduce the production of sex hormones, which can matter, by golly. So that's how statins work.
- 6.122** I co-authored a paper looking at the efficacy of statins. Typically, most clinicians will only diagnose a statin based purely on the LDL level. They will simply say, "You have a high LDL; you must go on a statin right now before you leave my office. Here's your prescription." However, we found that whether LDL levels were high or low didn't really predict the efficacy of the statin. In our paper that we published, we found that if a person had high LDL levels, the statin only helped if their triglyceride levels were also high in the blood, that glycerolipid that we've talked about.
- 6.123** Because when the liver makes LDL, yes, it puts cholesterol on there, but it also puts a lot of triglyceride on there. And if a person has high LDL and high triglycerides, then the statin appeared beneficial, but, for example, if someone adopts a low-carb or ketogenic or carnivore diet, they may have high LDL, but they will likely have very low levels of triglycerides. Based on our review of the evidence, a statin will provide them no benefit.
- 6.124** And then finally, one of the best predictors of longevity in humans—there are a handful of variables over the years, across the countries that have studied this, that have found these characteristics that tend to go along with people who live long and live well. One of the most persistent characteristics is high LDL cholesterol. So the irony here is that in our war that we've waged on LDL cholesterol in particular— the irony being it appears to be that the lower LDL gets, the more likely someone is to die.
- 6.125** For cancer cells, I was a little bit interested—you mentioned that they primarily eat glucose. Are all glucose molecules the same or whether it was produced by the liver

versus what we ate? Is that different? Also, if someone does have cancer and is trying to lower maybe the glucose that goes into the body, how will these cancer cells die, or how can we eliminate them?

- 6.126** Yeah, it's a great question. Yeah, to answer the first question, it does not matter where the glucose comes from. The glucose will be a fuel for cancer and, indeed, the primary fuel. Cancer cells use glucose at about 200 times the rate of a normal cell. Indeed, one of the ways that you can identify where a cancer is, where the tumor is, is by giving the person glucose, infusing glucose through IV with a glucose that has a slight radiated molecule on it. Not enough to do radiation damage, but you can see where the glucose goes. You can say, "Oh, it's going to lymph nodes in the breast, or it's going to the testicle in the male," because you can image this through what's called a PET scan.
- 6.127** The cancer cells pull in the glucose so fast that wherever that goes- and wherever you see the signal, that's where the tumors are. So, it doesn't matter where the glucose is coming from. But that does lead into your next question, which is, what about the role of carbohydrate restriction? Even though the body will always be making glucose because there are some cells that need it, even if you don't eat any glucose at all, the liver will be making enough to keep it in a generally normal range. But in that instance, the glucose will be at a very low end of the normal range rather than constantly being at a high end like it is in prediabetes or type 2 diabetes or just the average individual who's eating carbs all the time.
- 6.128** Because globally, about 71, 72% of all calories are coming from carbohydrate. We have a very carbohydrate-obsessed global diet, and we've been giving the perfectly asinine device to eat five or six times a day. So, people often have constantly elevated glucose. It's never given a chance to come even remotely down. Nevertheless, by restricting carbohydrates, we rely on any glucose, only glucose, that can be made from the liver. Thus, glucose levels are generally low.
- 6.129** At the same time, in that low insulin state, we're making a lot of ketones. This comes to answering your question, then. There are entire clinical studies that are going on now exploring the efficacy, and many that have already been done, of ketogenic diets.
- 6.130** Dr. Thomas Seyfried, who's a scientist at Boston University, has studied cancer and the role of ketones and ketogenic diets as a cancer therapy. They created the glucose ketone index, where you basically want your glucose level to be as close as possible to your ketone level. That index, if it's like one-to-one, that's a really good, aggressive way of starting to starve the cancer cell. They have found and published these reports that in some brain-based cancers like glioblastoma, which are very aggressive and often inoperable tumors, ketogenic diets can not only stop the growth of the tumor but even shrink these otherwise incurable cancers.
- 6.131** Other groups have found that ketogenic diets, by lowering the glucose and increasing the ketones, thereby starving the cancer cell or making it hungry, at least you can

increase the efficacy of other chemotherapeutics. So, other cancer-based drugs, these poisons that are trying to kill the cancer by putting the body into ketosis and getting that glucose to ketone level as close to one-to-one as possible, you make the cancer cell—the combination of hunger with the poison kills it faster.

6.132 Thanks, guys.